TFPT — E_8 Audit, Cascade Bridge and Bootstrap

The seven E_8 slices as an audit raster, the old cascade as the same even-integer spine, the Möbius self-consistency loop — and the thirty-two-step celestial/twistor route with the measure chain derived

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What this document is about

E_8 as an *audit container*, not a mystery: the seven maximal slices of 248 as a falsification raster (every load-bearing number must appear in at least one projection), the bridge showing the old E_8 orbit cascade $D{=}60{-}2n$ is the same even-integer spine as the compiler, and the Möbius bootstrap in which $g_{\mathrm{car}} = 5$ and the "8" in c_3 are overdetermined E_8-closure fixed points (only π irreducible). Since the celestial/twistor round the note also carries the **thirty-two-step continuum story** (§17) — from the μ_4 clock to the $(E_8)_1$ boundary shadow — in which the **measure chain of the celestial route is derived, not declared**: single-valuedness from F-independence + the Quillen pairing (v520), the $1/\det_j$ normalisation the Atiyah–Bott fixed-point factor computed from three independent sources with $\psi = 64$ reproduced (v523), the residual **[O]** narrowed to the global BCOV integral beyond the fibre zero-mode factor; plus the Woit OS-bridge stages $\alpha/\beta_1/\beta_2/\beta_3$ (v519/v522/v524/v565 — the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed: the OS cut and the $(2, 2)$ signature forced, the kill branch empty by algebra, the induced member = the Θ_{phys} carrier), the thermal seam (v526), the ten-test side-blind scoreboard with the twist-class definition of the bit (v528) and the first firing interacting kill test under its typed fence (v529) — whose straddle filter, since run as a *selector*, keeps exactly *one* member alive: reflection positivity dynamically selects the alignment bit $\delta = \pi/2$ with positive coupling (v534, toy level). `SEAM.EQUIV.01` and `WOIT.OS.TWISTOR.01` stay **[O]**; no marker moves.

What E_8 is *not* (and why the known no-go results do not apply)

E_8 here is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures — **not** an unbroken physical gauge group. The Standard Model is a *readout* after projection (not a direct E_8 gauge theory), the Lorentz signature appears only after projection, and fermions are not "everything in the adjoint". Consequently the standard objections to literal E_8 world-formulas do *not* bite: Distler–Garibaldi (no chiral fermions / representation obstructions for E_8 as a 4d gauge group) and Coleman–Mandula (no nontrivial mixing of spacetime and internal symmetry) both constrain E_8 as a *spacetime gauge symmetry*, which TFPT does not claim. The defensible statement is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull* — a discrete classifier, attacked on its own (combinatorial) terms, not on gauge-theoretic ones.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\mathrm{geo}}, G_{\mathrm{net}}, F_{\mathrm{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

You are reading document 3 (E_8 Audit & Bootstrap).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2| \chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2) e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\rm geo}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\rm net}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit $v_{\rm geo}$ plus the typed $F_{\rm transfer}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\rm wall}, G_{\rm metric}, F_{\rm frontier}$) are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven **[C]** conditional **[O]** open / axiom **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Part I

The E_8 secondary-branching atlas

Purpose and discipline. This note does *not* introduce new physics. It builds an *audit raster*: the claim that *every load-bearing TFPT number should appear in at least one E_8 branching projection.* A number that appears in none is suspect; a number appearing in several independent projections becomes structurally strong. All arithmetic identities below are exact [E]; the *physical readings* are audit-level [O] unless tied to a derivation. (All 77 numeric claims were machine-checked with exact integer/rational arithmetic.)

1 The compiler core (where this note sits)

The strongest current form of the theory is a five-layer compiler:

L1 Inputs $c_3=\frac{1}{8\pi}$, $g_{\rm car}=5$ (3+2) → **L2 Geometry** $C^+=D_5$, $\mathbb{P}^1\setminus\mu_4=A_3$, $(D_5\oplus A_3)+\mu_4=E_8$ → **L3 Flavor** $E_\bullet=\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$, $a=(1,1,2)$, R, L → **L4 Seed** $\varphi_0^{\rm ret}$, α, ρ_Λ, $M_{\rm scal}$ → **L5 Readout** SM, CKM, PMNS, A_s

E_8 is now a *closure* object, not an input: the carrier is the $D_5=\mathfrak{so}_{10}$ half-spinor, the four punctures $\mathbb{P}^1\setminus\mu_4$ give $A_3=\mathfrak{su}_4$, and the common discriminant group $\mathbb{Z}_4$ glues them, with the norm test $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$. This note charts the *secondary* maximal subalgebras of E_8 as audit windows onto the TFPT modules.

2 The $(1,1,2)$ anchor microcode

The parabolic anchor $a=(1,1,2)$ (splitting type of $E_\bullet=\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$) is a miniature compiler. Its power sums $p_n(a)=1^n+1^n+2^n=2+2^n$ generate the discrete budget:

n	$p_n=2+2^n$	**TFPT reading (verified)**
1	4	$\lvert\mu_4\rvert=-\deg E_\bullet$ (glue index)
2	6	$\lvert R^+(A_3)\rvert$ (positive A_3 roots) $=\mathbb{Z}_6$ cusp order
3	10	$\mathcal{A}_\Lambda=\binom{5}{2}=\dim V_{D_5}$ (action ladder / pair sector)
4	18	$N_{\rm fam}\,\lvert R^+(A_3)\rvert=3\cdot 6$
5	34	$2h(D_5)+3\lvert R^+(A_3)\rvert=16+18$; and $34-\lvert R(A_3)\rvert=22=\sum R$

and the budget compresses to products of power sums:

$$q(A_3)=\frac{p_2}{2p_1}=\frac{3}{4},\quad q(D_5)=\frac{p_3}{2p_1}=\frac{5}{4},\quad \sum L=p_1p_3=40,$$

$$\Omega_{\rm adm}=2p_1p_2=48,\quad D_{\rm start}=p_2p_3=60,\quad \lvert R(E_8)\rvert=4p_2p_3=240.$$

The first three power sums alone reach E_8: $p_1p_2p_3=4\cdot6\cdot10=240=\lvert R(E_8)\rvert$, and $p_1p_2p_3+(p_4-p_3)=240+8=248=\dim E_8$.

The anchor difference ladder: the binary carrier spine [E]

While the *products* of the power sums give the large budgets, the *differences* give the binary carrier spine in one line:

$$\boxed{p_n(a)=2+2^n\implies\Delta p_n=p_{n+1}-p_n=2^n}:\quad(2,4,8,16)=\big(\lvert\mathbb{Z}_2\rvert,\,\lvert\mu_4\rvert,\,h(D_5),\,\dim S^+\big).$$

So the single anchor $a=(1,1,2)$ generates the big numbers multiplicatively and the doubling spine $2,4,8,16$ additively — one vector, two readings.

The anchor reconstructs the carrier rank — and the 41-chain

The characteristic polynomial of the anchor vector is $(t-1)^2(t-2) = t^3 - 4t^2 + 5t - 2$, with elementary symmetric data

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\text{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|_{\text{sheet}}.$$

So the parabolic flavor geometry reads the carrier rank $g_{\text{car}} = 5$ *backwards*: $D_5 \Rightarrow a$ but also $a \Rightarrow g_{\text{car}}$. Moreover $3^2 + 4^2 = 5^2$ sits as $(p_0, e_1, e_2) = (3, 4, 5)$, giving the *anchor form* of the abelian index

$$\boxed{10\, b_1 = e_1(a)^2 + e_2(a)^2 = 4^2 + 5^2 = 41}$$

alongside the known $41 = 40 + 1 = \sum L + N_\Phi$. [E]

Pascal closure: $g_{\text{car}} = 5$ is the unique solution (kills alternatives) [E]

The carrier rank is not merely a good fit — it is the *unique* root of an exact closure. The even-exterior carrier dimension 2^{g-1} must equal the truncated Pascal sum $\binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ (the $1+g+\binom{g}{2}$ that builds $\dim S^+$):

$$\boxed{2^{\,g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2} \iff g = 5}$$

($g{=}3 : 4{\neq}7$; $g{=}4 : 8{\neq}11$; $\mathbf{g{=}5 : 16{=}16}$; $g{=}6 : 32{\neq}22$; $g{=}7 : 64{\neq}29$). At $g = 5$ the row is $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$, the same Pascal triple that reads family, action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. This is the fourth independent lock on $g_{\text{car}} = 5$ (with rank-fill, Coxeter-match, and integer-glue/norm), and unlike those it explicitly *kills* the neighbours $g = 4, 6$.

The Pascal Ladder Theorem (`[verification/v108_pascal_ladder.py]`). The closure generalises exactly: for *every* truncation degree K,

$$\boxed{2^{\,g-1} = \textstyle\sum_{k \le K} \binom{g}{k} \iff g = 2K + 1}$$

(odd g solves by the binomial midpoint identity; even g is excluded by the strict straddle $\sum_{k<g/2} < 2^{g-1} < \sum_{k \le g/2}$). **Consequence (zero slack): the carrier selection $g = 5$ is *equivalent* to the truncation degree $K = 2$** — the Pascal-*selection* residual of the red team (Target B) is therefore exactly the one named hypothesis *Quadratic Boundary Locality* (the seam certifies code degrees ≤ 2; the architecture document). Neighbour worlds on the ladder: $K{=}1 \to g{=}3$ (a *one*-family world), $K{=}2 \to g{=}5$ (three families), $K{=}3 \to g{=}7$ (*nine* families), $K{=}4 \to g{=}9$ *inconsistent* ($N_{\text{fam}} = 255/9 \notin \mathbb{Z}$). Overdetermination: among the ladder worlds, only $K = 2$ also satisfies the independent rank closure $g + N_{\text{fam}} = 8 = \text{rank}\, E_8$ — two independent selections agree on the same world. The hypothesis itself stays [C].

The seed is an anchor expression

The retained seed is exactly the two-term anchor combination of c_3:

$$\boxed{\varphi_0^{\mathrm{ret}} = \frac{2p_1}{p_2}\,c_3 + 2p_1p_2\,c_3^4 = \frac{4}{3}c_3 + 48\,c_3^4 = \frac{1}{6\pi} + \frac{3}{256\pi^4}}$$

(verified to 30 digits). The Cabibbo base is $\lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}$, the hierarchy engine of the whole fermion spectrum. [E]

3 The residue and length matrices as spectral objects

With the residue matrix R and the full word-length matrix $L = R + 6W$ (W the rank-one winding),

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \qquad L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

the spectral invariants are pure compiler numbers (all verified):

Invariant of R	**value**	**Invariant of L**	**value**
$\operatorname{tr} R$	$9 = N_{\mathrm{fam}}^2$	$\operatorname{tr} L$	15
$\det R$	$8 = h(D_5)$	$\det L$	$20 = \lvert R(A_4)\rvert = 2\mathcal{A}_\Lambda$
$\chi_R(t)$	$t^3 - 9t^2 + 10t - 8$	$\chi_L(t)$	$t^3 - 15t^2 + 40t - 20$
$\mathrm{PrinMin}_2(R)$	$(2,3,5)$	$\mathrm{PrinMin}_2(L)$	$(5,14,21)$
$\lVert R\rVert_F^2$	$78 = \dim E_6$	—	—

The three principal 2×2 minors of R are $(2,3,5)$ — sheet, family, carrier — with product $2\cdot3\cdot5 = 30 = h(E_8)$ and sum $10 = \mathcal{A}_\Lambda$. The wrong down-branch $\{1,3,4\}$ fails exactly these invariants.

Bilinear anchor forms. With $\mathbf{1} = (1,1,1)^\top$ and $a = (1,1,2)^\top$ (all verified):

R	$\mathbf{1}$	a	L	$\mathbf{1}$	a
$\mathbf{1}^\top$	$22 = \sum R$	$27 = 3^3$	$\mathbf{1}^\top$	$40 = \lvert R(D_5)\rvert$	$45 = \dim D_5$
$a^\top$	$32 = 2^5$	$40 = \lvert R(D_5)\rvert$	$a^\top$	$56 = \dim \mathbf{56}_{E_7}$	$64 = \dim S^+ \cdot \lvert\mu_4\rvert$

The corner value $a^\top L\,a = 64 = (16,4)$ reads exactly one spinor-glue sheet of the E_8 branching. And the rank-one winding update shows up in the inverses:

$$\boxed{R^{-1}\mathbf{1} = \tfrac{1}{4}(1,1,-1)^\top, \qquad L^{-1}\mathbf{1} = \tfrac{1}{10}(1,1,-1)^\top}$$

i.e. the residue response reads the glue denominator $\lvert\mu_4\rvert = 4$, the full-length response reads the decuple denominator $\mathcal{A}_\Lambda = 10$. [E]

Anchor-plane Plücker coordinates: scattered hits become oriented areas [E]

The strongest anti-numerology tool is to read *orientation invariants* of the anchor plane $\langle \mathbf{1}, a\rangle$ rather than loose integers. For $M \in \{R, Q, K, L\}$ the left block $B_M = \big(\mathbf{1}^\top M;\, a^\top M\big)$ (a 2×3) and the right block $C_M = (M\mathbf{1} \;\; Ma)$ (a 3×2) have the 2×2 Plücker minors

	$\mathrm{Pl}(\cdot)$ (left)	$\mathrm{Pl}_R(\cdot)$ (right)
R	$(-6, 2, 14)$	$(8, 12, 4) = (h(D_5), \lvert R(A_3)\rvert, \lvert\mu_4\rvert)$
Q	$(3, 6, 3)$	$(0, 4, 5) = (0, \lvert\mu_4\rvert, g_{\mathrm{car}})$
K	$(-1, 6, 4) = (-N_\Phi, \lvert R^+(A_3)\rvert, \lvert\mu_4\rvert)$	$(12, 12, -2) = (\lvert R(A_3)\rvert, \lvert R(A_3)\rvert, -\lvert\mathbb{Z}_2\rvert)$
L	$(6, 26, 14)$	$(20, 30, 10) = (2\mathcal{A}_\Lambda, h(E_8), \mathcal{A}_\Lambda)$

Three load-bearing flavor numbers are now single oriented areas, not coincidences:

$$\boxed{\|\mathrm{Pl}(K)\|_1 = 11, \qquad \|\mathrm{Pl}_R(K)\|_1 = \mathrm{Pl}(L)_{13} = 26, \qquad \sum \mathrm{Pl}_R(L) = 60 = D_{\mathrm{start}}}$$

(11 is the mass-power anchor-plane norm; 26 the c_t/c_b denominator; 60 the E_8 cascade start). The left-null-space response complements the right one above: the column sums of $\det L \cdot L^{-1}$ and $\det R \cdot R^{-1}$ are $(-5, 1, 6)$ and $(1, -5, 6)$, the same magnitudes $\{N_\Phi, g_{\mathrm{car}}, |R^+(A_3)|\} = \{1, 5, 6\}$. Finally the sector rows themselves are anchor-shifted: $K_e - K_u = (1, 1, 2) = a$ and $L_e - L_u = (1, 2, 3) = \mathrm{Exp}(A_3)$ — leptons are up-quarks displaced by the family roots. All entries are exact [E]; this is an audit layer, not a closure (the physical quark amplitudes stay [C]). [`verification/v37_plucker_anchor.py`]

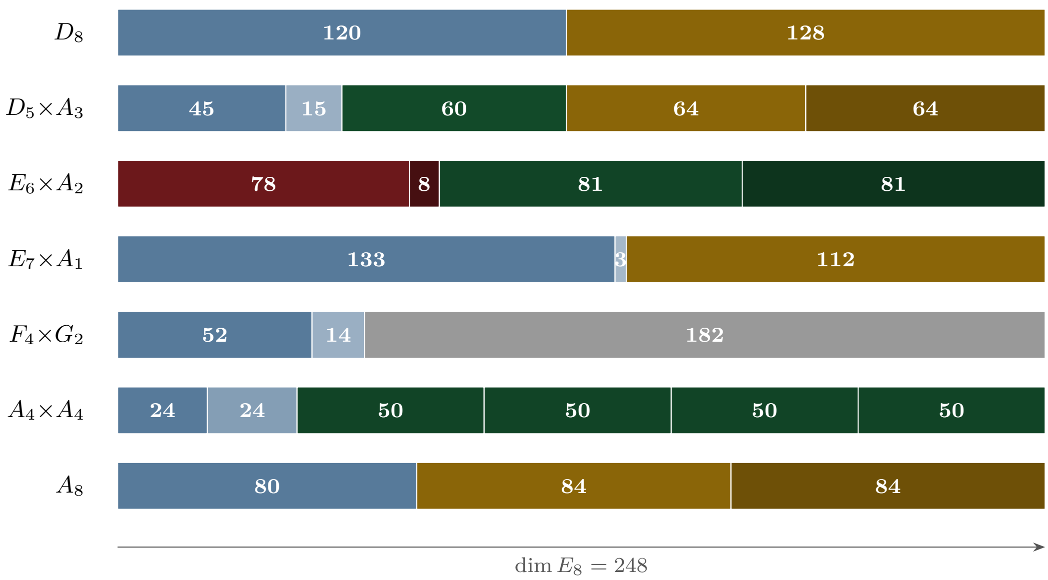


Figure 1: The E_8 slice atlas: seven maximal-subalgebra cuts of 248. Gold = spinor-glue (64=(16, 4), 112=7·16, 84=7·12, 128); red = flavor-residue (78=$\|R\|_F^2$, 8= det R); green = family/action blocks; blue = Lie adjoints.

The numerology null test: the look-elsewhere burden quantified [E]

The raster above disciplines *which* integers may appear; the complementary statistical question — *could a random "theory" of equal formula complexity land this many data hits?* — is answered by an explicit, fully declared null model. For each of the 13 scored frozen-registry observables the *entire* grammar class is enumerated exactly (no sampling): one-term expressions $\frac{p}{q}\,(\varphi_0^{\text{ret}})^a\lambda_C^c(1{+}\lambda_C)^d\pi^b e^{u/w}$ with $p, q \le 130$, $|a| \le 2$, $|c| \le 3$, $|d|, |b| \le 1$, $|u| \le 6$, $w \le 6$, and two-term sums with coefficients ≤ 12 — a class that provably contains every TFPT formula it scores (machine-checked fairness condition). Data windows are conservative: the *maximum* of the achieved deviation, the documented experimental σ, the quote resolution and the 5% quark scheme spread, so the known tension observables ($\sin^2\theta_{13}$, s_{23}, δ_{CKM}, m_μ/m_τ, β) enter with their honest *large* windows and contribute almost nothing. Result of the census: joint formula-fishing probability $\prod_i p_i = 10^{-25.8}$, stable to < 2 orders of magnitude when the coefficient bound is varied 80 . . . 200 and the plausibility window 5× . . . 20×. Monte-Carlo cross-check (deterministic seeds): $2 \times 2{\cdot}10^5$ pseudo-theories — random rational dials inside TFPT's own formula skeletons, with the seed φ_0^{ret} fixed resp. drawn log-uniformly — reach at most 5/13 resp. 4/13 hits versus TFPT's 13/13. The test has *power*: perturbing φ_0^{ret} by 1%/10%/30% collapses TFPT's own scorecard to 9/6/4 hits, and shuffling which measured value is assigned to which formula collapses it to a mean of 1.2. Separately, all 94 500 complexity-matched variants of the $F_{U(1)}$ fixed-point equation (coefficients, M, transport exponent, resolvent power) are root-solved: exactly *one* — the TFPT instance $(2, \frac{4}{5}, M{=}41, e^{-2\alpha}, -\frac{5}{4})$ — lands inside the achieved $4{\cdot}10^{-8}$ CODATA window, so $p_\alpha \le 1.1{\cdot}10^{-5}$. Headline:

$$\boxed{P\big(\text{null reproduces the scorecard incl. } \alpha\big) \;\le\; 10^{-30.7} \quad (\approx 102 \text{ bits})}$$

conditional on the declared grammar — a look-elsewhere-corrected null-model rejection, never "certainty". Complementary to the frozen registry: the freeze (v84) prevents *future* look-elsewhere drift, this test quantifies the *retrospective* look-elsewhere burden. [`verification/v100_numerology_null_mc.py`]

4 The secondary branching atlas

Each maximal subalgebra of E_8 mirrors a TFPT module. The dimension identities are exact [**E**]; the readings are audit-level [**O**]. The seven *slices* of dim $E_8 = 248$ are shown below: the same total cut seven ways, each cut a different TFPT module.

Subalgebra	$248 =$	**TFPT reading**
D_8	$120 + 128$	family-transition adjoint + two spinor-glue sheets ($128{=}2\dim S^+\lvert\mu_4\rvert$)
$D_5 \times A_3$	$45 + 15 + 60 + 64 + 64$	**main split**: $\dim D_5$, $\dim A_3$, $\mathcal{A}_\Lambda\lvert R^+(A_3)\rvert$, two $(16, 4)$ glue sheets
$E_6 \times A_2$	$78 + 8 + 81 + 81$	**flavor residue**: $\lVert R\rVert_F^2$, $\det R{=}\dim A_2$, cubic family blocks
$E_7 \times A_1$	$133 + 3 + 112$	**scalaron**: $112{=}7 \cdot 16$ (exponent × one family)
$F_4 \times G_2$	$52 + 14 + 182$	**occupancy/gauge**: $\lvert R(F_4)\rvert{=}48{=}\Omega_{\text{adm}}$, $\lvert R(G_2)\rvert{=}12{=}\dim \mathfrak{g}_{\text{SM}}$
$A_4 \times A_4$	$24 + 24 + 4 \cdot 50$	**action ladder**: $24{+}24{=}48{=}\Omega_{\text{adm}}$; off-diagonal $(5, 10)$ reps $\mathcal{A}_H, \mathcal{A}_\Lambda$
A_8	$80 + 84 + 84$	rank-decuple + scalaron: $80{=}8 \cdot 10$, $84{=}7 \cdot 12$

The D_8 split is a magnitude/phase channel [E]/[C]/[O] ([`verification/v227_degree_exponent_channel_split.py`])

The D_8 row $248 = 120 + 128$ is the canonical *exponent/degree* split of E_8: the exponents sum to $120 = \lvert R^+(E_8)\rvert$ (a **magnitude** channel — positive roots, root transitions, the dimensionless ratio/product readouts), the fundamental degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum to $128 = 2^{\text{rank}-1} = \text{rank}\, E_8 \cdot \dim S^+$ (a **phase/glue** channel — the spinor/Ramond sheets). This gives the red-team Target D CP residual a clean home: the (ratios, product) $\Rightarrow v_{\text{geo}}$ magnitude bijection lives in the 120 channel, the un-covered CP phases in the 128 channel (cf. the hexagonal phase fiber, red team Target D). Whether 128 also hosts a dark sector is Frontier [O] — this is the honest replacement for the over-strong "S^- is dark matter" reading, not an assertion of it. The conjugate sheet S^- and the double-cover deck sit in this *matched* channel ($\lvert\mathbb{Z}_2\rvert = 2$ and $128 = \sum \deg$), forced-disjoint from the five unmapped degrees $\{12, 14, 18, 20, 24\}$ (v430): the seam's "other side" is not the hull overhead.

E_8 Slice Compression Lemma [E] (audit)

The seven slices are not seven separate narratives — they are *one* projection of two small alphabets. The anchor power sums $p_n(a) = 2 + 2^n$ give $P = (p_0, p_1, p_2, p_3) = (3, 4, 6, 10)$, whose six K_4 edge products are exactly the carrier blocks $(p_0p_1, \dots, p_2p_3) = (12, 18, 30, 24, 40, 60) = (\lvert R(A_3)\rvert, N_{\text{fam}}\lvert R^+(A_3)\rvert, h(E_8), \lvert W(A_3)\rvert, \lvert R(D_5)\rvert, D_{\text{start}})$; the six Sheet-Diamond operators carry determinants $(\det Q, \det K, \det R, \det C, \det L, \det F) = (3, 4, 8, 14, 20, 32)$ with $\sum \det = 81 = N_{\text{fam}}^4$, the total determinant charge of the flavour space. All seven 248-slices, the row-budget cross $\text{row}(C + sU + tV) = (7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ (democratic winding + A_3 exponents), and $\dim E_6 = 78 = p_2(p_0 + p_3)$ then follow from P, those determinants and $\Delta = p_0 + p_3 = 13$. The atlas is therefore a closed grammar over admissible invariant classes (power sums, determinants, branching dimensions), not a shelf of trophies ([`verification/v170_diamond_slice_compression.py`]).

Theorem 1 ($E_6 \times A_2$ flavor-residue shadow [O])**.** The principal flavor branching reads the residue matrix:

$$\lVert R\rVert_F^2 = 78 = \dim E_6, \qquad \det R = 8 = \dim A_2 = h(D_5), \qquad \mathbf{1}^\top R\, a = 27,$$

and the full $E_6 \times A_2$ dimension count is the residue identity

$$\boxed{248 = \lVert R\rVert_F^2 + \det R + 2\,(\mathbf{1}^\top R\, a)\, N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3.}$$

Here E_6 reads the Frobenius norm of the residue matrix, A_2 the pure three-family symmetry, and the **27** the cubic family block. (Strongest new E_8 flavor fingerprint.)

Theorem 2 ($E_7 \times A_1$ scalaron shadow [O])**.** With the scalaron exponent $7 = \Omega_{\text{adm}} - 10b_1 = -\deg E + \text{rk}\, E$,

$$248 = 133 + 3 + 112, \qquad 112 = 7 \cdot \dim S^+ = 7 \cdot 16, \qquad \lvert R(E_7)\rvert = 126 = 7 \cdot 18,$$

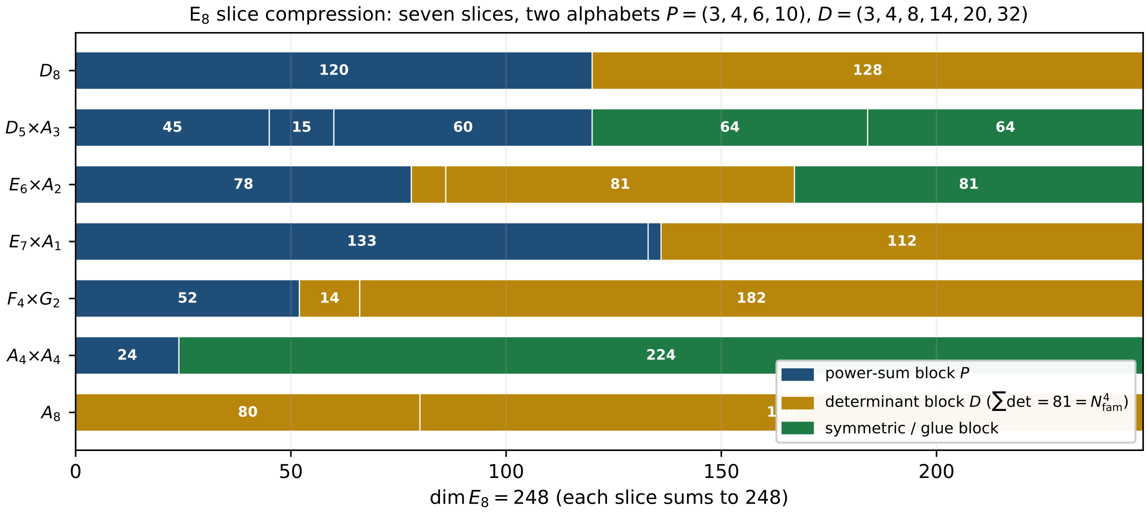


Figure 2: The same seven 248-slices, now coloured by their *alphabet* source rather than by physics module: blue = anchor power-sum blocks ($P = (3,4,6,10)$), gold = Sheet-Diamond determinant blocks ($D = (3,4,8,14,20,32)$, $\sum \det = 81 = N_{\mathrm{fam}}^4$), green = symmetric/glue data. Every slice is one projection of the two alphabets ([`verification/v170_diamond_slice_compression.py`]).

where $18 = N_{\mathrm{fam}}|R^+(A_3)|$. The scalaron exponent 7 (the seam power $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2 = c_3^7$) is thus exactly the E_7 off-block coefficient. *Cross-check:* the centred E_8 exponents give $\sum_{m\in\mathrm{Exp}(E_8)} |m-15| = 56 = \dim \mathbf{56}_{E_7}$ (exponents $\{1,7,11,13,17,19,23,29\}$, the primitive residues mod 30).

Proposition 1 ($A_4 \times A_4$ action-ladder and $F_4 \times G_2$ occupancy [O]). $248 = 24+24+4(5\cdot 10)$ with $24+24 = 48 = \Omega_{\mathrm{adm}}$ and off-diagonal reps $(\mathbf{5},\mathbf{10}) = (\mathcal{A}_H, \mathcal{A}_\Lambda)$ — the action ladder appears inside E_8 (an audit, *not* an $SU(5)$ GUT). And $F_4 \times G_2$: $|R(F_4)| = 48 = \Omega_{\mathrm{adm}}$, $|R(G_2)| = 12 = \dim \mathfrak{g}_{\mathrm{SM}}$, rank $F_4 = 4 = |\mu_4|$, rank $G_2 = 2 = |\mathbb{Z}_2|$.

Flavor-matrix and prime-table fingerprints (audit-level, machine-verified). Five exact linear-algebra readouts of R, L and the prime atoms, kept here under the no-free-pattern rule.

- **Row norms ladder.** $\|R_u\|^2, \|R_d\|^2, \|R_e\|^2 = (10,30,38) = (\mathcal{A}_\Lambda,\ h(E_8),\ 2\cdot 19)$ with $10{+}30{+}38 = 78 = \dim E_6$ ($19 \in \mathrm{Exp}(E_8)$).
- **Residue inventory.** The entries of R on C_6 are $m = (1,2,2,2,0,2)$: the *only* gap is $4 = |\mu_4| = \sum a$. Moments $\sum R = 22$, $\sum R^2 = 78 = \dim E_6$ — so $\|R\|_F^2 = 78$ is the second moment of the C_6 inventory with the μ_4 hole.
- **First-column norm $\to$ $E_6{\times}A_2$ off-blocks.** $L_{\cdot 1} = (7,7,8)$, $\|L_{\cdot 1}\|^2 = 162 = 81+81$ — the two **81** off-blocks of the $E_6{\times}A_2$ slice are exactly the wound first-generation column.
- **Lepton exponent polynomial.** The φ_0^{ret}-ladder lepton powers $(k_e, k_\mu, k_\tau) = (5,3,2)$ are the Coxeter primes $30 = 2\cdot 3\cdot 5$ in hierarchy order: $(t-5)(t-3)(t-2) = t^3 - 10t^2 + 31t - 30$ ($e_1 = \mathcal{A}_\Lambda$, $e_3 = h(E_8)$). The lepton c-ratios obey $\frac{(16/7)(7/6)}{4/3} = 2 = |\mathbb{Z}_2|$ and $\frac{(16/7)(7/6)}{(4/3)^2} = \frac{3}{2}$ (the near-Koide factor).
- **The** $2,3,5$ **multiplication table** (the shortest integer-landscape compression):

$$(2,3,5) \xrightarrow{\times} (6,10,15,30) \to (8{=}3{+}5,\ 16,\ 240{=}8\cdot 30,\ 248{=}240{+}8).$$

5 Group-theoretic audits (optional)

Weyl completion index. Over the main stabiliser,

$$\frac{|W(E_8)|}{|W(D_5)|\,|W(A_3)|} = \frac{696729600}{1920\cdot 24} = 15120 = 63\cdot 240 = (2^6-1)\,|R(E_8)|.$$

A group-theory machine-check, not physics: it quantifies the Weyl symmetry created beyond the $D_5 \times A_3$ stabiliser. [E] (arithmetic) / [O] (reading)

E_8 theta series as a higher-shell raster. $\Theta_{E_8} = E_4 = 1+240\sum_{n\geq 1}\sigma_3(n)q^n$ has shell degeneracies $240,\ 2160,\ 6720,\ 17520,\ldots$ The first is $|R(E_8)| = 240$; the second is $2160 = 9\cdot 240 = N_{\mathrm{fam}}^2|R(E_8)|$. The prime shells read the sheet-generator bilinears of v410: $\sigma_3(2) = 9 = a^T V\mathbf{1} = N_{\mathrm{fam}}^2$, $\sigma_3(3) = 28 = \mathbf{1}^T V^3 a = \det(I+R)$ and $\sigma_3(5) = 126$ — the local E_7 root shell $\{0{:}126\}$ of lemma (v) below. If later one-loop or CMB shell degeneracies touch these multiplicities, the theta series is the right place to type them — as an audit raster, not a claim. [O] ([`verification/v410_sheet_generator_binary.py`])

Hecke from geometry on the glue census (`HECKE.GEOM.01`). The same E_8 theta/q-series channels that feed the atlas raster now carry an *in-suite* Hecke mechanics, machine-checked as one module ([`verification/v535_hecke_from_geometry.py`]). *(A)* The classical Kneser p-neighbour correspondence (isotropic lines of $(E_8/pE_8, q)$) factors as $\#\mathrm{iso\_lines} = \sigma_3(p)\,\#\mathbb{P}^3(\mathbb{F}_p)$ identically and by full enumeration at $p = 2,3,5,7$ (135/1120/19656/137600) — the Eisenstein eigenvalue $\sigma_3(p) = 1+p^3$ is a lattice-native correspondence count. *(B)* The marked neighbour-sum operator is an affine Hecke element $\nu_p = a\,\mathrm{Id} + b\,T_p$ with $b = \sigma_3 + a_p$ and cusp rule $a_p = b - \sigma_3$, recovering $a_3 = -4$, $a_5 = -2$ (and $p = 7$ consistency $\lambda_{\mathrm{odd}} = 19840$). *(C)* The seven-dimensional census form space is exactly the 2-adic oldform hull of two weight-4 systems (5 copies of E_4 at levels $1,2,4,8,16$ plus 2 copies of f_8 at levels $8,16$), with recovery projectors $\pi_{\mathrm{cusp}} = (28-T_3)/32$ and newform via $(1-V_2U_2)$. Classical theorems (Kneser, Hecke, Atkin–Lehner, multiplicity one) are named classical; the TFPT content is the compiler identification. Honest fence: both new systems stay weight 4 — the GL(1)/weight-drop transport remains [O]; no RH claim. [E] (mechanics) / [O] (weight drop)

Eichler trace layer at the E_8 lattice (`HECKE.GEOM.EICHLER.01`). Building on the Hecke package without duplicating it, the census channels now carry a machine-checked Eichler/Witt layer ([`verification/v536_eichler_trace_layer.py`]; 23 checks, $\sim 30\,$s). Witt $\lambda_{\mathrm{Eis}} = L - \sigma_3^2$ is closed; the identity $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$ is two-sided at $p = 3$ (live; $352 = 364 - 12$) and at $p = 5$ via a T36-validated freeze ($3784 = 3906 - 122$), with assembler $R = \sigma_3 - 1 - c(p^2)/8 = a_p^2$. Closed Type-A/B densities $N_A = \min(240(1+p^3), \#\mathrm{iso} - 1)$ give the $p = 7$ prediction 82560/743040 live; signed $a_p = -c(p)/8$ recovers $(-4, -2, +24)$. Classical Eichler/Siegel/Witt named classical; $\mathrm{Shell}(p^2)$ for $p \geq 7$ out of scope; weight-4 $\to$GL(1) stays [O]; no RH claim; no marker moves. [E] (mechanics) / [O] (weight drop)

Half-integral bridge (`HECKE.GEOM.HALFINT.01`). Companion package on the same f_8 channel ([`verification/v537_halfintegral_bridge.py`]; 20 checks, $\sim 90\,$s). Unique signed Shimura preimage $\mathrm{Sh}_{t=2}(g) = -8\,f_8$ in the 70-element weight-5/2 compiler theta-monoid; exact $T(p^2)$-equivariance with eigenvalues $a_p(f_8)$; $U_4(g) = 0$ and level $32 = 4\cdot 8$ outside Kohnen 1982 (plus-route closed); AFE Waldspurger quotient $R(d)$ constant on ten $d \equiv 1 \pmod 8$ ($R = 23.1873585645\ldots$, spread $\leq 10^{-12}$). Classical scaffolding named classical; GL(2) centre $s = 2$ (not ξ); no RH; no marker moves. [E] (Shimura/Hecke/Kohnen fence) / [C] (measured R)

Compiler relative-trace identity (`HECKE.GEOM.RTF.01`). Synthesis of the three Hecke/Eichler/half-integral packages as projections of one finite relative-trace identity ([`verification/v538_relative_trace_identity.py`]; 18 checks, $\sim 14\,$s). Bilateral $\mathrm{Tr}_V(\nu_p \circ \pi)$ equality at $p = 3,5,7$ (geometry without modular input = spectrum without geometry); Eichler $\lambda_{\mathrm{Eis}} + a_p^2$ as the RTF orbit dictionary; period side = signed quaternary lattice counting with measured $R = 23.1873585645$. Verdict `ONE-FORMULA`. Classical Jacquet RTF named classical; finite identity; infinite/archimedean extension open; no RH; no marker moves. [E] (R1/R2) / [C] (R3)

Weil structure of the compiler family (`RTF.GNS.WEIL.01`). Route-B synthesis: fully identified Weil structure up to two isolated obstructions ([`verification/v539_weil_structure_family.py`]; 25 checks, $\sim 6\,$s). GNS fibre isolation (twist-mix $\leq 5.037\times 10^{-16}$); GL(1) core $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$ with $\sum 2^{\omega(n)} n^{-u}$; $Q_{\mathrm{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_\flat) + \mathrm{Arch} + \mathrm{Corr}$. Obstruction 1: doubling enters with a minus (family positivity $\not\Rightarrow Q_\zeta \geq 0$). Obstruction 2: Plancherel Corr non-automorphic ($e^{-\sum p^{-u}}$-type). Not "almost RH"; dense-class positivity open; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (structure/obstructions) / [C] (finite-class Q_{fam})

Amplitude route and positive linear carrier (`RTF.GNS.AMP.01`). The route out of the square plane behind the two v539 obstructions ([`verification/v540_amplitude_linear_carrier.py`]; 34 checks, ~ 3 s; T67–T72). Amplitude Dirac $D^2 = \mathrm{diag}(VV^\mathsf{T}, K)$ exact and Hecke-equivariant; polarisation $b = N_+ - N_-$ with $\Theta = N_+ + N_-$ a pure σ_3-eigenform and Cohen seed $\Theta(d) = -48\,L(-1, \chi_d)$ (exact-rational, 159 live d); Cauchy–Littlewood deletion pair-independent at $st = uv = p^3$ *and* equal to the square-class double counting (fam $+ 2\flat + \delta_{k1} = [2, 2, 2, 2]$); positive linear carrier $\ell^2(d, 48|L(-1, \chi_d)|\,|d|^{-a})$ with full weights $[1, 1, 1, 1]$ and plus balance $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$; FE $\Lambda_\Theta(s) = 8^{1-s}\Lambda_{\Theta^\dagger}(5/2 - s)$ exact (rel $\sim 10^{-40}$, Fricke closed). Open boundary as content: Euler-region positivity only; the residual distance to the Weil cone is the FE-covariant gap functional λ^* on $n \equiv 6 \pmod 8$ (Farkas-certified). Not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (A–E exact) / [C] (sampled cone facts; λ^* named open)

Matching lemma and transport ledger package (`RTF.GNS.LEDGER.01`). The T78–T85 proof package, recomputed as one module ([`verification/v541_matching_lemma_ledger.py`]; 33 checks, ~ 10 s). Window proof: $7S < 40A$ at every $j \le 10^6$ (0 violations, 939 870 clash atoms; exact margin $X = 0.082159$, $\rho_{\mathrm{crit}} = 1.1440$; structure laws 8-law / seed-tower / ψ w-table / Cohen seeds $\{-48, -80, -8, -8\}$ at 0 tolerance). Transport ledger closes exactly: $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ with $\Delta_{\mathrm{pole}} \equiv \Delta_{\mathrm{conv}} \equiv 0$ proven (residual 4.4×10^{-16} on 24 rows); odd-prime side = certified plus combination (rel 3.9×10^{-16}). Signed envelope character-exact ($-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8})\Theta$; signed certificate 0 violations, margin 8.86×; confinement set equality). Arch internal via Legendre duplication ($k_\zeta = K_{\mathrm{fam}} - K_{\mathrm{shift}}$ pointwise 7.9×10^{-31}; $\Delta_{\mathrm{arch}} = A_{\mathrm{fam}} - A_{\mathrm{shift}}$ at 9.9×10^{-16}). Coherent class closed by the λ-equivariant CM channel (g_λ two routes exact; $\mu_1 \in [-1, 1]$ exact; λ-certificate 0.0653 vs 0.2365, 3.6×). Frontier facts [C]: coherent-chain crossing $k^* = 14$ ($10^{23.1}$); Robin tail factor 6.16, constant route open. **Named limits as content:** one classically-shaped open lemma (correlated cancellation, non-coherent uniform tail; provably-shaped $\neq$ formal proof) + I5 in one-family form ($Q_{\mathrm{cert}} + \Delta_2 + A_{\mathrm{fam}} - A_{\mathrm{shift}} \ge 0$), by the closed ledger equivalent to Weil positivity $\iff$ RH (equivalence typing, no progress claim). Classics named classical (Gronwall, Robin 1983 unconditional, Cohen 1975, Hecke 1918–1920, Cornacchia, Legendre duplication, Weil 1952, Alaoglu–Erdős). No RH-evidence language; not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. [E] (window proof / ledger / character / duplication / λ-channel) / [C] (frontier facts; two named limits open)

The margin-chain identities of phase 2 (`PRIME.MARGIN.IDENT.01`). The identity/theorem block of T128–T131 as one narrow module ([`verification/v542_margin_chain_identities.py`]; 44 checks, $\sim$0.2 s) on 24 small frame-A seam instances ($n = 3 \ldots 97$, $\rho = 1.25 \ldots 4.00$, every inverted matrix ≤ 300), 43 graded/fine grid pairs and 400 randomised profiles, each identity with a preregistered tolerance and a mutation control. Interval geometry: $\sum_i (1 - g_i) = t_l + t_r$ (5.5×10^{-13}) $\Rightarrow \kappa_{\mathrm{flat}} = 1$ exactly; the (T) identity $\tau = \|y\|^2 - m_{\mathrm{prot}} + m_{\mathrm{fill}} - V_{\mathrm{bord}}$ (2.2×10^{-16}, two independent routes). Profile identity $p_{N-1} = 2 - p_0 + 2E$ with the equivalence $2E > p_0 \iff p_{N-1} > 2$ (0 exceptions); the Abel/Hölder curvature chain per instance (0 violations; without the curvature term violated 88/400). Assembly $= J^\mathsf{T} Q J$ and graded overlap $= J$ (5.2×10^{-16} / 5.2×10^{-14}); the Céa/Strang defect identity $S_{\mathrm{graded}} - S_{\mathrm{uniform}} = R^\mathsf{T} X^{-1} R$ with PSD defect on all 43 grids (2.1×10^{-14}) $\Rightarrow$ one-sided compression. The secular sandwich on all 48 sections where $A > 0$ and $\varepsilon > 0$ are *checked* (width ≤ 1.4448) with Albert's sign equivalence in both directions; $S^{-1} > 0 \Rightarrow$ sign-constant simple ground vector as an *implication*; the counting chain $n_{\mathrm{run}} \le n_{\mathrm{cross}} + 1 \le s_2 + 2$ and $\sum_j |\Delta p_j - \overline{\Delta p}| = \mathrm{TV}(P)$. **Named limits as content:** per instance on small windows only, nothing uniform in the zone index; the κ law itself is *not* claimed; positivity of the pole-free section at depth stays open; the Perron hypothesis is measured; the defect identity says nothing about the defect's size. Abel / Hölder / Haynsworth / Albert 1969 / Céa–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski named classical. [E] (per-instance identities; no marker moves)

The lumped M-matrix pair (`PRIME.MMATRIX.IDENT.01`). The identity block of T136 as a second narrow module ([`verification/v543_lumped_pair_identities.py`]; 35 checks, $\sim$1 s) on 20 small frame-A seam windows ($n = 4 \ldots 59$, $\rho = 1.25 \ldots 4.00$, $h = 25 \ldots 299$, every factorised matrix

≤ 300), 312 shifted ladder rungs and 8 whitened two-grid rungs, each statement with a preregistered tolerance and a mutation control. The lumped pair $A_B = A + L_\Delta$ is Stieltjes and row-sum preserving (1.6×10^{-15}) with $A_B \succeq A$ — an *upper* bound on $\lambda_{\min}(A)$, never a floor; the congruence $A = A_B^{1/2}(I - W)A_B^{1/2}$ holds as a matrix identity (4.4×10^{-15}) with $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ and the *declared* equivalence $\lambda_{\max}(W) < 1 \iff A \succ 0$ checked in both directions, so the Loewner sandwich is circular by itself. The M-matrix anchor $\lambda_{\min}(A_B) \geq 1/\max_r x_r$, $x = A_B^{-1}\mathbf{1} \geq 0$, holds on all 39 rows (sharpest ratio 0.9070) from one solve and one sign check; lumping commutes with diagonal shifts and M-matrix inverses are entrywise monotone, so the shifted ladder is structurally empty (0 of 312); Varga's $\rho(J) = \tau/(1+\tau)$ is an identity (4.8×10^{-15}) and the Collatz–Wielandt bracket at the anchor vector gives an a-priori $\rho(J) \leq 0.9861 < 1$ with no eigenvalue. One *negative* result: the k-scanned M18 majorant is valid at every k (39 pairs) and still stays above its bar 1/2 (best 3.080), the bar is not moved. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the T136 D-exponents are fits and stay in the sandbox. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Collatz–Wielandt named classical. [E] + [X] (one closed route; no marker moves)

The long-lag support structure (`PRIME.LONGLAG.SUPP.01`). The structure block of T137 as the companion module ([`verification/v544_long_lag_support.py`]; 24 checks, ∼0.2 s) on the same 20 windows, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons. The kernel splits exactly as $c = c_{\text{arch}} - c_{\text{comb}}$, $c_{\text{comb}} \geq 0$, and the archimedean section alone has *all* off-diagonals ≤ 0 on 20/20 windows, so every positive off-diagonal is comb-generated; all 20555 positive edges have their Hankel index inside the atom-hat index set (per-window fraction exactly 1.000000), so the support is a union of anti-diagonal stripes at prime-power atoms, each a perfect matching with exactly orthogonal edge vectors; the amplitude certificate $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$ holds on every edge (sharpest 0.99998532) while the Toeplitz lag fails on 20415. The Green function of the lumped comparison is bracketed entrywise from below by the Neumann partial sum *because the discarded terms are nonnegative* and from above by the anchor-weighted tail. And one *death certificate*: since Gershgorin, all row-sum norms and all Collatz–Wielandt quotients at positive weights are upper bounds for $\rho(|E|)$, the single lower bound $\rho(|E|) \geq 1.3141 \geq 1$ on 20/20 rows kills the whole absolute-value envelope family — while the *signed* $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$. **Named limits as content:** per instance on small windows, no decay law and no amplitude law; the support statement is one inclusion; the structure bound assembled from it does not beat the margin (0 of 35 windows in T137) and is not promoted. Gershgorin 1931 / Neumann series named classical; Demko–Moss–Smith 1984 cited and *not* applied (the comparison is dense). [E] + [X] (one family closed; no marker moves)

The Hardy-core identities (`PRIME.HARDY.IDENT.01`). The identity blocks of T140 and T141 as the third companion module ([`verification/v545_hardy_core_identities.py`]; 36 checks, ∼1 s) on 11 frame-A windows ($n = 8 \dots 139$, $h = 58 \dots 285 \leq 300$, D spanning a factor 3.15). With $C = \operatorname{diag}(\sqrt{\Delta})M$ the weighted edge–interval incidence, Gram $= CHC^{\mathsf{T}}$ (3.0×10^{-15}), so the signed Gram has rank $\leq h - 1$; the covering kernel in closed geometric form $K_{rs} = W([r \wedge s, r \vee s])$ equals $M^{\mathsf{T}}\Delta M$ (1.3×10^{-15}), and $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ is an exact eigenvalue confirmed by three independent routes (5.4×10^{-14}, 7.8×10^{-16}), with $H = \operatorname{diag}(s) + L_N$ (4.7×10^{-16}). Four identities put the residue in classical two-weight shape: the K^+ Rayleigh form (6.6×10^{-13}), the signed crossing kernel $L_N = D^{\mathsf{T}}BD$ (4.9×10^{-16}), $Q = B_+\mathbf{1}$ (1.4×10^{-15}) and $DKD^{\mathsf{T}} = L_\Delta$ on the interior nodes (2.7×10^{-14}), whose diagonal is the endpoint edge mass; the crossing kernel of $|N|$ does not reproduce L_N, so the signs are load-bearing. The closed Muckenhoupt quantity $A_{M0} = \max_k Q_k(\Delta\mathbf{1})_{k+1}$ agrees on two routes (8.3×10^{-15}) and its size is reported, not promoted; the checkerboard split is entrywise exact with three dominating Weyl steps. Both certified shapes — the additive Hardy chain and the joint $\Lambda(H,Y) \cdot \Omega(Y)$ — are *valid* upper bounds on 11/11 windows, and only that validity is promoted. One *negative typing*: the exactly bounded object spreads 1.36× over the D range while every absolute variant spreads $\geq 54.94\times$. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; all exponents are fits;

neither shape beats the target, and the zone-uniform discrete Hardy inequality stays open. Abel summation / Cauchy–Schwarz / Weyl 1912 / Gershgorin 1931 / Muckenhoupt 1972 / Opic–Kufner 1990 / Gantmacher–Krein named classical. [E] + [X] (one route typed; no marker moves)

The capacity-chain identities (`PRIME.CAPCHAIN.IDENT.01`). The identity spine of the capacity cycle T142–T145 as the fourth companion module ([`verification/v546_capacity_chain_identities.py`]; 26 checks, ∼0.5 s) on 12 frame-A windows ($n = 4 \ldots 139$, $h = 26 \ldots 285 \le 300$; covering kernel positive definite on 11). The capacity decomposition $K^{-1} = D^\mathsf{T} J^{-1} D + xx^\mathsf{T}/\mathrm{cap}$ is a matrix identity (1.4×10^{-13}; $J = DKD^\mathsf{T}$ entrywise 3.8×10^{-15}) whose Dirichlet half is an orthogonal projection under the congruence — $\Omega = 1$ exactly (5.1×10^{-12}), where T141 had guessed 20.7–2724; the exact capacity–Rayleigh identity for $1 - \rho(W)$ assembles against the direct quotient for every v (9.5×10^{-14}) and reproduces the eigenvalue under the declared 1/gap conditioning; $\mathrm{cap}_E([a, a+j))$ is a prefix sum of one forward substitution (3404 interval capacities at 2.0×10^{-14}, exhaustive on the median window, the reversed ordering failing at $\ge 4.6 \times 10^{-1}$); the whitening certificate $\kappa_{\mathrm{up}} = 1.2245$–$1.4236$ sits under the Gershgorin ceiling 2.1213, direction-correct; and the T145 M4 split — the pointwise mass-half theorem, $\sigma_{\mathrm{tot}} = 0.2197$–$0.4425 < 1$ per window, licence 4 with $|R|$ and its built-in witness $[[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$ $(3 > 2)$, the ordered sign-free chain — certifies S1′ per window on both routes (better route $c_0 = 2.5154$–4.2265 against the Dirichlet value 4), with Charikar's cited density constant bracketed by exhaustive enumeration of all 4095 subsets of a small block. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the c_0 is measured at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open; Maz'ya's strong-type half is not used as a theorem anywhere. Maz'ya 1985 / Muckenhoupt 1972 / Fukushima–Ōshima–Takeda 1994 / Miclo 1999 / Charikar 2000 / Goldberg 1984 named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The level-lemma identities (`PRIME.LEVEL.LEMMA.01`). The a-priori core of T146 as the fifth companion module ([`verification/v547_level_lemma_identities.py`]; 20 checks, ∼0.4 s) on 12 frame-A windows ($n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$), 9 random positive definite forms and a stress ladder $m = 64 \ldots 288$ — the a-priori side of the constant the capacity-chain module certified with a *measured* input. The union identity for nested cakes $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T\|\psi_t\|_1 - \|\psi_t\|^2$ is exact (0.0) against the $K \times K$ double sum and against union sizes computed with no nestedness used (controls 3.2×10^{-2} / 4.8×10^{-3}); the counting lemma $G(b) \le 2b^2 \|v\|_\infty \|v\|_1 / \|v\|^2 + \varepsilon$ holds for any vector at all 6 preregistered bases (the classical dyadic 8 at $b = 2$ exactly, the base free, gain 3.7612; the one-level-down cake breaks the domination on every draw); the resolvent delocalisation bound $\psi = \lambda R \psi \Rightarrow \|\psi\|_\infty \le \lambda_{\mathrm{up}} \max_j \|Re_j\|$ holds within 1.061–1.337 of the truth with Rayleigh–Ritz λ_{up} and residual-paid certified column norms — the columns sitting 2.85–10.17 below the trivial ceiling, $\Gamma = 1.8356$– 2.2797, no Davis–Kahan step anywhere; and the composite level lemma $c_0^{\mathrm{ap}} = 2b^2 \Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042$–$4.8488$ certifies $\mathrm{cert\_lam\_max}(R) \le c_0^{\mathrm{ap}} \Psi_{\mathrm{abs}}$ on 12/12 windows (chain loss 0.0423–0.1272 of the exact gap), with the built-in no-go discriminator (Γ grows $3.676 \to 6.798$, $c_0^{\mathrm{ap}} = 11.10$ exceeds every window value) and the flat Dirichlet control (1.0063). **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the asymptotic delocalisation of the Green columns stays open, typed open; the energy-route σ_{tot} stays measured. Maz'ya 1985 / Rayleigh–Ritz / Miclo 1999 / Davis–Kahan 1970 (cited, deliberately not used) / Charikar 2000 / Goldberg 1984 named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The Green/Szegő identities (`PRIME.GREEN.SZEGO.IDENT.01`). The identity/certificate core of T147+T148 as the sixth companion module ([`verification/v548_green_szego_identities.py`]; 21 checks, ∼0.4 s) on 12 frame-A windows ($n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$), 7 random positive definite forms and a weight-class model ladder $m = 64 \ldots 288$ — the two scalars the level-lemma constant is made of, certified. The factorisation identity $\Gamma = \sqrt{m}\, \lambda_{\mathrm{up}} \max_j \|Re_j\| = \sqrt{Q^\star} \cdot S_w$ ($S_w = \lambda_{\mathrm{up}} \|R\|_F$ spectral, $Q^\star = m \max_j (R^2)_{jj} / \mathrm{tr}(R^2)$ geometric) is exact at 2.2×10^{-16} on all windows and random forms, the certified factorisation dominating everywhere ($S_w = 1.1186$–1.5777,

$Q^\star = 2.0879$–2.8634); the bottom-block trace identity $\sum_j (\Pi_B)_{jj} = |B|$ is exact (3.3×10^{-16}, $\theta = 10$, $|B| = 3$–5) with the measured equidistribution $Q_B/|B| = 1.5091$–1.7980 inside the preregistered $[1, 2.2]$ against the Szegő/Widom reference $2|B|$ (coordinate-projector control overshoots ≥ 3.9); the second-difference ℓ^1 ceiling $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^p\psi\|_1/\mu_k^p)$ holds against the true coefficients everywhere with the m-free closed form $\|a\|_1 \leq 2\sqrt{\nu}+1$ and the Dirichlet calibration $\|a\|_1 = 1$ exact, $\nu \to \pi$; the layer-cake S_w certificate from completed LDL^T inertia counts (inertia = direct count on all 96 rungs) gives $S_w \leq 1.3288$–1.8220 (true 1.1186–1.5777, the bottom-layer drop breaking it everywhere); and the Liouville transform $m\|\psi\|_\infty^2 \leq 2\kappa_\Lambda\,\mathrm{ceil}(\hat\varphi)^2$ ($\varphi = \Lambda^{-1/2}\psi$, a-priori $\kappa_\Lambda = 1.3868$–2.5805) holds stepwise, with the weight-class discriminator at matched κ_Λ (mismatch 0.0013): BV flat (1.003) versus rough growing 19.9 (TV split 162.6) — the roughness, not the conditioning. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the one open lifting input ($\nu_L \leq F(\text{form})$ m-free / bounded $\mathrm{TV}(\log\Lambda)$) stays open, typed open; the symbol-level order-2 hypothesis is refuted ($f(0) < 0$ on every surface window) and enters nowhere as a theorem; σ_{tot} stays retired as a route. KMS 1953 / Widom 1958 / Basor–Ehrhardt 2009 / Sylvester 1852–Bunch–Kaufman 1977 / Euler 1735 / Davis–Kahan 1970 (cited, computed vacuous, not used) named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The gauge/parity identities (`PRIME.GAUGE.PARITY.IDENT.01`). The identity/certificate core of T149+T150 as the seventh companion module ([`verification/v549_gauge_parity_identities.py`]; 21 checks, ∼0.3 s) on 12 frame-A windows ($n = 4\ldots139$, $m = 26\ldots285 \leq 300$), 48 gauge assemblies, the full-section parity items on the 6 windows with $M \leq 300$, and a parity control model at $m = 2001$ — the gauge freedom around the Green/Szegő machinery and the parity structure underneath it, certified. For any positive diagonal $\tilde\Lambda$ the chain $\lambda_{\min}(A) \geq \mathrm{cert}\,\lambda_{\min}(\tilde E)\cdot\min_j\tilde\Lambda_j$ is a valid certified lower bound (identity plus one Rayleigh step; family maximum valid), $\Gamma = \sqrt{Q^\star}S_w$ holds gauge-invariantly at 2.4×10^{-16}, and the const gauge has $\mathrm{TV}(\log\tilde\Lambda) = 0$ exactly with certified sandwich $\sigma = 1.3868$–2.5805 (the completed-Cholesky floor refusing on the shifted indefinite form, 12/12); the two-regime discriminator: the flutter smoother removes TV by $\geq 2.88\times$ while the Liouville-transported $\nu_{\tilde L}$ moves ≤ 0.0088 (rough re-gauge ≤ 0.0046), the untransported functional jumping 0.41–9.76 under the same rough gauge — the roughness is in the form, not the multiplier; the explicit zone diagonal $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s\neq r}\max(c_{|r-s|} - c_{M-1-r-s}, 0)$ at 5.9×10^{-16} with the exact split $c = c^{\text{arch}} + c^{\text{atom}}$ ($c^{\text{atom}} \leq 0$) and the closed atom budget $\|c^{\text{atom}}\|_1 \leq 4B\sqrt{N}$ ($B = 1.038821$ verified on the table), the archimedean section itself indefinite on 12/12 — the atoms co-responsible for the positivity, the additive perturbation reading structurally empty; the parity compression $U_-^\mathsf{T} T_M(c)\,U_- = T - H$ exact (cross block 1.4×10^{-16}) with the negative inertia of the sign-changing symbol localised entirely in the *even* sector (odd sector zero, LDL^T) — $f(0) < 0$ explained rather than worked around; and the parity-sine calibration $\nu_k^P = \pi k^2$ on the exact model (3.0×10^{-5}) with the per-window ladder $\nu_k^P \leq Ck^2$, certified $C = 11.70$–17.28. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the one open input (an m-free C in the odd-sector ladder for a sign-changing symbol) stays open, typed open; $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis; σ_{tot} stays retired as a route. KMS 1953 (parity sector) / Widom 1958 / Basor–Ehrhardt 2009 (the sign-changing symbol is exactly what it does not cover) / Sylvester 1852–Bunch–Kaufman 1977 / Chebyshev 1852 (verified on the table) / Abel / Weyl 1912–Bauer–Fike 1960–Davis–Kahan 1970 (cited, computed dead, not used) named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The odd-sector identities (`PRIME.ODD.SECTOR.IDENT.01`). The identity/certificate core of T151 as the eighth companion module ([`verification/v550_odd_sector_identities.py`]; 19 checks, ∼0.2 s) on 12 frame-A windows ($n = 4\ldots139$, $m = 26\ldots285 \leq 300$), parity control models at $m = 241$ and $m = 2001$, and a no-go size ladder $m = 64\ldots256$ — the reroute off the quadratic ladder, certified: the grid step-over ($\mu_k^P = 2 - 2\cos\theta_k$ exact on the shifted grid $\theta_k = 2\pi k/N$; the symbol's negative window satisfies $\theta_c/\theta_1 = 0.371$–$0.485$ on every window — odd-sector positivity is a grid fact, with the symbol-only floor certified vacuous on 12/12: the local model is a matrix statement, not a symbol one); the certified bottom ladder $\lambda_k(A) \le S\mu_k^P$ for $k \le 8$ by completed $\mathrm{LDL}^{\mathsf{T}}$ counts ($S = 1.1019$–6.4725, $f(0) < 0$ absorbed by the Rayleigh floor); the discrete Sobolev step $\|\psi\|_\infty^2 \le 2\|\psi\|_2\|\nabla\psi\|_2$ licensed by the odd sector's virtual node, with the model pencil $(1,1)$ certified and the per-mode price exactly π (5.1×10^{-8}); the closed archimedean minimum $\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}}$ exact and attained (residual 0.0); and the linear per-mode bound $b_k = 2m\sqrt{K_{\mathrm{bot}}\mu_k^P/\kappa} \le C_S k$ with certified $\kappa = 0.1939$–0.3599 and $C_S = 12.081$–$26.953 \le 2\pi\sqrt{K_{\mathrm{bot}}/\kappa}$ — linear where the quadratic ladder grew, the T145 no-go's bottom pencil ratio exploding $\times 15.8$ while the parity model holds 1. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the one open input (the m-freeness of the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa = 3.7362$–$18.6607$) stays open, typed open; the symbol route stays certified vacuous. KMS 1953 / Widom 1958–Basor–Ehrhardt 2009 (the computed answer: the pointwise symbol language does not extend to the bottom mode) / Weyl 1912 / Parlett / Hardy–Littlewood–Pólya 1934–Agmon 1965 / Sylvester 1852–Bunch–Kaufman 1977 / Chebyshev 1852 named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The fixed-size Ritz ceiling certificate (`PRIME.RITZ.CEIL.01`). The certificate core of T154 as the ninth companion module ([`verification/v551_ritz_ceiling_certificate.py`]; 16 checks, $\sim$0.4 s) on the 12 deepest in-cap frame-A windows ($n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$; a no-go size ladder $m = 48 \ldots 288$; parity controls $m = 64 \ldots 256$; 3 Dirichlet control windows) — the fixed-size replacement for the ceiling step, certified: the direction lemma ($\lambda_k(A) \le \theta_k$ for every $k \le \dim$ — Ritz values are upper bounds of the same-index eigenvalues, Courant–Fischer 1920 / Cauchy 1829, verified against exact spectra; the reversed reading refuted on 12/12 — so the ceiling needs no residual argument); the sixteen-column certificate $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ (dimension 16 fixed, independent of m; 8 completed Choleskys $\le 8 \times 8$) with $K^F = 1.1019$–1.5845 attaining the inertia-certified K_{bot} to $\le 9.9 \times 10^{-7}$ against the declared cap 10^{-4} on 12/12 — the size-m $\mathrm{LDL}^{\mathsf{T}}$ counts retired from the ceiling step (sines alone overshoot 41.6–739; corrupted Green columns break the attainment); the one-direction anatomy (median of the seven aligned angles 0.03–0.14° against the largest 83.79–89.70°; the collapse price $\lambda_1(A)/(t\,\mu_1^P) = 4.408$–$7.042$ recovered in full per window by one Cholesky, size m, labelled as such); and the no-go discriminator (the floor survives at $t = 0.05$ while K_{bot} explodes $\times 35.4$ and the ceiling with it $\times 57.5$; the Dirichlet control certified non-positive on 3/3 — the reflection half is load-bearing for positivity). **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the attainment is not promoted to a theorem (the no-go overshoots), and the two open inputs after T154 (R1′ the m-free block floor, R2′ the m-free bottom-mode floor on the Ritz complement, arithmetic-free and worth 91–100% of the price) stay open, typed open. Courant–Fischer 1920–Cauchy 1829 / KMS 1953 / Schur 1917–Haynsworth 1968 / Sylvester 1852–Bunch–Kaufman 1977 / Temple 1928–Kato 1949 (a floor device, not used here) / Kaniel 1966–Paige 1971 (quoted, never a bound) / Björck–Golub 1973 named classical. [E] (per-instance theorems and certified inequalities; no marker moves)

The four fixed-size angle instruments (`PRIME.ANGLE.INSTR.01`). The instrument cores of T155 and T157 as the tenth companion module ([`verification/v552_angle_instruments.py`]; 18 checks, $\sim$0.4 s) on the same declared surface (the 12 deepest in-cap frame-A windows, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$; no-go ladder $m = 48 \ldots 288$; parity and Dirichlet controls at $m = 64, 128, 256$) — none of the four closes an angle: the 12×12 complement-floor certificate ($\min_{v \perp W} v^{\mathsf{T}} L_P v \ge \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^{\mathsf{T}})M^{1/2})$, an identity plus one certified Rayleigh bound; never above the exact complement bottom, agreement $\le 1.01 \times 10^{-3}$ against the declared cap 2×10^{-3} on this surface, both closed-form controls to 2.0×10^{-11} including the worthless configuration); the sine-block confinement ($\|\gamma_H\|^2 \le (S/t)/\rho_{17} = 0.0153$–$0.0245$ from the certified floor and ladder alone — the bottom eigenvector is 97.5–98.5% inside the first 16 parity sines by a per-instance theorem; κ in place of S: vacuous, 17.6–477); the Schur-entry identity ($s = \mu_1^P\, t_1^{\mathsf{T}} A^{-1} t_1 = (B^{-1})_{11}$ to 1.1×10^{-11}, $(B^{-1})_{LL} S_L = I$ to 5.8×10^{-11} — $1/s = 4.0460$–5.8962 is an $O(1)$ number carried by one entry of the fixed-size 16×16 Schur complement, while the inversion-free Cauchy–Schwarz ceiling misses by 250–7.27×10^3); and the adaptive Lipschitz ceiling (an *executed* certificate $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \geq 0.25$ on 12/12 in 519–1201 evaluations, refusing false targets on 12/12). The no-go floor survives while the confinement goes vacuous ($\times 52.0$) and $(S_L)_{11}$ explodes ($\times 60.0$); parity exact to 4.9×10^{-14}; Dirichlet hits $2/\sqrt{N}$ to 7.4×10^{-5}. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — nothing about the angles is closed; the two open terms after T157 (an m-free upper bound on the one Schur diagonal entry; an m-free R1″ domination with a non-shrinking margin) stay open, typed open. KMS 1953 / Schur 1917–Haynsworth 1968 / Cauchy–Schwarz (refuted as a ceiling, wired as a control) / Kantorovich 1948 / Sylvester 1852–Bunch–Kaufman 1977 / Szegő 1915–Widom 1958 (a named address, never a licence) named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The exact-form identities (PRIME.EXACT.FORM.IDENT.01). The theorem cores of T158 and T159 as the eleventh companion module ([verification/v553_exact_form_identities.py]; 25 checks, ∼0.2 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$; no-go ladders $m = 48 \ldots 288$ on both T145 forms; parity and Dirichlet controls at $m = 64, 128, 256$) — the algebra of the sixteen-form itself, closing neither open term: the Thomson dual form $s = (B^{-1})_{11} = \max_x(2x_1 - x^\mathsf{T} Bx)$ with the direction *executed* and the positive Cholesky ladder tight to 1.0642–1.1522 (a 5% corruption makes the Cholesky refuse); the two-dimensionality $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ attaining s to 9.0×10^{-13} (KMS 1953); the gauge identity $\sum_d w_d = 0$ *entrywise-exact* in its mechanism (odd_toeplitz($\mathbf{1}$) = 0, zero tolerance — the form is blind to the lag mass, exactly where the h^2 halves cancel) with the Toeplitz-minus-Hankel signature to 1.4×10^{-16}; the exact lag sum $x^\mathsf{T} B_{LL} x = \sum_d c_d w_d$ to 2.7×10^{-16} of the absolute scale (relative to the cancelled total only 1.4×10^{-11} — the honest price, measured) with the two closed scalars $w_0 = \sum_k x_k^2/\mu_k^P$ and $2w_0 - w_1 = \|x\|^2$ exact; and the Z-matrix law — B_{HH}^{arch} symmetric Z-matrix *raw* on 12/12 with the closed sign-based Collatz–Wielandt floor 0.9093–1.1284 ≥ 0.25 verified against the exact $\lambda_{\min} = 0.9138$–1.1432, while the atom block obeys no sign law (a coin toss) and the full-block criterion is vacuous. The T145 no-go breaks on three axes; on the $c(l) = 1/(1+l)$ reconstruction the ladder theorem survives while $1/g_{16}$ explodes ($1187 \to 7.11 \times 10^4$); parity control: for $A = L_P$ the full machinery returns $s = 1$ with every ladder partial sum constant in K. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D, and explicitly no statement for all D — the two open terms after T159 (the one explicit pairing inequality for R2″, needing correlation instead of sizes; a third device for the atom off-band block for R1″) stay open, typed open. Maz'ya 1985 / Miclo 1999 / Schur 1917–Haynsworth 1968 / KMS 1953 / Dirichlet 1829 / Perron 1907–Frobenius 1912 / Collatz 1942–Wielandt 1950 / Gantmacher–Krein 1950 named classical. [E] (per-instance identities and certified inequalities; no marker moves)

The sampling/harmonics identities (PRIME.SAMPLING.HARM.01). The closed theorem cores of T160 and T161 as the twelfth companion module (21 checks, ∼0.5 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$) plus a declared Λ-battery of six log-spaced cut-offs $X = 10^3 \ldots 3.2 \times 10^5$ (finite von-Mangoldt sums only, no matrix anywhere) — what the sixteen-form *pairs against*, closing no open term: the sampling identity — the atom half of the pairing equals $-\frac{1}{2}\sum_{n \leq X}(2\Lambda(n)/\sqrt{n})\,\widehat{w}(\log n)$ to 5.6×10^{-15} of the absolute atom scale on 12/12, identically a finite combination of sums $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies; the closed moment laws $m_0 = 0$, $m_1 = -[S_0^2 + 2\sum_j P_j^2] \leq 0$, $m_2 = -[2S_1 - (M-1)S_0]^2 \leq 0$ (both strictly negative) with the total-variation theorem from *checked* hypotheses, $\mathrm{TV}(c^{\mathrm{arch}}) \leq |c_0| + 2\sup < 6$ closed and window-independent; the harmonic-frequency theorem $t_j(2\alpha) = 2\pi j$ *exactly* — the 32 frequencies are the Fourier harmonics of the log-window — with the closed parameter-free PNT prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12 (wrong-pole and fake-measure mutations fail loudly); the head split $c_d^{\mathrm{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$ closed, D-free and m-free, exact to 3.2×10^{-16} at every lag, plus the scale-free Bernstein rate $\rho^* = (3 + \sqrt{5})/2 =$

2.618034; and the certified off-diagonal fraction bound $|\text{off}| \le \frac{1}{4}|Q^{\text{arch}}|$ (0.171–0.212 on 12/12) with the *refuted* sign law wired as the negative control (arch half negative, off-diagonal block positive, 176/240 pairs violate) — nothing single-signed promoted. The head split misses the T145 no-go reconstruction by 2.39; the Dirichlet cosine-sum and KMS controls are exact. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; finite Λ-sums allowed and used, *no* RH statement, the δ triage of T161 a documentation item and not a module claim — the open terms after T161 (R-A′ the prime-free log-moment; R-B′ the m-free 16×16 Gram fraction bound; R-C′ one split with $\delta_{\text{eff}} < \frac{1}{2}$; R-D a fifth R1″ device) stay open, typed open. Dirichlet 1829 / Abel 1826 / Fejér 1915–Schur 1917 / Bernstein 1912 / Chebyshev 1852–Mertens 1874 / de la Vallée Poussin 1896 / Clausen–Lerch / KMS 1953 named classical. Machine-checked: [`verification/v554_sampling_harmonics_identities.py`]. [E] (per-instance identities and certified inequalities; no marker moves)

The Pareto/total-variation identities (`PRIME.PARETO.TV.01`). The closed theorem cores of T162 and T163 as the thirteenth companion module (23 checks, ∼0.4 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \le 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *price structure* of the pairing, closing no open term beyond the negative closure the floor inequality itself is: the exchange law $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16}\text{TV}(x)/P(x))/\log X$ as an identity at all 612 grid points of the declared Fejér knob (5.1×10^{-16}), the crossing criterion $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\text{TV}$ an exact equivalence at every point — price and demand two coordinates of one object; the total-variation floor $w_0 = \|a\|^2$, $\text{TV} \ge |w_0|$ (telescoping, $w_M = 0$), $x_1 = 1 \Rightarrow \mu_1^P\text{TV} = 5.00$–$28.84 \ge 1$ on all 708 trial vectors, the closed floor $1/(4\sin^2(\pi/N)) \sim h^2$, hence every sub-$\frac{1}{2}$ point pays $P > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ — a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface (R-C‴ closed negatively), the inadmissible vector undercutting the floor as the wired negative control; the chain-derived price axis ($\sigma = 1$ exactly the $K = 1$ rung, 4.1×10^{-16}; $\sigma = \infty$ the $K = 16$ top rung, 1.2×10^{-12}; the knob monotone in both coordinates — its curve *is* the Pareto front; a corrupted sixteen-block makes the Cholesky refuse); the closed Mellin cell moments $C_d(-(2k + \frac{1}{2}))$ against an independent quadrature (2.8×10^{-13}, $k = 1 \ldots 4$) and the boundary-free Abel step at levels 1–3 (8.3×10^{-17}) with the closed loss reason $32\pi/\alpha = 40.6$–$59.4 > 1$ and the gauge break equal to its closed prediction (1.2×10^{-15}); and the Lerch/Frullani log-moment on three routes (Wronskian form 2.8×10^{-14}, integral with the $d = 1$ peel 1.8×10^{-15}, $2\log 2$ to 2.2×10^{-16}, $\Psi_1 = -\log 2$ exact), plus the monotone no-go staircase ($1.00 \to 3.49 \to 10.81 \to 24.98$, exact null control) and exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the T163 h-exponents are sandbox fits, *not* consumed; finite Λ-sums allowed and used, *no* RH statement — the open terms after T163 (R-E the successor, both arms prime-free: a growing $1/s$ ceiling for the downstream chain, or a sector whose gap does not vanish like h^{-2}; R-B‴ the h-uniform positivity margin; R-D a fifth R1″ device) stay open, typed open. Fejér 1915 / Mellin 1896 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Legendre / Dirichlet 1829 / KMS 1953 / Frullani 1828–Lerch 1887–Clausen 1832 / Schur 1917–Haynsworth 1968 named classical. Machine-checked: [`verification/v555_pareto_tv_identities.py`]. [E] (per-instance identities and certified inequalities; no marker moves)

The gauge/P_{pr} identities (`PRIME.GAUGE.PPR.01`). The closed theorem cores of T164 and T165 as the fourteenth companion module (23 checks, ∼0.3 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \le 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *gauge structure* of the pairing, not closing the one remaining quantifier: the gauge theorem — $x_1 = 1$ fixes only the scale of a, Q and TV are homogeneous of degree two, so the demand ratio Q/TV of the same a-direction is the same number in *every* sector, unchanged to 2.9×10^{-11} on 2520 sector-by-trial-vector combinations, with the pure shift scaling the certified value by exactly $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$ and the full space settled strictly worse ($\mu_0 = 0$ exactly; the wrong transport map breaks by 2.0×10^2–5.4×10^3); the h^2 conservation law floor × transfer $= 1/\mu_1^P$ *identically* at every shift (2.2×10^{-16}; the wrong transfer misses by 0.9995); the power-one spend as an identity, $d\log r/d\log U = +1$ exactly (1.1×10^{-14};

the quadratic consumption reads 2); the $P_{\rm pr}$ identity $P_{\rm pr} = g_{16} \cdot R \cdot (\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ exact to 4.3×10^{-16} on all 192 battery vectors with the predicate equivalence exact everywhere and every crossing vector paying $P_{\rm pr} > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ above the preregistered tolerance 10 on 12/12 — the successor question R-F closed negatively by an identity plus a floor (the $(t_1 v)$-unsquared mutation breaks the gauge invariance loudly); and the Schur-cascade direction $s \geq g_{16} \geq g_1$ with $P_K = \hat{a}\, s \geq 1$ (Kantorovich 1948) and the exact exponent closure of the lists (1.3×10^{-15}; the g_8 closure breaks by 0.10–0.24, a corrupted sixteen-block makes the Cholesky refuse), plus the monotone no-go staircase ($1.00 \to 3.49 \to 10.81 \to 24.98$, exact null control) and exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the T164/T165 h-exponents and the measured anatomy (the free ascent, the η(Cap) profile, the decoupled ν surface, the retired constant $U_{\rm ref} = 4.90 \to 5.7327$) stay sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement — the one open object after T165 ($\inf_m g_{16}(m) > 0$, equivalently a lower bound on the sixteen-step Schur-cascade gain g_{16}/g_1 uniform in m — a quantifier over a certified flat list) stays open, typed open; R-B$'''$ open as narrowed. Schur 1917–Haynsworth 1968 / KMS 1953 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Kantorovich 1948 / Fejér 1915 / Dirichlet 1829 named classical. Machine-checked: [`verification/v556_gauge_ppr_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The cascade/vector identities (`PRIME.CASCADE.VECT.01`). The closed theorem cores of T166 and T167 as the fifteenth companion module (23 checks, $\sim$0.1 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96 \ldots 291 \leq 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *anatomy of the cascade* and the *exactness of the closed vector at* $K = 2$, not closing the one remaining scalar: the cascade identities — the prefix property (one Cholesky, all sixteen rungs; 1.1×10^{-12} on 192 comparisons), the minor form $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ on the raw arithmetic Gram block (2.7×10^{-12}; Schur 1917–Haynsworth 1968) and the regression form $g_K/g_1 = 1/(1 - R_K^2)$ (8.8×10^{-13}; the wrong minor misses by 0.99–1.45); the diagonal invariance of the gain under $B \to DBD$ (2.7×10^{-12} on 144 random positive diagonals — the gain belongs to the arithmetic Gram block alone; a non-diagonal congruence moves it by 2.7–3.8); the exact $K = 2$ identity — the closed vector $(1, -Q_{21}/Q_{22})$ attains $1/g_2 = Q_{11}(1 - r_{12}^2)$ exactly (1.1×10^{-12}) and $B_{11} g_2 = 1/(1 - r_{12}^2)$ identically (1.2×10^{-12}): the fast-null-vector is free at $K = 2$, the third dress collapses onto the second (the wrong-sign vector pays $4r_{12}^2/(1 - r_{12}^2) = 6.7 \times 10^2$–$1.5 \times 10^4$); the pivot identity $u^{\mathsf{T}} Q_K u = 1/g_K + \delta^{\mathsf{T}} Q_K \delta$ for $\delta_1 = 0$ (2.0×10^{-12} on 252 evaluations, every candidate excess an exact number; the $\delta_1 = 0.1$ mutation breaks by exactly the closed cross term $2\delta_1$) with the exact entry threshold (sign-aligned attainment 3.8×10^{-16}) and the unification $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ (1.2×10^{-12}) — one inequality, not three; and the trial theorem (288 random trials with $u_1 = 1$ all bound g_K from below, min ratio 15.14; Schur 1917) with the L_P no-go (cascade gain exactly 1, deviation 0.0) and the scramble type-change (B_{11} itself loses positivity on a majority of draws on 12/12 windows — a change of type, not a factor) as must-breaks, plus exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in D; the T166/T167 h-exponents ($1 - r_{12}^2 \sim h^{-2.921}$ on frame A, the certified gain exponent $h^{+2.921}$, the threshold monotonicity in K, the $K = 6$ separation $h^{+1.138}$) and the measured anatomy (the rung profile, the polarisation split, the Kato radius, the headroom) stay sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement — the one open object after T167 (R1, the single scalar: an m-free upper bound on $1 - r_{12}^2 = 1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C\, h^{-3+\varepsilon}$, three closed lag sums and nothing else) stays open, typed open; the routes T167 closed off (perturbation theory of every order — the Kato series converges to the wrong object; position-blind surrogates) stay sandbox negatives; R-B$'''$ open as narrowed. Schur 1917–Haynsworth 1968 / KMS 1953 / Rayleigh 1877–Courant–Fischer 1920 / Cauchy 1829 / Kato 1949 (cited as the closed-off route, never used) / Chebyshev 1852–Rosser–Schoenfeld 1962 / Dirichlet 1829 named classical. Machine-checked: [`verification/v557_cascade_vector_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The bilinear/rank identities (`PRIME.BILINEAR.RANK.01`). The closed theorem cores of T168, T169 and T170 as the sixteenth companion module (27 checks, $\sim$0.2 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96\ldots291 \le 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table — the *exact shape of the one remaining scalar*, not closing it: the Lagrange/Wronskian anatomy — the lag-sum identity $\hat{a}_{ij} = t_i^{\mathsf{T}} A_h t_j = \sum_d c_d W_d^{ij}$ (1.6×10^{-15}), Euclidean orthogonality $t_1 \cdot t_2 = 0$ (4.4×10^{-16} — the near-parallelism is created entirely by the arithmetic metric), the Wronskian telescope with closed window edge (1.9×10^{-17}) and $\frac{1}{2}\sum M^2 = 1/(\mu_1\mu_2)$ exactly (Lagrange 1773); the exponent account $1 - r_{12}^2 = \nu_1\nu_2/(\hat{a}_{11}\hat{a}_{22})$ (2.5×10^{-13}) with the unconditional Gershgorin certificate $\nu_1 \le \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$ (12/12, loss 1.19); the T169-TH7 det-collapse $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}}\,\det\hat{A}/[(\hat{a}_{11}+\hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}}+\hat{a}_{12})]$ (8.7×10^{-13} — the loop closes as an identity) with $\nu_2 \le 2\det\hat{A}/(\hat{a}_{11}+\hat{a}_{22})$ tight to < 2 exactly; the exact bilinear von Mangoldt form $\hat{A} = B - S$ (1.7×10^{-14}), $\det\hat{A} = \det B - D(B,S) + \det S$ (1.0×10^{-10}) and $\det S$ as an explicit double Λ-sum against the wedge kernel $K = \frac{1}{2}D(X_n, X_m)$ (7.5×10^{-16}; Cauchy–Binet 1812–1815); the rank-3 theorem (polarisation-matrix eigenvalues $\{-2,-1,+1\}$, $\sigma_4/\sigma_1 \le 1.6 \times 10^{-16}$: the form *is* $S_{11}S_{22} - S_{12}^2$ in three linear Λ-sums) with the Type II collapse (Vaughan coefficient-exact, kernel rank $[2,2,2,2]$ against generic $[20,15,11,9]$); and the R4-free chain identity (the Weil fence never approached) with the L_P no-go ($\hat{a}_{12} = 1.1 \times 10^{-17}$ identically) and the scramble discriminator (rank invariant on all 60 draws, value moves $\times$834.8 median — the arithmetic sits in the joint values of (S_{11}, S_{22}, S_{12}) against B) as must-breaks, plus exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T168–T170 h-exponents and route-table deltas stay sandbox, *not* consumed; finite Λ-sums allowed and used, *no* RH statement — the one open object after T170 (R1, classified as a joint near-degeneracy of three explicit linear Λ-sums, not a size) stays open, typed open. Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Rayleigh 1877 / Montgomery–Vaughan 1973–Vaughan 1977 (the priced route) / KMS 1953 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Dirichlet 1829 named classical. Machine-checked: [`verification/v558_bilinear_rank_identities.py`]. [**E**] (per-instance identities and certified inequalities; no marker moves)

The phase-2 capstone (`PRIME.PHASE2.CAPSTONE.01`). The capstone *verifies the map and claims nothing new*: the whole sixteen-link reduction chain of phase 2 — from the I5 floor to the R1 shape — as *one* machine-checked pass (35 checks, $\sim$0.3 s) on the same declared surface (12 deepest in-cap frame-A windows, $m = 96\ldots291 \le 300$; $\kappa = 0.038821$ verified at every jump point; low block positive definite on 12/12, a numerical fact, never routed through the Weil criterion), 13 links THEOREM / 3 links CERT: the T152 Schur two-block criterion sharp at 0.5λ vs 1.02λ; the T151 Sobolev bound (ratio 0.4816); the T153 Ψ-collapse with the Z-law (6.3×10^{-16} / 9.0×10^{-17}); the T154 Ritz ceiling from below ($g_{32}/s = 0.9520$–0.9849); the sharp T155 12×12 floor; the T156 $F(P,r)$ chain; the T158 Thomson/ladder duality (36/36 trials below); the T160 sampling identity; the T163 Abel swap with the TV floor; the T164 gauge/quantifier collapse $\Psi(x^*) = 1/g_{16}$; the T165 P_{pr} identity; the T166 cascade $1/(1 - R_{16}^2)$, diagonal-invariant; the T167 $K = 2$ exactness; the T169 det collapse, R4-*free* (no positivity of A_h consumed); the T170 rank-3 theorem ($\{-2,-1,+1\}$ exact); and the T170 R1 shape ($\sin\angle = 1.169 \times 10^{-4}$–$2.467 \times 10^{-3} \le 10^{-2}$ per window, rate and quantifier typed OPEN); plus the eight-route no-go battery failing exactly as classified (size budget, symbol infimum, sector invariance, perturbation scale, band-local, sieve operator norm, scramble $\times$679 median with rank-3 untouched, L_P gain $= 1$) and the precision ledger (needed 2.752×10^{-4}, finer than the RH yardstick by 215$\times$ and the best unconditional by 13$\times$ on this surface; both self-similarity identities $\le 6.8 \times 10^{-13}$), each load-bearing identity with a mutation control. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in D; the T171 trend fits and full-surface numbers stay sandbox, *not* consumed; *zero* new uniform-in-m statements — which is exactly why the capstone is promotable; the one open phase-2 object (R1, the collinearity / near-degeneracy rate) stays open, typed open; *no* RH statement. Schur 1917–Haynsworth 1968 / KMS 1953 / Cauchy 1829–Courant–Fischer 1920 / Rayleigh 1877 / Abel 1826 / Dirichlet 1829 / Lagrange 1773 / Cauchy–Binet 1812–1815 / Kato 1949 / Montgomery–Vaughan 1973–Vaughan 1977 / Chebyshev 1852–Rosser–Schoenfeld 1962 /

Fejér 1915 named classical; Weil 1952 / Bombieri 2000 cited, never a criterion. Machine-checked: [`verification/v559_phase2_capstone.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The frame/deficit identities (`PRIME.FRAME.DEFICIT.01`). The gauge route of phase 2, closed as algebra: the theorem cores of T172 (`FRAME.BEYOND`), T173 (`FRAME.RATE`) and T174 (`CANCEL.IDENTITY`) as *one* machine-checked module (24 checks, $\sim$0.5 s) on a small declared surface (a 4×3 (anchor, h) rectangle of gap-blind frame-B cells with a complete comb, positive definite on 12/12; one non-prime-power and one truncated-comb instance; $\kappa = 0.038821$ verified at every jump point): the identity transfer — the KMS/Schur pair, the lag-sum identity, the R4-free det collapse and the additive split re-derived exactly on a frame-B *and* a non-prime-power instance, with the sieve-horizon discriminator (truncated comb $\Rightarrow$ indefinite, $\lambda_{\min} = -9.6 \times 10^4$; complete comb $\Rightarrow$ positive definite on 12/12; the identity links hold either way — none uses positivity); the $q = 1 - s$ identity ($hD = \alpha$ forces the OLS slopes to obey $q = 1 - s$ to 2.9×10^{-14}) with the integer gauge-degree account ($G2$ and the delivery both $(0, 0)$ — the unique doubly gauge-invariant arithmetic member; $G3 = \mu_1^P G2$ identically, the offset-2 explained; $(D/\alpha)^3 h^3 = 1$ a tautology); the blindness theorems (the amplitude gauge over twelve decades at 0.026 of the conditioned bar; *the entire* $\mu_1^P = h^{-2}$ *channel of the demand cancelling exactly on both sides*; the diagonal-group asymmetry — delivery blind, demand keeping the KMS shape) with the additive limit (0/12 + 0/12 single-term Schur floors against 12/12 for the sum: no factorisation exists, P6 not a theorem); the unconditional KMS shape band (≤ 0.312 of $2 \log((1 + \varepsilon)/(1 - \varepsilon))$, $\varepsilon = (K\pi/(2h + 1))^2/6$); and the placement discriminator (24/24 scramble draws destroy the floor), each with a mutation control. **Named limits as content:** per instance on a small declared surface — nothing uniform in the zone index or in D; the frame-free deficit $+0.1111 \pm 0.0222$ (5.0σ) is a sandbox MEASURED number, *not* consumed; the one open phase-2 object (R1, now stated frame-free) stays open, typed open; *no* RH statement. Schur 1917 / KMS 1953 / Dirichlet 1829 / Rayleigh 1877 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 / Bombieri 2000 cited, never a criterion. Machine-checked: [`verification/v560_frame_deficit_identities.py`]. **[E]** (per-instance identities and certified inequalities; no marker moves)

The dense-limit identities (`PRIME.DENSE.LIMIT.01`). The exact cores of the phase-2 endgame, T176 (contract `DENSE.LIMIT`, probe 24/24, verdict SITS-AT-ZERO): the twentyfold sieve raised the density ceiling $361 \to 6120$, both new ratio-4 bins came out consistent with zero (pooled $+0.0496 \pm 0.0372$), the plateau window narrowed by a factor 3.6, and *the phase-2 measurement programme closed as planned*. This module carries the identities per instance (16 checks, $\sim$0.5 s, deterministic, small sieve — the identities are sieve-size-free): *R1 closed* — the Feynman–Hellmann identity $d \log R/d\delta = v^\mathsf{T} d\tilde{A} v/\lambda_{\min} - w^\mathsf{T} d\tilde{A} w/\lambda_{\max} - d\mathrm{DEL}/\mathrm{DEL}$ with the exact lag derivative (piecewise linear at 7.0×10^{-12} of the increment) reproduces the full rebuild on 6/6 instances at the U-minimum of a declared step ladder (2.3×10^{-7} at $\delta = 10^{-6}$; mutations miss by 4×10^1), retiring T175's measured intervention slope as a separate claim, with the near-degeneracy channel $d\mathrm{DEL}/\mathrm{DEL}$ carrying the largest share; *the phase pole* — positive definiteness survives only 2% of the period, $\delta_{\mathrm{PD}} \cdot |d \log R/d\delta| = 7.2$ median (the declared $O(1)$ band), the two lowest modes already within 0.02–0.17 at $\delta = 0$: the $\sim$300$\times$ harmonic anomaly is retired; *sieve consistency at construction level* — complete-comb lag vectors bit for bit, R exact, the bin deficit identical, the truncated must-break moving as it must; and *the estimator with the ladder caveat as a check* — deterministic, exactly reproducible, and ladder-dependent (shift 0.036 fine vs coarse; up to 0.20 on T176's surface): any quoted bin deficit must cite its rung ladder, and the ledger row `PRIME.DENSE.LIMIT.01` does. **Named limits as content:** the SITS-AT-ZERO verdict is a sandbox MEASURED number on T176's 2.5×10^7 surface with its rung ladder, *not* consumed; true zero, power-law approach and low plateau stay indistinguishable under any resource-reachable sieve (one more decade of window costs `ATOM_MAX` $\sim 2.5 \times 10^9$); the closure is of the measurement programme, not of the question; *no* RH statement. Feynman–Hellmann (Hellmann 1937, Feynman 1939) / Schur 1917 / KMS 1953 / Dirichlet 1829 / Chebyshev 1852–Rosser–Schoenfeld 1962 named classical; Weil 1952 cited, never a criterion. Machine-checked: [`verification/v562_dense_limit_identities.py`]. **[E]**

(per-instance identities and certified inequalities; no marker moves)

The Paper-II readout closure (`PRIME.PAPER2.READOUT.01`). The check closure of the classification paper (Paper II, `parity_toeplitz_classification`): every number that paper quoted from an exploration probe without a recomputing module is recomputed on the identical declared surface (24 checks, $\sim$0.5 s) — the T170 reference window bit for bit, the ℓ^1-to-signed ratio 40.6×, the tightness envelope with both ends attained, the density-curve endpoints — and the declared rung-ladder caveat is exhibited as a structure fact: the shift is *anchor selection* (13 vs 12 anchors in the same bin; the anchor part carries the whole shift, curvature transport is small and a wrong curvature is 11.9× louder). Pure readout/identity checks; every measured quantity stays MEASURED with its rung ladder; no RH statement. Machine-checked: [`verification/v563_paper2_readouts.py`]. **[E]** (readout/identity checks with mutation controls; no marker moves)

The relative pencil, one-mode on the declared surface (`PRIME.PENCIL.ONEMODE.01`). The parabolic self-code module proved symbolically that Paper II's rank-3 polarisation is (2+1)-Lorentz geometry with a relative pencil $Z = B^{-1}S$; this module executes that lens on the real declared surface (6 checks, $\sim$2 s; the v563 T170 frame-A scan imported read-only, bit for bit). The three pencil identities are machine-tight on all 70 windows (worst residual 2×10^{-14}); the *locking* mode is λ_2, hugging 1 (median 1.000021), while the spectator λ_1 stays uniformly separated — $\min |1 - \lambda_1| = 3.53$, always *above* 1 (the motivating review guessed below): the two-dimensional cancellation of Paper II reduces to the single scalar $1 - \lambda_2$ on the whole reachable surface (verdict ONE-MODE); the $h = 540$ reference window reads $1 - \lambda_2 = -2.27 \times 10^{-5}$, and the position scramble explodes it by $\times 1.9 \times 10^6$ while the identities keep holding. Declared finite surface only; Problem 7.1 of Paper II is untouched — typed, not bounded; every measured quantity stays MEASURED; no RH statement. Machine-checked: [`verification/v569_lambda_pencil_onemode.py`]. **[E]** (identities and controls exact; eigenvalue readouts measured on the declared surface; no marker moves)

The separation floor, certified from closed scalars (`PRIME.PENCIL.SEPFLOOR.01`). The open question the pencil module named — does the spectator eigenvalue stay separated from 1? — is answered on the reachable surface by reduction (8 checks, $\sim$13 s): two one-line theorems ($P = \det S / \det B > 0,\ T = \operatorname{tr} Z > 0 \Rightarrow \lambda_1 \geq \sqrt{P};\ P \leq 0 \Rightarrow \lambda_1 \geq T$) turn the separation into a closed per-window floor, and on all 70 declared windows the floor clears the margin, $\min F = 2.130 \geq 1.05$ — the separation *follows* from $\det S/\det B$ and $\operatorname{tr} Z$ alone. The dominance behind it is large and grows with depth (median 30.1, $h^{0.90}$ on the surface ladder); the honest negative travels with the result: the cheap two-atom witness route is dead ($\det S$ is itself a pair-level cancellation, negative pair mass up to 13.7× the total), so the h-uniform dominance stays open — typed as pair-level size-vs-cancellation, strictly weaker than Paper-II Problem 7.1 but not free. Declared finite surface only; no rate, no bound, no RH statement. Machine-checked: [`verification/v570_separation_floor.py`]. **[E]** (floor lemmas symbolic; floor certified per window; dominance growth measured with its ladder; no marker moves)

The dominance is long-range (`PRIME.PAIRBAND.01`). Where does the certified dominance live? The pair-band anatomy of $\det S$ on the declared surface answers it (6 checks, $\sim$6 s; a declared kernel subset): *every* near band (<16 lag cells) and the diagonal drag *negative* on every selected window — any near-diagonal mass budget aims at the wrong sign — while the far band (≥ 16 cells) carries the *whole* positive excess with a stable net (far/$\det S \in [1.55, 1.91]$) even as its gross mass grows $4.1 \to 13.0 \times \det S$: the dominance is long-range coherence, not local mass. Sharper still: the excess is *scale-locked* (pairs $\geq h/2$ apart carry it, bands 16..256 stay negative) and *weight-concentrated* — the top-10%-weight pairs alone net 0.85–$1.00 \times \det S$ on every window, essentially $\det S$ itself, while the lag-grid phase plays no role (flat profile, shrinking with h): the certificate target is *heavy-pair coherence at window-scale separation.* The scramble control cuts both ways: random placement at the same masses *explodes* $\det S$ by $\sim \times 49$ (the real placement holds it $\sim$50× below random) and collapses the far-band share — arithmetic placement again. Declared surface and subset only; the v570 open question is typed, not bounded; no rate, no RH statement. Machine-checked:

[`verification/v573_pair_band_structure.py`]. **[E]** (band decomposition and scramble exact; band signs and net measured on the declared subset with their ladders; no marker moves)

The Chebyshev–Loewner edge structure (`PRIME.CHEBLOEWNER.01`). A second external review proposed closed formulas for the parity read block; all of them audit EXACT (9 checks, $\sim$6 s): the polarized cross-weight formula closes the whole $K = 2$ read block analytically; the reflection-edge read is a scaled Loewner matrix of U_{s-1} (rank $\leq s-1$; the first edge read rank one by theorem); the leading modes-$(1,2)$ profile has Lorentz image $(5,-3,4)$ — null exactly, the same Pythagorean triple as $(g_{\mathrm{car}}, N_{\mathrm{fam}}, |\mu_4|)$ via Euclid, typed compression; the normalized edge defect is the exact series coefficient $-\frac{3\pi^4}{175}(s^2-4)(11s^2+6)N^{-4}$; and the pair-kernel polynomial $P(s,z)$ ($P(2,3) = 0$ exact; sign transition $s/z \approx 0.503/1.987$) is the analytic prototype of the measured near-negative/far-positive anatomy. Three-regime split: edge closed at N^{-4}, macro and transition open; Problem 7.1 untouched; no RH statement. Machine-checked: [`verification/v576_cheb_loewner_edge.py`]. **[E]** (exact formulas and series coefficients; the null bridge typed compression; no marker moves)

The null-ray locking census (`PRIME.NULLRAY.01`). The review's Priority-2 prediction, tested verbatim: the pencil locking mode aligns with the exact null ray $(2,-1)$ on all 70 declared windows (23.5–33.8°, below random), the decline is real but slow ($h^{-0.10}$ — the clean success bar honestly not met), and the scramble destroys the alignment (+21°): a real *partial* carrier; the macro regime bends the locking direction, exactly as the three-regime split predicts. Declared surface only; measured with ladders; no RH statement. Machine-checked: [`verification/v577_nullray_census.py`]. **[E]** (the ray exact; angles measured; honest partial verdict; no marker moves)

The two-scale kernel signs (`PRIME.MACROKERNEL.01`). The macro object gets its global test (6 checks, <1 s): sign $K_\infty(\sigma,\tau)$ matches the sign of the real two-lag kernel on *every* resolved grid cell at $h = 300$ (332/332); the diagonal $\det W^\infty(\sigma)$ is negative on $\sim$72% of the range — same-scale pairs couple negative, analytically: the macro origin of the measured near-diagonal drag — and the wrong polarisation breaks the match loudly. The *geometric* half of Regime B closes; the arithmetic half (the Λ-weighted occupation of the cells by actual prime pairs) is the remaining open content; no RH statement. Machine-checked: [`verification/v579_macro_kernel_signs.py`]. **[E]** (kernel geometry exact and sign-verified; the arithmetic half open; no marker moves)

The arithmetic occupation map (`PRIME.OCCUPATION.01`). The lag-side polarization identity $\det S = \frac{1}{2}\sum c_{\mathrm{at}}(d_1)c_{\mathrm{at}}(d_2)D(W(d_1),W(d_2))$ is exact, and every significant cell of its binned occupation map sign-matches K_∞ on every tested window: the primes *populate* the geometric sign map — the open Regime-B content is the amounts, not the signs; scramble control clean. Machine-checked: [`verification/v580_occupation_map.py`]. **[E]** (identity exact; occupation measured with ladders; no marker moves)

The multilevel transport census (`PRIME.TRANSPORT.01`). The second review's transport lemma, measured on 40 doubling pairs and killed honestly: the locking-direction drift is small (0.3–6.7° per octave) but grows ($h^{+0.36}$) against the proposed summability bar $h^{-1-\delta}$ — the review's own kill criterion fires; any future transport must carry the window-dependent macro direction. Machine-checked: [`verification/v581_locking_transport.py`]. **[E]** (census exact; drift measured; an honest negative; no marker moves)

The density-dominance reduction (`PRIME.DENSITYDOM.01`). The distilled quantitative question receives its first-level answer (5 checks, $\sim$2 s): the smoothed comb reproduces $\det S$ to 99.8–99.9% (2% smoothing) on every regular window through the exact lag-side identity, while on the scrambled window the smooth model captures only 70% — the real comb is anomalously *smooth* at the $\det S$-relevant scales: the dominance is carried by the deterministic density profile, and the h-uniform question relocates to PNT-level regularity of the smooth functional (plus a $\sim$1% fluctuation correction needing only a crude bound); the anomalous $h = 1292$ boundary named; no RH statement. Machine-checked: [`verification/v582_density_dominance.py`]. **[E]** (identity exact; ratios measured with ladders; the smooth functional not yet bounded; no marker moves)

The prime-free closed form (`PRIME.PNTMODEL.01`). The named next step lands the same day (8 checks, ∼4 s): the two-term classical law $\mathrm{mass}(u) = 4e^{u/2} - 2(\zeta'/\zeta)(\frac{1}{2})$ — pole term plus the $s = \frac{1}{2}$ constant, *no zeros* — reproduces $\det S$ on all 69 regular windows *parameter-free* to 1.016–1.143 (entries 0.995–1.001; dominance correlation 0.998); the anomalous $h = 1292$ sign flip is typed as genuine prime-fluctuation content (prime-free +1844 vs real −1465). The h-uniform dominance splits into a deterministic prime-free two-integral inequality plus a zero-oscillation bound; no RH statement. Machine-checked: [`verification/v583_pnt_model.py`]. **[E]** (two-term law classical; theoretical constant, no fit; the inequality not yet proven; no marker moves)

The interacting Ext-source response (`DIAMOND.BIT.EXTSOURCE.01`). The v572 remaining route is executed (8 checks, ∼1 s): rank-one Ext-source coupling on the Gibbs pencil has exact rational flip thresholds $J_c(s) = -1/(v^T K(s)^{-1}v)$ whose numerators are exactly the twins' pencil factors; in the crossed phase positive coupling $J_c \le 15/16 < 1$ heals exactly the compiler twin, while the alt twin cannot be flipped by any positive coupling in the open unit box — the healing direction is strictly positive (consistent with v534); the response pole sits on the golden polynomial $s^2 - s - 1$ (observed). A response-level selection, not a positivity theorem; the alignment-bit contract stays open. Machine-checked: [`verification/v584_ext_source_response.py`]. **[E]** (thresholds exact; response level only; no marker moves)

The two-layer locking split (`PRIME.LOCKSPLIT.01`). The complementary side of the prime-free closed form (6 checks, ∼3 s): the same model that reproduces $\det S$ to 1.02–1.14 *misses* the Paper-II locking defect by 2–4 orders of magnitude (median 1350×, growing $h^{1.08}$) — the dominance is density, the *depth* of the lock is the primes; the one-mode geometry is density-level (coarse lock 10^{-2}–10^{-4} without primes); layers $h^{-1.43} \times h^{-1.08} = h^{-2.51}$, with real sign overshoots at the deep windows $h = 1219, 1445$. The Problem-7.1 budget splits into a classical density part and a zero-oscillation part; no RH statement. Machine-checked: [`verification/v585_pnt_locking_split.py`]. **[E]** (model verbatim from v583; ladder slopes on the declared surface; no marker moves)

The density-fixed locking direction (`PRIME.LOCKDIR.01`). The direction side of the split (7 checks, ∼3 s): the real locking eigenvector agrees with the prime-free model's to a median 0.030° (max 0.331°, decaying $h^{-1.16}$) while the depth stays 1350× apart — the primes are a pure depth amplifier along a density-fixed direction; the deterministic direction certifies the real defect within a factor ≤ 2.9; the v577 slow-decay puzzle dissolves as a comparator artifact (drift limit −0.551 under $1/\log h$, consistent with −0.5, not settled); scramble must-break fires (6–25°). No RH statement. Machine-checked: [`verification/v586_pnt_lock_direction.py`]. **[E]** (census with ladders; deterministic asymptotics open; no marker moves)

The exact diagonal weight formula (`PRIME.WCLOSED.01`). A new exact formula (6 checks, ∼2 s): the corpus diagonal lag weight is the single closed expression $W_{kk}(d) = \frac{2}{N}[(N-1-d)\cos(\omega_k d) + \sin(\omega_k(d+1))/\sin\omega_k]$ — machine-exact for all lags ($h = 7$–1000, $k = 1$–3); the apparent piecewise break cancels, the edge suppression is an exact two-term cancellation; the exact cell-sum reproduces the prime-free model's entries and determinant to 0.03% — the S-side of the deterministic layer is closed-form (the first executed step of the v585/v586 classical program); the v579 kernel is the edge-anchored version identically. Machine-checked: [`verification/v587_w_closed_form.py`]. **[E]** (formula machine-exact; model comparison at declared tolerances; no marker moves)

The closed deterministic defect (`PRIME.CLOSEDDELTA.01`). Both sides in one function (6 checks, ∼5 s): the closed geometric-trig sums $G_{ij}(\beta)$ carry the S side at the pole exponent and the B side as the *trivial-zero ladder* ($c_{\mathrm{ar}}(d) = -Df(dD)$, $f = \sum_n e^{-(2n+1/2)w}$); closed B to 2×10^{-5}, the closed defect to 0.2% against the deterministic layer on all 70 windows, census decay $h^{-1.40}$ — the deterministic half of Problem 7.1 is the explicit formula's skeleton (pole vs Γ-factor) in closed form; elementary asymptotics remain for a theorem; no RH statement. Machine-checked: [`verification/v588_closed_delta.py`]. **[E]** (G machine-exact; near field declared; no marker moves)

The zero-comb identification (`PRIME.ZEROCOMB.01`). The arithmetic layer gets its object (4 checks, ∼16 s): the atom table's residual mass oscillation is the explicit-formula zero comb (correlation 0.82, slope 1.28 with the first 200 damped zeros; pointwise at low u); the comb recovers a factor 5.7 of the lock depth at the smallest window ($176 \to 31$) while the deep windows stay honestly out of reach — the zero-oscillation bound is a quantified explicit-formula target (resolution π/D, amplitude accuracy = lock depth); no bound claimed, no RH statement. Machine-checked: [verification/v589_zero_comb.py]. [E] (zeros from mpmath; damping declared; no marker moves)

The sheet-involution existence (QGEO.INVOL.01). The involution-existence input of `GATE.QGEO.01` is derived (7 checks, < 1 s): the integral V-commuting involutions are exactly $(\mathbb{Z}/2)^3$ (rigidity, both twins); the v574 type theorem plus seam reversal plus zero-mode fixing single out the explicit S_0 of exact Σ-signature uniquely — existence unconditional; the corpus RP reflection is a different structure (anti-automorphism, does not commute with V); the two demands' continuum derivation stays the named open. Machine-checked: [verification/v590_involution_existence.py]. [E] (exact sympy; demands declared; no marker moves)

The rank-one pole term (PRIME.POLERANKONE.01). The formal expansion of the closed entries delivers a theorem-grade identity (5 checks, $\sim$2 s): the pole part is *exactly* separable rank-one ($S^{\text{pole}} = -32\pi^2 a e^a\, gg^T$, $g_k = k/(a^2+4\pi^2k^2)$, $\det \equiv 0$ symbolically), and its null direction gives the closed locking-direction law $v_2/v_1 = -(a^2+16\pi^2)/(2(a^2+4\pi^2))$ with limit exactly $-\frac{1}{2}$ — deriving the v577 null-ray conjecture, explaining the v586 drift, and matching the measured directions to 0.04–1.4%; the density layer becomes the correction theory of a rank-one matrix. Machine-checked: [verification/v591_pole_rank_one.py]. [E] (symbolic exact; ladder match measured; no marker moves)

The continuum determinant law (PRIME.DETLAW.01). The correction theory of the rank-one pole term begins (6 checks, $\sim$2 s): $\det S(a)$ is closed with the e^{2a} coefficient identically zero and the exact leading law $640\pi^2 e^a/a^3$ (one full exponential order suppressed, derived); the real dominance growth tracks the closed formula at correlation 0.9985 (the v570 growth derived at the continuum level); the finite-N gap is typed and the entry-level arithmetic deviation is certified by $\sup|\mathrm{Osc}|\times\mathrm{TV}$ with margin 47.6×, conditional on the measured oscillation sup. No RH statement. Machine-checked: [verification/v592_continuum_det_law.py]. [E] (symbolic exact; conditional certificate declared; no marker moves)

The cutoff completion and the first unconditional bound (PRIME.CUTOFF.01, PRIME.UNCONDCERT.01). Two modules close the round ($4+5$ checks): the v592 finite-N determinant gap was the integration *cutoff* — with the σ_0-corrected closed integrals the entries match to 0.7–1% and the determinant to 0.2–1% (the density layer of Problem 7.1 is closed end-to-end, residual $O(1/N^2)$); and via partial summation plus the published bound $|\psi(x)-x| < 0.94\sqrt{x}$ (Büthe 2018, unconditional to 10^{19}) the entry-level arithmetic deviation is bounded UNCONDITIONALLY (sup 17.4 vs measured 1.474; certificate margin 565×) — the program's first unconditional statement; the determinant-level lock stays with the zero-oscillation theorem. The algebraic cores of the round are additionally formalized in Lean 4 (`TfptCarrier/ParityWeightLaws.lean`: branch merge, rank-one determinant, direction law, sheet involution). Machine-checked: [verification/v593_cutoff_completion.py], [verification/v594_unconditional_cert.py]. [E] (closed forms exact; one cited external bound; no marker moves)

The mapping completion (PRIME.MAPCLOSE.01). The residual bookkeeping lands (4 checks, $\sim$4 s): the v593 residual was the σ mapping — with the exact dictionary ($a_{\text{eff}} = ND/2$, limits $[u_0/(ND), 2\alpha/(ND)]$) the closed entries and determinant match the deterministic model at the identity level (1.00000–1.00017; kernel piece 9×10^{-6}); honest amendment of the v593 attribution (first-order in D via the mapping, not $O(1/N^2)$ kernel terms); the density layer of Problem 7.1 is a finite set of exact elementary evaluations, and the determinant law is formalized in Lean (`det_S_closed`). Machine-checked: [verification/v595_mapping_completion.py]. [E] (closed forms exact; declared windows; no marker moves)

The 1D lock projection (PRIME.LOCKPROJ.01). The arithmetic layer in one dimension (6 checks, $\sim$35 s): along the closed v591 direction the zero-side functional cancels the closed density value in a near-identity (median $|q_{\text{real}}/q_{\text{model}}| = 7.7 \times 10^{-4}$, decay $h^{-1.01}$; scramble breaks by 10^6); the coupling spectrum is collective (max share 5.9%, $N_{50} = 56$) — the arithmetic content of Problem 7.1 is one explicit identity family, equidistribution- flavored; no bound claimed, no RH statement. Machine-checked: [`verification/v596_lock_projection.py`]. **[E]** (closed direction parameter-free; declared floor; no marker moves)

The μ_3-cover candidate (`QGEO.COVER.01`). Strategy I/II, first executed step (6 checks, < 1 s): the cover $y^3 = x^4 - 1$ hits four compiler fingerprints — rank-3 Eisenstein H_1 (Chevalley–Weil $(1, 2)$), projective D_4 via the deck ($r^4 = \omega$, Burau at $t = \omega$ with verified braid relations), cusp-class group $(\mathbb{Z}/3)^4$ of order 81 (the saturation index with the per-mark mechanism), and the unique invariant hermitian form of LORENTZ signature $(1, 2)$ (matching Paper II's rank-3 polarization — Strategy II's first prediction). The Q/V dictionary and the RP map are the named next steps; the gate does not move. Machine-checked: [`verification/v597_mu4_cover_model.py`]. **[E]** (exact sympy; convention validated; no marker moves)

The cover dictionary, round 2 (`QGEO.DICT.01`). Three compiler objects located exactly (4 checks, < 1 s): $U \sim G_2$ (single-cusp Gram, char $x^2(x - 3)$); $Q_- \sim G_1 - G_3$ (the μ_4 sheet coupling as the *diamond-diagonal* Gram difference, char $x(x^2 - 3)$); the binary AnchorLadder $\sim 1 + G_1 + G_3$ (char $(x - 1)(x - 2)(x - 4)$); opposite-twist products nilpotent; the grading half $(Q_+/V/H/Q)$ absent from 7296 scanned canonical combinations — the real structure is the named suspect; the Strategy-I kill criterion is not triggered. Machine-checked: [`verification/v598_cover_dictionary.py`]. **[E]** (exact entries; quantified negatives; no marker moves)

The real structure (`QGEO.REAL.01`). The v598 suspect constructed (18 checks, $\sim$3 s): the anti-holomorphic D_4 reflection acts semi-linearly, $R(v) = M\overline{v}$, with a *one-dimensional* intertwiner space (R unique up to scale), $R^2 = +1$ exactly, and all four twist intertwinings $RT_kR^{-1} = T_{5-k}^{-1}$ exact. Its grading functor $\Gamma(A) = M\overline{A}M^{-1}$ closes the three dictionary gaps at char level: $V \sim G_2 + G_3 - \Gamma(G_3)$ (char $x(x - 1)(x - 2)$), $H \sim 1 + G_2 - \Gamma(G_3)$ (char $(x - 1)^2(x - 2)$ — *the anchor spectrum*), $Q_+ \sim 1 + G_2 + G_3 - \Gamma(G_3)$ (char $(x - 1)(x - 2)(x - 3)$); the Γ-even/odd cusp elements carry $\{0, 1, 5\}$ ($g_{\text{car}} = 5$), $\{0, 3, 9\}$, $\pm\sqrt{5}$, $\pm\sqrt{27}$ and one nilpotent. Honest joint negative, quantified: 4×6 candidates, zero pairs pass $\operatorname{tr}(UV) = 3$ and $\operatorname{char}(U+V) = \operatorname{char} Q$ — char-matching is not yet an algebra embedding (kill not triggered). The invariant hermitian form is unique up to scale with $\det J = 8$ — a sheet-diamond determinant-list member — and Lorentz signature $(1, 2)$: Strategy II's Hodge prediction pinned at form level. Machine-checked: [`verification/v599_real_structure.py`]. **[E]** (exact existence/uniqueness; quantified joint negative; no marker moves)

The joint embedding (`QGEO.EMBED.01`). The compiler pair is *realized* on the μ_3-cover (15 checks, $\sim$4 s): in the Γ-even degree-2 sector the pair $U^* = -2E_1 + (G_1\Gamma(G_1) + \Gamma(G_1)G_1) + 3$, $V^* = -E_3 + \frac{1}{2}(G_2\Gamma(G_2) + \Gamma(G_2)G_2) + 1$ is $\mathbb{Z}[\omega]$-integral with compiler-exact spectra and joint traces ($\operatorname{tr}(U^*V^*) = 3$, $\operatorname{tr}(U^*V^{*2}) = 6$, $\operatorname{char}(U^*+V^*) = \operatorname{char} Q$); the seven words have rank 7 (the full 2|1 parabolic) and an explicit invertible g solves $gU^*g^{-1} = U_c$, $gV^*g^{-1} = V_c$ exactly — the v566 self-code algebra lives on the cover. On the explicit saturated $\operatorname{Fix}(R)$ lattice (SNF certificate $(1, 1, 1)$) the restrictions are *integer* matrices whose seven-word $\mathbb{Z}$-order has Smith divisors $(1, 1, 1, 3, 3, 3, 3)$ — saturation index $81 = 3^4$, exactly the v566 self-code index. Must-break control passes (the degree-1 pair fails the joint criterion); canonicity is open and typed open (existence, not uniqueness); the gate does not move. Machine-checked: [`verification/v600_joint_embedding.py`]. **[E]** (exact realization; canonicity open; no marker moves)

The canonical equivariant realization (`QGEO.CANON.01`). The canonicity question answered (14 checks, $\sim$12 s): the Burau image preserves J, not the standard hermitian form, so the canonical building blocks are the J-adapted cusp projectors $GJ_k = (1 - T_k)J^{-1}(1 - T_k)^\dagger J$ — idempotent rank-1 J-self-adjoint projectors rotating *exactly* under D_4 ($rGJ_kr^{-1} = GJ_{k+1}$; opposite cusps J-orthogonal). The pair family $U_k = 3\,GJ_k$, $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ is $\mathbb{Z}[\omega]$-integral, compiler-exact (spectra, joint data, seven-word rank 7) and D_4-*equivariant by construction*: the mark choice is the compiler's winding choice. The duality surprise: the equivariant pair is *not* conjugate to

(U_c, V_c) (the intertwiner is identically singular — must-fail documented) but exactly conjugate to the *transposed* pair — the equivariant presentation realizes the DUAL (plane-stabilizer) parabolic, the v600 pair the line-stabilizer: the cover carries both types, exchanged by duality. And the v590 Klein four-group lives *integrally* on the curve: all eight spectral-sign involutions of V_1 are $\mathbb{Z}[\omega]$-integral, the two demands select the explicit S_0 (trace -1), and $AB = S_0$ closes the group. Honest: the naive J-word Gram has inertia $(3, 2, 2)$, not the v572 pattern $(4, 0, 3)$ — the Hodge/RP comparison needs the Weil-twisted polarization (named open); the seam identification stays open. Machine-checked: [`verification/v601_equivariant_dual.py`]. **[E]** (exact equivariance/duality/Klein; typed opens; no marker moves)

The duality-and-forms round (`QGEO.DUALFORMS.01`). The duality mechanism identified (17 checks, $\sim$8 s): U_1 is J-*self-adjoint*, the J-adjoint of V_1 is the exact product reversal, and (U_1, V_1^J) is exactly conjugate to the compiler line-stabilizer pair while (U_1, V_1) realizes the transposed dual — the J-adjoint exchanges the two parabolic types inside one equivariant structure. The dictionary completes in the J-projector language: $H = 1 + V(V-1)/2$ carries char $(x-1)^2(x-2)$ — *the corpus anchor formula holds verbatim on the curve*; $Q_+ \sim 2 - GJ_1 + GJ_2GJ_3$ and the binary ladder $\sim 2 - GJ_1 + 2\,GJ_2GJ_3$ are exact. Weil positivity holds exactly: the twisted form JC is positive definite with eigenvalues $\{2\sqrt{2}-2,\, 2,\, 2+2\sqrt{2}\}$ — the Riemann bilinear relations on the ω-sheet; and the canonical bilinear form $B = M^\dagger J$ (real structure $\times$ polarization) is symmetric with $\det B = 8$. Honest negatives, typed: $g^T g \neq \lambda B$ (the corpus transpose is not the (R, J)-form); the naive mark-local cokernel map fails; the seam normal is explicit but not a deck eigenvector. The v600 integer core is Lean-certified (`FORM.QGEO.04`, `TfptCarrier/CoverEmbedding.lean`: determinant-free Smith certificates, kernel `decide`). Machine-checked: [`verification/v602_duality_forms.py`]. **[E]** (exact duality/dictionary/positivity; quantified negatives; no marker moves)

The seam-and-marks round (`QGEO.SEAMMARKS.01`). The seam line found, and it is *real* (15 checks, $\sim$7 s): the line-stabilizer pair has a unique common eigenline v ($Uv = 0$, $Vv = v$ — the transported compiler seam), explicitly $v = (0, (\sqrt{3}-i)/(\sqrt{3}+i), 1)$; the real structure *fixes* this line and the two-demand involution *reverses* it ($S_0 v = -v$) — the v590 seam-reversal demand holds geometrically on the curve. The infinity cusp separates in the unreduced Burau module (invariant column $(1, 1, 1, 1)$; full twist eigenvalues $\{1, \omega^{\times 3}\}$). And the first per-mark $\mathbb{Z}/3$ witness lands: E_1 restricts to an integer matrix lying in the seven-word algebra with denominator lcm exactly 3 — an order-3 class in the word-order cokernel (the v566/v597 mechanism's first exact witness); E_4's class is trivial and E_2/E_3 lie outside the mark-1-anchored algebra (equivariant per-mark framing typed). The corpus-transpose translator negatives are quantified ($F_2 = g^T \Sigma g$ matches no canonical menu form); named open. Machine-checked: [`verification/v603_seam_marks.py`]. **[E]** (exact seam/infinity/witness; quantified negatives; no marker moves)

The equivariant order and the infinity bridge (`QGEO.EQORDER.01`). Three integral statements (12 checks, $\sim$4 s). First, a correction with a payoff: R *is* an anti-isometry of the polarization — the correct compatibility equation $M^\dagger J M = J^T$ holds exactly — so J restricts to a real symmetric *integer* form on the saturated fixed lattice, $J_{\text{fix}} = [[16, 2, 4], [2, -2, 2], [4, 2, -2]]$ with $\det = 72 = 8 \cdot 9$. Second, the four mark frames together generate a *full* order: the joint lattice of all 28 rotated words has rank 18 in $M_3(\mathbb{Z}[\omega])$ with Smith divisors $(1^{15}, 3, 3, 3)$ — index $27 = 3^3$ (one frame: the 7-dim parabolic of index $81 = 3^4$; all four: the full algebra of index 27). Third, the 3-torsion *localizes at infinity*: $\operatorname{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$ while every finite cusp twist is torsion-free — three objects share $(\mathbb{Z}/3)^3 = 27$ (infinity monodromy, equivariant order, $\det_{\mathbb{Z}}(1 - \text{deck})$): the compiler constants localize, 81 at the finite marks and 27 at infinity. Translator negatives extended (J_{fix}-menu and algebra twists fail; the $\bar{\omega}$-sheet is the named candidate). Machine-checked: [`verification/v604_equivariant_order.py`]. **[E]** (exact integral statements; typed localization reading; no marker moves)

The translator round (`QGEO.TRANSLATOR.01`). The corpus RP anti-automorphism is *found* on the curve (11 checks, $\sim$11 s): pulling $\theta(X) = \Sigma X^T \Sigma$ back along the compiler conjugation and composing with the J-adjoint gives an algebra automorphism φ — and φ is *inner*: the intertwiner space is one-dimensional, its generator c is invertible, and $\varphi(A) = cAc^{-1}$ holds exactly. Hence $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(c)$ — the RP structure is fully expressed in curve language (polarization plus one explicit element), with $14c$ integral over $\mathbb{Z}[\omega]$ (denominator $14 = 2 \cdot 7$, the parabolic dimension); the single-sheet negatives of v603/v604 are thereby resolved (the missing object was inner, not a new form). The seam is a *nonzero pure 3-torsion class at infinity*: its lift has zero free coordinates and torsion coordinates $(2, 2, 0)$ in $\text{coker}(1 - T_\infty)$ — the integral seam/infinity identification. The 81 *survives* frame-joining: the chain $O_1 \subset O_{\text{joint}} \cap \text{span}_1 \subset P_1$ has indices 3^5 and $3^4 = 81$. And the J_{fix} duality is certified with an explicit integer dual ($C_V^T J_{\text{fix}} = J_{\text{fix}} C_{V\text{dual}}$; Lean addendum green). c's independent geometric identity stays open. Machine-checked: [`verification/v605_translator.py`]. **[E]** (exact translator/bridge/chain/duality; typed opens; no marker moves)

The mode-separation round (`QGEO.MODESEP.01`). The zero-mode selection gains its carrier (7 checks, ∼2 s): all three eigenlines of the sheet operator V (eigenvalues $0, 1, 2$), lifted to the unreduced integral module along the *unique* intertwiner, are *pure 3-torsion classes at infinity* (zero free part), and the three classes are *pairwise distinct* elements of $(\mathbb{Z}/3)^3$ (fixed Smith frame $(2, 0, 2)/(2, 2, 0)/(1, 2, 0)$; distinctness frame-invariant, none trivial) — the infinity-cusp torsion distinguishes the zero mode from the top mode, exactly the separation the real structure could not provide. The canonical rule singling out the "+1" class stays open. And the arithmetic-identity route for the translator element is closed honestly: $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$ has discriminant $-783 = -3^3 \cdot 29$, not a square — Galois S_3, not abelian, no cyclotomic identity (Kronecker–Weber). Machine-checked: [`verification/v606_mode_separation.py`]. **[E]** (exact separation; typed frame caveat; honest arithmetic negative; no marker moves)

The selection rule (`QGEO.SELRULE.01`). The canonical "+1" rule found (10 checks, ∼2 s): the infinity torsion is a genuine $\mathbb{F}_3$ space (ω-multiplication acts as the identity), and the deck rotation acts on it with characteristic polynomial $(x + 1)(x^2 + 1) \bmod 3$ — $x^2 + 1$ irreducible over $\mathbb{F}_3$, so there is *exactly one* projective eigenline (eigenvalue -1), and it *is* the top-mode class; the zero mode and the seam are not projectively fixed (nor under r^2). With v603/v605 the three modes carry mutually exclusive canonical labels — seam = the *R-real* line, top = the unique *deck-eigen* torsion class, zero = the only *unlabeled* mode — and the two-demand involution flips exactly the labeled pair: *both* v590 demands are now expressed through canonical cover structure (typed reading; the continuum derivation from the raw seam stays the gate residual). Machine-checked: [`verification/v607_selection_rule.py`]. **[E]** (exact $\mathbb{F}_3$/rigidity/labeling; typed selection reading; no marker moves)

The free gamma slice (`WOIT.GAMMA.FREE.01`). The first executed step of the WOIT γ milestone (9 checks, ∼1 s, on the explicit v524 net): the mirror (anti-chiral) state's OS Gram splits *sector-exactly* — even $(8, 0, 0)$ positive definite, odd $(0, 8, 0)$ *strictly negative definite* — so every mirror fermion vector has strictly negative OS norm and *no mirror mode survives OS reconstruction*: kill test (3) cannot fire at the free level (the v519 shadow upgraded to the sector level). The chiral generation is multiplicity-free (clock spectrum exactly $\{1, \sqrt{2} - 1\}$ on the fermion domain), and the mark incidence holds at the free level: the two mark algebras have PD Grams of dimension $4 = |\mu_4|$, jointly generate the full 16-dim H_{phys}, and the quarter-turn transfer τ_2 *is* the mark-A→B transport (Hermitian PD) — kill test (7) does not fire there. The γ milestone itself stays open on $\mathcal{A}_{\text{hol}}$ (kill tests live; (6) untouched). Machine-checked: [`verification/v608_gamma_free_slice.py`]. **[E]** (exact sector/spectrum/incidence on the explicit net; free slice only; no marker moves)

The translator factorization (`QGEO.CFACTOR.01`). The geometric identity of the translator element is *closed* (8 checks, ∼8 s): the mirror intertwiner d (defined by $dA = \Gamma(A)d$ on the realized pair, Γ the real-structure conjugation) is *unique* (1-dim intertwiner space), *unimodular* ($\det d = 1$), *integral* over $\mathbb{Z}[\omega]$, and Γ-*involutive* ($\Gamma(d)d = 1$ exactly); and the v605 element factorizes exactly, $c = \lambda K_1 d$ with $K_1 = J^{-1}\overline{F_2}^T M^{-1}$ — so $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$: polarization × RP metric × real structure × canonical mirror intertwiner. With this, all three v605 residuals are resolved (seam reversal geometric v603; zero-mode rule canonical v607; c-identity factorized v609); the v606 arithmetic negative stands as the reason the cyclotomic route had to fail. Honest scope: F_2

carries the corpus datum Σ by definition of the translator; the gate does not move. Machine-checked: [`verification/v609_translator_factorization.py`]. **[E]** (exact factorization; closed-identity reading; no marker moves)

The curve landscape (`QGEO.LANDSCAPE.01`). The inverse question — what *else* does the cover carry? — answered (9 checks, ∼1 s). The Jacobian *splits*: the three holomorphic differentials carry (deck, rotation) characters of combined orders $(12, 12, 6)$, so by CM theory $\mathrm{Jac}(C) \sim A_2(\mathrm{CM}\ \mathbb{Q}(\zeta_{12})) \times E(\mathrm{CM}\ \mathbb{Q}(\omega))$ — with $\mathbb{Q}(\zeta_{12}) = \mathbb{Q}(i, \sqrt{3})$ the field generated by the deck *and* the mark rotation together, and $E : y^3 = u^2 - 1$ the $x \mapsto -x$ quotient. Point counts confirm the splitting: $a_p(C) = a_p(E)$ *exactly* for all ten tested $p \not\equiv 1\ (12)$ (the ζ_{12} surface is supersingular there), with nonzero surface traces exactly at $p \equiv 1\ (12)$. And the integer-spectrum census honestly relativizes single-char hits: integer characteristic polynomials are *common* in the declared pool (7894 entries) — the anchor class is the most frequent nontrivial family (286), non-compiler spectra exist ($\{1, 1, 3\}$: 254) — the compiler's distinction is the *joint* algebra structure (v600), not individual spectra. Two canonical period resources named for future RP/Hodge rounds. Machine-checked: [`verification/v610_curve_landscape.py`]. **[E]** (exact characters/point counts/census; CM reading typed; no marker moves)

The periods round (`QGEO.PERIODS.01`). The analytic layer of the cover is *classical* (6 checks, ∼1 s): the five basic period integrals $I_{k,j} = \int_0^1 x^k(1 - x^4)^{-j/3}dx$ equal $\frac{1}{4}B(\frac{k+1}{4}, 1 - \frac{j}{3})$ *exactly* (symbolic), with all Gamma denominators on the ζ_{12} grid $(\Gamma(11/12), \Gamma(7/12), \Gamma(5/12), \Gamma(1/12))$ — the $12 = |\mu_4|\cdot|\mu_3|$ structure appears in the analytic layer; the μ_4 rotation acts on actual periods with the exact characters i^{k+1} (40-digit certificates); the quotient curve's real period is the classical ω-CM value $\Omega_E = B(1/3, 1/6)/2 = \Gamma(1/6)\Gamma(1/3)/(2\sqrt{\pi})$ exactly; and the cross-factor period ratio admits *no* low-degree integer relation (PSLQ null) — the two CM factors are period-independent, exactly as the v610 splitting demands. The period dictionary (every cover period a Beta/Gamma monomial on the 12-grid) is the named analytic resource for the Hodge/RP comparisons. Machine-checked: [`verification/v611_periods.py`]. **[E]** (symbolic Beta identities; certificates; PSLQ-null consistency; no marker moves)

The polarization frame (`QGEO.POLFRAME.01`). The form-level bridge (9 checks, ∼2 s): the braid rotation has three *simple* eigenvalues $\zeta_{12} \cdot \{1, -1, -i\}$ ($\det r = -1$), and in its left-eigenvector basis the polarization diagonalizes *exactly*, $(W^\dagger)^{-1}JW^{-1} = \mathrm{diag}((1+\sqrt{3})/2, (1-\sqrt{3})/2, -1)$ (honest typing: diagonality follows from J-unitarity with simple unimodular eigenvalues; the content is the exact values and the signature distribution). The centerpiece: the unique *positive* direction of J — the $h^{1,0} = 1$ Hodge line of the ω-sheet — is *exactly* the eigenspace of the primitive twelfth root $e^{i\pi/6} = \zeta_{12}$: the analytic Hodge structure and the finite rotation agree on where positivity lives. With v611 (periods = Beta values on the ζ_{12} grid) the analytic bridge is one step from closing; the canonical period normalization (Beta weights for the diagonal) is the named next step. Machine-checked: [`verification/v612_polarization_frame.py`]. **[E]** (exact spectrum/diagonalization/signature; bridge reading typed; no marker moves)

The canonical period normalization (`QGEO.PERNORM.01`). The analytic bridge *closes* (19 checks, ∼6 s; executes v612's named step). Geometry first: the quarter rotation $x \mapsto ix$ maps the four branch-point segments of $y^3 = x^4 - 1$ cyclically with *no* deck correction (all 12 wrap factors exactly 1; 40-digit certificates), the single ω-sheet homology relation is $\mathrm{seg}_1 + \mathrm{seg}_2 + \mathrm{seg}_3 + \mathrm{seg}_4 = 0$, and a must-fail control documents that the *raw* period rows are *not* r-eigenvectors (the naive ansatz falsified, residual 0.42). The exact dictionary: $\mathrm{char}(r) = \mathrm{char}(\omega M)$ *exactly* for the induced companion rotation M ($M^4 = 1$) — the reduced Burau rotation at $t = \omega$ is GL_3-conjugate (explicit conjugator G) to the geometric quarter rotation composed with *exactly one* deck step, explaining $r^4 = \omega$ structurally; the deck power is unique among the four scalar twists with $(cM)^4 = \omega$. The period rows (hol dx/y^2, hol $x\,dx/y^2$, antihol $\overline{dx/y}$) are M-eigenvectors with characters $(i, -1, -i)$ and transport through G to r-eigenvectors with $\omega \cdot (i, -1, -i) = \mathrm{spec}(r)$; character match: antihol $\leftrightarrow \zeta_{12}$. The Hodge weights are explicit Gamma monomials (complex Beta formula, verified to 10^{-25}): $h_1 = (3\sqrt{3}/16\pi)\,\Gamma(1/3)^2\Gamma(1/6)^2$ and the two Γ-quotient companions;

ω-sheet signature $(2, 1)$ with the unique *negative* on the antiholomorphic line. The sign bridge: all three character-matched sign products are negative — $J = -h$ *per character*: the compiler polarization is minus the analytic Hodge form up to per-line positive scales, and J's unique positive line (ζ_{12}, v612) is h's unique negative line — Riemann–Hodge positivity realized. Residue named: the canonical fork-basis identification (basepoint bookkeeping for G). Machine-checked: [`verification/v613_canonical_periods.py`]. **[E]** (exact dictionary/uniqueness; certificates; Gamma monomials; sign bridge; no marker moves)

The Gauss–Manin transport (`QGEO.TRANSPORT.01`). The deck step is the boundary monodromy (12 checks, $\sim$7 s; closes v613's residue F1 at the transport level). Transporting the segment cycles along the rigid quarter-rotation loop of the four branch points ($p_k(\tau) = e^{i\pi\tau/2}p_k$, the rotation braid), the discriminant $\lambda(\tau) = e^{2\pi i\tau}$ winds *once* around 0, and the transported periods factorize *exactly*: $I_m(\tau) = e^{i\pi\tau(m+1)/2}\, e^{-2\pi i\tau j/3}\, I_m(0)$ (rotation substitution identity, symbolic). At $\tau = 1$ the deck factor is $\omega = t^4$ on the $t = \omega$ sheet, *uniform* across all three rows (the two holomorphic forms and the conjugated antiholomorphic row), and $\bar{\omega} = \bar{\omega}^4$ on the conjugate sheet: the deck step is t^4 — the *boundary monodromy* of the local system — on every sheet; must-fail controls confirm $t^3 = 1 \neq \omega$ and $t^5 \neq \omega$ (the exponent $4 = |\mu_4|$ is forced by the puncture count). Consequence: the canonical transport matrix in the segment basis is *exactly* ωM — no conjugator freedom left: the v613 dictionary $r \sim \omega M$ is *canonical*, and $(\omega M)^4 = \omega \cdot 1$ re-derives v597's $r^4 = \omega$. Certificates: direct branch-tracked integration at $\tau = 1/3, 1/2, 2/3, 1$ (dps 80) matches the closed factorization to $5 \cdot 10^{-28}$, and the transported segment at $\tau = 1$ equals $\omega\times$ the static neighbor to $2.7 \cdot 10^{-81}$. The reading: the braid fixes the boundary, the rigid rotation does not — their homological difference is one boundary loop of local-system monodromy; the rotation-braid convention is the typed literature identification, carried by the exact char identity. Machine-checked: [`verification/v614_transport_deck.py`]. **[E]** (exact factorization/deck identities; transport certificates; no marker moves)

The interacting gamma slice at toy level (`WOIT.GAMMA.TOY.01`). The gamma battery executed *on* the RP-surviving interaction class (11 checks, $\sim$2 s): the alignment bit $\delta = \pi/2$ with positive coupling — the unique member of the equivariant FK mark family that keeps reflection positivity (v534) — now carries the chirality data at $g > 0$. The two parents (chiral $I + iC$, mirror $I - iC$, both gap 1) are identified; the chiral interacting Grams stay PD $(37, 0, 0)$ at the forced twist $\eta = +i$ on all four cuts for $g \in \{1/32, 1/4, 1, 8\}$ (v534 regression); and the centerpiece: at the *same* twist the mirror state splits sector-exactly on *every* cut and *every* coupling — even $(29, 0, 0)$ PD, odd $(0, 8, 0)$ *strictly negative definite* with max odd eigenvalue $\leq -1.5 \cdot 10^{-3}$, bounded away from zero over the whole grid. Every mirror fermion vector has strictly negative OS norm at the interacting level: in the doubled vector-like system any OS-positive subspace meets the mirror odd sector in $\{0\}$ — kill test (3) cannot fire on the surviving interaction class (v608 upgraded from $g = 0$ to the full grid). The mark transport exists interacting: α_4 maps the mark-A quartet algebra exactly into the mark-B sites (16/16), and both the mark Gram and the transport form are Hermitian PD at every g — kill test (7) does not fire at the interacting toy level. The odd transfer spectrum stays multiplicity-free at every g (honest finding: a strong-coupling near-doubling at $g = 8$, gaps $\sim 2 \cdot 10^{-5}$, without closing; the $g = 0$ top 1.641438 is numerically the v524 compressed-clock top — the quarter-turn transfer datum recurs across N). Dead-member control: the RP-dead member ($m{=}2, s{=}+$) is indefinite on its straddled cut — off the survivor the gamma battery has no RP home: the dynamical bit selection and the chirality data *co-locate* on the same member. Fence: one toy, one interaction class; kill tests (3)/(6)/(7) stay formally live on $\mathcal{A}_{\mathrm{hol}}$; kill test (6) untouched. Machine-checked: [`verification/v615_gamma_toy_interacting.py`]. **[E]**/**[C]** (float certificates, declared tolerances; toy fence; no marker moves)

The internal $SU(2)$, kinematic separation (`WOIT.SU2.KIN.01`). The first executed slice of gamma kill test (6) (17 checks, $<$1 s). On $\mathbb{C}^4 = \mathbb{H}^2$ the right multiplications close the (opposite) quaternion algebra exactly, and Woit's Euclidean structure *is* an internal direction: $\rho_{\mathrm{tw}} =$ right-j verbatim in the v519/v565 conventions. The internal algebra right-sp(1) preserves *every* fiber of the twistor fibration $\mathbb{PT} \to S^4$ (trivial on spacetime; the left multiplications move a generic fiber), commutes *exactly* with the full spacetime quaternion action, and the two algebras intersect in $\{0\}$: the internal factor *separates* from the Euclidean rotation group — kill test (6)'s kinematic branch does not fire. The clock factorizes exactly: RHO4 = left-u $\circ$ right-u with $u = e^{i\pi/4}$ (neither factor alone; must-fail controls); the ALE deck is *purely* spacetime-side (DECK4 = left-i exactly) and RHO4^2 = DECK4$\circ$ right-i. The complex choice breaks the internal $SU(2)$ to $U(1)$ kinematically, and the v519 mark torsor is exactly the μ_4 subgroup of the internal $U(1)$ composed with the OS conjugation; σ_{std} fixes right-j and inverts the internal charge and both clock factors. The dynamical half of kill test (6) stays open and formally live on $\mathcal{A}_{\text{hol}}$. Machine-checked: [`verification/v616_su2_internal_kinematic.py`]. [E]/[C] (exact operator identities; kinematic fence; no marker moves)

The seam forcing round (`QGEO.SEAMFORCE.01`). A constructive slice of the bedrock premise `QGEO.SYM.01` (14 checks, <1 s): the conformal seam axioms *force* the μ_3-cover. $\mathbb{Z}_4$-Möbius rigidity: any 4-point configuration on $\mathbb{P}^1$ cyclically permuted by an order-4 Möbius map is Möbius-equivalent to μ_4 (the multiplier is forced to a primitive fourth root); the cross-ratio census of μ_4 is the harmonic orbit $\{2, 1/2, -1\}$ exactly (must-fail: 4/3, the disc-7 jet datum, is not in the orbit). The weight census: among all 81 μ_3-monodromy weight vectors exactly four are clock-equivariant, and the mark-equivalence demand kills the alternating pair — the weights are *forced* uniform, $j \in \{1, 2\}$, the conjugate sheet pair of v613. Uniform $j = 1$ gives $y^3 = (x-1)(x-i)(x+1)(x+i) = x^4 - 1$ *exactly*; the total finite monodromy forces full ramification at infinity (the v603 separated cusp); Riemann–Hurwitz gives genus 3 exactly. And the seam geometry matches: the marks lie on the unit circle, the doubling reflection fixes seam and marks, the OS-cut (bond) reflection $z \mapsto -i\bar{z}$ permutes the marks by exactly the $(k, 5-k)$ pattern of the v599 real structure and meets the circle at the two v519 cut points. The bedrock residue narrows to two named steps: the cyclic-3 choice ($N_{\text{fam}} = 3$, anchored in the E_8 glue) and the physical-seam $\leftrightarrow$ conformal-seam identification. GATE.QGEO does not move. Machine-checked: [`verification/v617_seam_cover_forcing.py`]. [E]/[C] (exact rigidity/census/genus identities; residue typing; no marker moves)

The uniform constant round (`PRIME.UNIFC.01`). The equidistribution constant $C = 1$, frozen (6 checks, ~3 s). On the declared frame-A surface (69 floor-passed windows, $h = 142..1445$) the model value q_{model} keeps *one* sign on the whole ladder (0.039..0.112 — no model zero crossing anywhere), and on *every* lock-sign window (67 of 69) the rate $|q_{\text{real}}/q_{\text{model}}| \cdot h \le 1$ holds — the conjecture's explicit constant $C = 1$, measured max 0.982; the tertile medians are non-increasing (0.61/0.45/0.39). The sharpened typing: exactly *two* windows carry a q_{real} sign flip ($h = 1219$ at $9.2 \cdot 10^3$, $h = 1445$ at 3.5), and the violator set of the $C = 1$ bound *equals* the flip set exactly — the sign predicts the bound violation window-sharp. Honest correction of the earlier diary note: the blow-up mechanism is the q_{real} flip (not a q_{model} crossing), and the note missed the edge window $h = 1445$. Scrambled combs break the bound by more than 10^4. No uniformity proof, no RH statement; Problem 7.1 untouched. Machine-checked: [`verification/v618_uniform_constant.py`]. (MEASURED) + [E] controls (declared surface and floors; no marker moves)

The flip mechanics (`PRIME.FLIPMECH.01`). The v618 dichotomy upgrades to a *mechanism* (6 checks, ~35 s). The atom demand of a window is $u \le 2\alpha$ and the prime-power data cap is $U_{\max} = 12.899$: *exactly* the two flip windows are truncated ($h = 1219$ with gap $\Delta u = 0.677$, $h = 1445$ with 0.089; all 67 other windows complete) — the flip set *equals* the truncation set. The coupling spectrum at the flips is regular-collective (share 5.8%, $N_{50} = 57$ — no zero resonance), and truncation *injection* into complete windows reproduces the flips with matched sign and magnitude at both scales ($\Delta u = 0.089$ gives $-5 \cdot 10^{-5}/-1 \cdot 10^{-4}$ vs the observed $-1.0 \cdot 10^{-4}$ at 1445; $\Delta u = 0.677$ gives $-0.18/-0.22$ vs -0.29 at 1219), monotonically in the depth. On the complete-comb surface ($2\alpha \le U_{\max}$: 67 windows) the $C = 1$ bound holds with *zero* exceptions (max $\varepsilon h = 0.982$): the sign-flip windows are *retired* as data-boundary artifacts — extending the surface requires extending the prime-power data, not the theory. Machine-checked: [`verification/v619_flip_mechanics.py`]. (MEASURED) + [E] (saturation census; injection mechanism; no marker moves)

The cyclic-N census (`QGEO.NCENSUS.01`). The first v617 bedrock residue made *conditional* (9 checks, <1 s). For uniform-weight cyclic covers $y^N = x^4 - 1$ over the forced μ_4 base: full ramification everywhere iff $\gcd(N, 4) = 1$, i.e. $N \in \{3, 5\}$ among $N = 2..6$; the Riemann–Hurwitz genus table is $1, 3, 3, 6, 7$; the primitive character sheets have the compiler's rank 3 exactly for $N \in \{3, 5, 6\}$. The Chevalley–Weil signatures decide: $N = 3$ has *both* primitive sheets Lorentz ($(1, 2)/(2, 1)$ — the v599/v613 signature); $N = 5$ has a Lorentz pair; $N = 6$ has *no* primitive Lorentz sheet (its Lorentz content sits on the order-3 characters — $N = 6$ reduces to $N = 3$): the seam-admissible orders with a primitive Lorentz sheet are exactly $\{3, 5\}$. The pinning: $\det_{\mathbb{Z}}(1 - \text{deck}) = \Phi_N(1)^3 = N^3$ for prime N — 27 for $N = 3$ (*exactly* the machine-checked v597 constant) versus 125 for $N = 5$; the saturation analogue $81 = 3^4$ versus 625: among $\{3, 5\}$ the compiler's integral constants select $N = 3$ *uniquely*. The residue moves one level down: why the compiler is 3-adic is $N_{\text{fam}} = 3$, anchored in the E_8 glue ($240 = 16 \times 5 \times 3$). GATE.QGEO does not move. Machine-checked: [`verification/v620_cyclic_n_census.py`]. [E]/[C] (exact census/pinning; conditional typing; no marker moves)

The interacting Klein–Landau semigroup (`WOIT.GAMMA.SEMI.01`). The gamma net-dynamics slice (6 checks, ∼2 s). At $g = 0$ the v524 Klein–Landau pattern reproduces exactly at $N = 16$ on the admissible cut: all seven steps τ_k Hermitian (domain dims $29, 22, 16, 11, 7, 4, 2$), even steps PD, odd steps $k = 1, 3, 5$ indefinite with *exactly zero* one-particle diagonal (the free chirality datum), $k = 7$ parity-trivial. At $g > 0$ (the RP survivor) exact Hermiticity survives *precisely* on the steps $\{4, 6, 7\}$ and is lost on $\{1, 2, 3, 5\}$ — the semigroup *contracts* to the μ_4-commensurate window; the clock step $k = 4$ stays Hermitian and PD ($(11, 0, 0)$) over the whole grid $g \in \{1/32..8\}$: the μ_4 clock is a positive transfer operator on the interacting net. The mechanism is exact: the survivor interaction is α_4-invariant *exactly* and not α_1/α_2 invariant (translation invariance breaks to the quartet stabilizer $\{0, 4, 8, 12\}$); $k = 5$ shows quartet containment alone does not protect (even parity *and* alignment are needed). The reading: the reconstructed rotation group of the interacting toy is the *clock tower*, not the full circle. One toy, one interaction class; gamma proper stays open on $\mathcal{A}_{\text{hol}}$. Machine-checked: [`verification/v621_interacting_semigroup.py`]. [E]/[C] (float certificates; exact combinatorics; toy fence; no marker moves)

The seam identification (`QGEO.SEAMID.01`). The second v617 bedrock residue closes at the kernel + dictionary level (10 checks, ∼1 s): *the physical seam is the conformal seam.* The kernel identity: the v519 chiral seam kernel $c(d) = (2/N)/\sin(\pi d/N)$ for odd d and 0 for even d is *exactly* the antiperiodic (NS) chiral mode sum on the 16-site circle, for all $d = 1..15$ *including* the even-distance zeros — the flat-band structure *is* the NS spectrum (closed form $\sum_{j<8} \sin((2j+1)x) = \sin^2(8x)/\sin(x)$, exact); the Ramond (periodic) sum fails as a must-fail control — the NS structure is forced, not conventional. The site map $\theta_a = (2a+1)\pi/16$ places the four mark-bond midpoints *exactly* at μ_4 and no site touches a mark (the v519 precision (ii), literal geometry); the group dictionary is exact (α_1 = rotation by $2\pi/16$, the clock $\alpha_4 = z \mapsto iz$, the RP reflections = diameter reflections about $(k+1)\pi/16$; the admissible cuts map to the axes $\{\pi/4, \pi/2, 3\pi/4, \pi\}$). And the v534 straddle law becomes geometry: the through-mark axes are *exactly* the straddled cuts $\{7, 15\}$, the between-mark axes the avoiding cuts $\{3, 11\}$; v617's bond reflection is the cut $k = 11$ carrying the v599 $(k, 5-k)$ mark permutation. The remaining continuum residue is the *covering-level* identification (the μ_3-cover of the seam double carrying the full interacting theory), named, not claimed. GATE.QGEO does not move. Machine-checked: [`verification/v622_seam_identification.py`]. [E]/[C] (exact mode-sum/dictionary identities; must-fail control; no marker moves)

The covered seam (`QGEO.COVERSEAM.01`). The lattice skeleton of the covering-level identification (9 checks, ∼4 s; the named v622 residue, first slice). Lifting the 16-site seam circle to the μ_3-cover (sheet shift per mark-bond crossing), the covered seam is *one* 48-site NS circle for the uniform weights $j = 1, 2$ — and *disconnects* into three circles for the trivial and the alternating weights $(1, 2, 1, 2)$: a new, independent kill of the alternating pair (seam connectivity; total monodromy $6 \equiv 0 \bmod 3$). The explicit walk closes after exactly 48 steps (three base turns, 12 mark crossings). The lifted clock: L^4 = deck and L^{12} = id on *all* 48 points with proper order 12 — v597's $r^4 = \omega$

and v614's boundary-monodromy transport are *lattice combinatorics* on the covered seam, and the ζ_{12} spectrum of the Burau rotation is the lifted clock's order. The v622 kernel identity generalizes verbatim to $N = 48$ (all $d = 1..47$). The 12-grid: the NS modes split under the deck into three classes of 16 modes with offsets $\{1/6, 1/2, 5/6\}$; the offset-1/2 class equals the *base* NS frequencies exactly (the untwisted sector *is* the base seam, verbatim); the fermionic deck lift satisfies $\text{deck}^3 = -1$ (the NS full rotation — the spin double, resonating with the v510/v519 $(-1)^F$ dichotomy), and the *bilinear* deck eigenvalue of the twisted pair is exactly ω — the $t = \omega/\bar{\omega}$ sheets live on the H^1/bilinear level. The 12 lifted mark crossings form a single 12-cycle under the lifted clock. The remaining residue is the interacting/CFT-level orbifold statement (twist-field OPE, the full $\mathbb{Z}_3$ orbifold of the seam CFT), named, not claimed. GATE.QGEO does not move. Machine-checked: [`verification/v623_covered_seam.py`]. [E]/[C] (exact combinatorics and mode-sum identities; orbifold residue typed; no marker moves)

The external-review lattice audit (`EXTREV.LATTICE.01`). Three claimed derivations of the third external review (2026-08-02), machine-checked (14 checks, <1 s); one follow-up retyped honestly. *The four-mark closure* [[E], conditional]: generalizing the glue to $D_{d+1} \oplus A_{d-1}$, three *independent* selectors each force $d = 4$ — the disc-group census ($\mathrm{disc}(A_{d-1}) = \mathbb{Z}_d$ cyclic; $\mathrm{disc}(D_{d+1})$ cyclic iff d even; $\mathbb{Z}_d = \mathbb{Z}_4$ forces $d = 4$; negative controls $d = 3, 5, 8$), the even-glue norm equation $(d{+}1)/4 + (d{-}1)/d = 2$ (unique positive solution 4; no other even target $2k$, $k \le 10$, admits an integer d), and the Pythagorean selector $(d{-}1)^2 + d^2 = (d{+}1)^2$ — the $(3, 4, 5)$ triple of (cycles, marks, carrier). With $(1, 1, 2)$ the unique 3-part partition of 4 and $c_3 = 1/(2e_1\pi) = 1/(8\pi)$: under the named architecture the two-input presentation reduces to *one* boundary degree $d = 4$ plus π; honest typing: conditional on the glue architecture — `AX.P2.01` keeps its marker, and the residue moves to why the physical boundary realizes the minimal cyclically-glueable genus-0 divisor. *The anchor bytecode* [[E]]: $\chi_a = (t{-}1)^2(t{-}2)$ carries $(4, 5, 2)$; the power sums $p_n = 2 + 2^n$ obey $p_{n+2} = 3p_{n+1} - 2p_n$; Newton recovers the e_k; $p_1p_2p_3 = 240$, $p_4 - p_3 = 8$, $240 + 8 = 248$ (the $H = 1 + V(V{-}1)/2$ realization is v602-checked). *The Lorentz congruence* [[E] + honest negative]: $P = [[3, 0, 0], [3, 0, 2], [-1, 1, -1]]$ with $\det P = -6$ satisfies $P^{\mathsf{T}} J_{\det} P = J_{\text{fix}}$ *exactly* — the prime-front determinant form ($\det 2$) and the v604 cover polarization lattice ($\det 72$) are the *same* rational quadratic form, the cover an index-6 sublattice ($36 = 6^2$, $6 = p_2(a)$); the review's follow-up test (integer R to $B = M^\dagger J$) is *ill-posed* as stated (B is complex symmetric, $\det B = 8$ verified); on the reformulated $\mathbb{Q}(\omega)$ level the third symmetric-Gauss ratio is an exact square; P-canonicity and the Hodge-chamber question are the named opens. The review's RH architecture is preregistered as the contract `PRIME.WEIL.OPERATOR.01` with verified citations (Suzuki `arXiv:2606.09096`; numerical realization `arXiv:2607.24830`). No marker moves; no RH statement. Machine-checked: [`verification/v624_external_lattice_audit.py`]. [E]/[C] (exact closure/bytecode/congruence identities; honest retype; citations verified)

The prime-shadow audit (`PRIME.SHADOW.01`). The checkable core of the external note's "primes are the shadow of the finished geometry" reading, machine-verified on the compiler's own glue chain (6 checks, <1 s). The E_8 theta from the glue decomposition $(\theta_2^8 + \theta_3^8 + \theta_4^8)/2$ has only even norms and shell counts $r(2n) = 240\,\sigma_3(n)$ for $n = 1..12$ *exactly* (the first shell is $240 = |R(E_8)|$): $\Theta_{E_8} = E_4$ — the finished lattice's counting function is a modular form. The 'address space' reading is unique factorization, exactly: shell counts factor over *coprime* addresses (σ_3 multiplicative; must-fail: $\sigma_3(4) = 73 \neq 81$). The 'independent check channels' reading is a theorem at the theta level: $T_p\Theta = (1{+}p^3)\Theta$ for $p = 2, 3, 5, 7$ and $[T_2, T_3] = 0$ — every prime is an independent *commuting* channel and the compiler's theta is a *simultaneous* eigenvector of all of them. And the zeta shadow: $L(E_4, s) = \zeta(s)\zeta(s{-}3)$ (verified at $s = 6$ to $5 \cdot 10^{-8}$) — the Riemann zeta appears as the *factorized shadow* of the E_8 counting function: the primes enter *after* the geometry, through its Euler decomposition. The speculative framings (compiler eigenfrequencies, synchronization code, complexity barriers, RH as maximal coherence) stay typed hypotheses, not adopted. Honest scope: classical Jacobi/Hecke facts, realized verbatim by the compiler's own objects — the content is the fixed direction of explanation (geometry first, primes as readout) within TFPT's narrative. No marker moves, no RH statement. Machine-checked: [`verification/v625_prime_shadow.py`]. [E]/[C]

(exact theta/Hecke identities; typed speculation fence)

The E_8 code round (`E8.CODE.01`). "E_8 is an error-correcting code" is a *theorem*, not a metaphor (7 checks, $\sim$2 s). The extended Hamming code $[8,4,4]$ (16 codewords, weights $\{0{:}1, 4{:}14, 8{:}1\}$, self-dual) under Construction A yields an even (weights in $4\mathbb{Z}$) unimodular (covolume 1) rank-8 lattice — by uniqueness, E_8 — with 240 minimal vectors and 2160 on the second shell (the `v625` theta identity on the code model). Error correction is literal: every single-bit corruption of every codeword has a unique nearest codeword equal to the original (16×8 exhaustive). The compiler ties are typed observations: $8 = $ rank, $4 = $ min distance $= d = |\mu_4|$ (the `v624` boundary degree), $16 = |C| = $ the carrier half-spinor $=$ the seam sites, $14 = 2 \times 7$ (the parabolic dimension; the `v605` denominator). No physics claim. Machine-checked: [`verification/v626_e8_code.py`]. [E]/[C] (exact code/lattice identities; typed ties)

The Hodge chamber round (`PRIME.HODGECONE.01`). The measured half of the `v624` geometric positivity route (5 checks, $\sim$2 s). Transporting every complete window's vector $y_h = (S_{11}, S_{22}, S_{12})$ through the exact Lorentz congruence, $z = P^{-1}y$ with $z^{\mathsf{T}} J_{\text{fix}} z = 2 \det S$, all 67 windows land in the *positive cone* of the cover polarization lattice ($\det S > 0$ everywhere, min 11.8) and on *one sheet* (the J-inner products with the positive eigendirection keep one sign, range 42..1216). Honest typing: the scramble control does *not* leave the chamber — membership is a density-layer property (`v582`: 98.7% density-driven), a coarse stability statement; the fine $C = 1$ arithmetic lives *inside* the chamber. P-canonicity and the chamber theorem stay open; no RH statement. Machine-checked: [`verification/v627_hodge_chamber.py`]. [E] (exact transport) + (MEASURED) (chamber/sheet)

The orbifold Casimir round (`QGEO.ORBCAS.01`). The first quantitative slice of the `v623` orbifold residue (5 checks, $\sim$1 s). The chiral vacuum energy of the ν-offset sector on N sites has the exact closed form $E_0(\nu, N) = -\cos(\pi(2\nu-1)/2N)/(2\sin(\pi/2N))$, whose symbolic $1/N$ expansion gives the finite-size coefficient exactly $\pi(6\nu^2-6\nu+1)/12$ — the fermionic twist formula. For the covered seam's deck classes $\nu = (1/6, 1/2, 5/6)$ the sector table is $(1/72, -1/24, 1/72)$: the twisted pair is degenerate (conjugate sheets), the twisted/untwisted ratio is exactly $-1/3$, and the $\mathbb{Z}_3$-twist gap is exactly $1/18$ (in π/N units) — the covered seam carries the $\mathbb{Z}_3$-orbifold Casimir data verbatim at the kinematic level. The interacting statement (twist-field OPE, crossing, RP) stays the named residue. Machine-checked: [`verification/v628_orbifold_casimir.py`]. [E] (exact closed forms and expansions)

The root incidence census (`E8.INCIDENCE.01`). The review's $48 \times 5 \to 240$ kill test, executed at the first hurdle (6 checks, $\sim$1 s). For the compiler-canonical order-12 element $g = J \circ \sigma$ (the μ_4 quarter-turn times the family 3-cycle) the 240 roots decompose as $19 \times 12 + 3 \times 4$ — *not* the 20 free ζ_{12} orbits a naive 48×5 incidence needs: the naive construction is *killed*; $48 \times 5 = 240$ stays an audit fingerprint. The positive residue is sharp: the μ_4 clock alone acts *freely* with exactly 60 orbits — $60 = D_{\text{start}}$ appears as the free clock-orbit count on the roots — and the family 3-cycle fixes exactly 12 roots (the `v623` lifted-mark count), which form exactly 3 free clock orbits. Machine-checked: [`verification/v629_root_incidence.py`]. [E]/[C] (exact census; honest kill; typed observations)

The Suzuki first contact (`PRIME.WEIL.CONTACT.01`). The first executed slice of the `PRIME.WEIL.OPERATOR.01` contract (4 checks, $\sim$55 s). The *atom layer* of the TFPT window form *is* Suzuki's prime measure, literally: the screw function g of `arXiv:2606.09096` (eq. 1.3) carries the prime term $\sum_{n \le e^{|t|}} \Lambda(n)/\sqrt{n}\,(|t| - \log n)$, whose distributional second derivative places atoms at $\pm \log n$ with weights $\Lambda(n)/\sqrt{n}$ — exactly the TFPT atom table (positions $U = \log$ prime powers, weights $\mathrm{MU}/2 = \Lambda(n)/\sqrt{n}$, verified exactly on 40 atoms), tent-projected exactly as in `v563`. The first numeric Galerkin comparison (hat basis, D^*GD) measures a *non-scalar* conversion profile against the TFPT symbol ($-0.064..-0.070$ over lags 3..9): the W1 identification requires a gauge/normalization dictionary beyond a scalar — the preregistered rank-one/normalization freedom, now with first data. W1's atomic half is closed; the smooth half stays open. No positivity claim, no RH statement. Machine-checked: [`verification/v630_suzuki_contact.py`]. [E] (literal atom identity) + (MEASURED) (Galerkin profile)

The W1 dictionary round (`PRIME.WEIL.DICT.01`). The v630 non-scalar residual is *resolved*: it IS the zeta pole term (7 checks, $\sim$3 s). The Lerch collapse — term-wise $(2n+\frac{1}{2})^2/(n+\frac{1}{4})^2 = 4$, hence $[e^{-t/2}\Phi(e^{-2t}, 2, \frac{1}{4})]'' = 4e^{-t/2}/(1-e^{-2t})$ exactly — turns the Hurwitz–Lerch block of Suzuki's screw function into a geometric series, and yields the structure theorem: off the atoms, $g''(t) = -2\cosh(t/2) - 4\,e^{-t/2}/(1-e^{-2t})$ — Suzuki's smooth layer is $-4\times$ the TFPT archimedean density *minus* the pole block $2\cosh(t/2) = e^{t/2} + e^{-t/2}$, the ζ pole weights at $s = 0, 1$ that TFPT tracks separately as the exact rank-one piece (`v591`). The closed per-lag ratio law $c_{\text{gal}}/c_{\text{arch}} = -D[4 + (e^t{+}1)(1-e^{-2t})]$ verifies with the declared d^{-2} discretization profile; after pole subtraction the smooth conversion is the *scalar* $-4D$ (monotone to 1.0006), and the atom-layer lattice constant is D^2 exactly (atoms $\log 2$, $\log 3$ agree to 10^{-3}). W1 closes at the measured level on both halves: $\widehat{A}_{\text{arch}} = -(1/4D)\,\text{Galerkin}_{\text{smooth}} + \text{pole rank-one}$; the theorem-level remainder (second-order term, $d \le 2$ boundary cells, L_0^2 projection) is typed. W2/W3 stay open; no positivity claim, no RH statement. *Erratum (2026-08-02, corrected the same day):* this round read eq. (1.3) with Lerch coefficient -1; Suzuki's own §2.2 data lock $+1/4$ ([`verification/v643_w1_theorem.py`], C0.1). All identities above are correct identities of the chain kernel $\tilde{g} = g - \frac{5}{4}\,\text{Lerch}$, and every number transfers verbatim via $c^{\text{sm}}_{\text{gal}}(\tilde{g}) = -4\,c^{\text{sm}}_{\text{gal}}(g)$ — on Suzuki's true g the smooth layer is $+\rho$ (not -4ρ) and the dictionary is the *single* scalar $+1/D$ on both layers, sign-compatible with positivity. Machine-checked: [`verification/v631_w1_dictionary.py`]. [E] (exact Lerch/structure identities) + (MEASURED) (scalar closure)

The F_{transfer} $\mathbf{PGL_2}$ round (`FTR.PGL2.01`). The `v533` preregistration executed in its *literal* group form (25 checks, $\sim$1 s). On the disc-(-7) norm line α_t (norms 2/4/14/32; base action $T : \alpha \mapsto \alpha + 2$, parabolic) the declared `v183` Koide reading induces the parabolic step $+9 = N^2_{\text{fam}}$ and is *exactly* intertwined with T (the reading map Φ is itself Möbius, $\Phi \circ T = g \circ \Phi$); the cross-ratio is 4/3 exactly and Möbius transport forces the F corner 53 fit-free. On the native continuous half the time-1 maps are explicit PGL_2 elements: g_{pole} hyperbolic with multiplier $(2/3)^6$, $g_{\text{Boltzmann}} = g_{\text{pole}}$ *identically* (the `v425` single-flow theorem), g_{QCD} parabolic and *explicitly* conjugate to T (translation $+7/2\pi$ in the $1/\alpha$ coordinate), $g_{\text{relic}} = \text{id}$ (degenerate); the `v575` uniflow degeneration and the `v578` own-clock equivalence reproduce. The kill criterion "four incompatible arithmetic actions" is *not* met; five must-fail controls have teeth (the raw determinant reading transports to $-16 \neq 32$). The residual — the external anchor clocks/jets — stays fenced by `CONTRACT.F.01`. Machine-checked: [`verification/v632_ftransfer_pgl2.py`]. [E]/[C] (exact PGL_2/intertwining identities; typed reading)

The $\mathbb{Z}[i]$-E_8 quotient round (`E8.ZIQUOTIENT.01`). The `v629` positive residue, one level deeper (17 checks, $\sim$3 s). The μ_4-quotient of the 240 roots (the 60 free clock orbits) *is* the hermitian-unimodular $\mathbb{Z}[i]$-E_8: the hermitian form $H(x, y) = \langle x, y\rangle + i\langle x, Jy\rangle$ has $H(v, v) = 2$ on all 60 line representatives, the $\mathbb{Z}[i]$-span is the full E_8 lattice, and a hermitian-*unimodular* $\mathbb{Z}[i]$-basis exists among the reps ($\det \text{Gram} = 1$). σ acts on the 60 lines with census $19 \times 3 + 3$ fixed (the fixed lines are exactly the J-orbits of the 12 σ-fixed roots), all 60 order-2 unitary reflections preserve the root set, and the line group has order $11520 = 46080/4$ — the ST31 fingerprint ($60 = 7{+}11{+}19{+}23$ hyperplanes; ST32 killed by its 40-hyperplane count). The $60 \to 6$ cascade route *on the quotient* is killed: the hermitian Gram has rank 4, both $|H|^2$ graphs are regular (any degree rule is ill-defined), and pair elimination has no canonical maximum — a typed negative. Machine-checked: [`verification/v633_orbit60_quotient.py`]. [E] (exact quotient/lattice identities; typed cascade kill)

The ST31 structure round (`E8.ST31.01`). Does the ST31 find carry structural weight? Yes — and the numerology does not (56 checks, $\sim$10 s). $G_{31} = \langle\text{the 60 reflections}\rangle$ has order 46080 and *is* the full unitary stabilizer of the μ_4-quotient structure (exact backtracking count); the degrees $(8, 12, 20, 24)$ are *unique* via Molien among all divisor quadruples with product 46080; Springer regular elements of orders 8/12/20/24 exist with centralizers 192/288/20/24. The compiler data embeds canonically: J, σ, $c = J \circ \sigma$ all lie in G_{31} with the exact clock identities $\sigma = c^4$ and $J = c^9$ (the μ_4 center is a *power* of the clock); the root stabilizer is $G(4, 2, 3)$ (Steinberg), the line stabilizer $Z_4 \times G(4, 2, 3)$. The computed structure is $G_{31} = (Z_4 \circ 2^{1+4}).\text{Sp}_4(2)$. The numerology kill: $|G_{31}| = |W(D_5)| \times |W(A_3)|$ carries *no* structure — not an isomorphism (centers 4 vs 1), not a direct or semidirect product (all small normal closures join to order 64), and $W(D_5)$ does not even *embed* (minimal faithful dimension $5 > 4$); $W(A_3) = S_4$ does embed. Honest sharpening: the `v629` compiler clock c is *not* the Springer-regular 12-element (census $19 \times 12 + 3 \times 4$, $|C(c)| = 72 \neq 288$) — the free $20 = 240/12$ comb belongs to the regular 12-clock of G_{31}, a different conjugacy class. The order identities are additionally machine-proved in Lean (`G31Orders.lean`, `lake build` green). Machine-checked: [`verification/v634_st31_structure.py`]. **[E]** (exact group/Molien/Springer census; numerology kill; honest clock typing)

The P-canonicity census (`PRIME.PCANON.01`). The first open step of `v627`, executed (18 checks, $\sim$3 s). The full $[-4, 4]$ box census finds *exactly* 40 integer solutions Q of $Q^{\mathsf{T}} J_{\mathrm{det}} Q = J_{\mathrm{fix}}$; the minimal-Frobenius class (norm^2 25, 8 solutions) *is* P's class modulo the natural order-8 prime-side symmetry group, and operator compatibility (integral transport of both C_U, C_V) holds on it; P is *not* in the $\mathbb{Q}$-span of the 7-word algebra of (C_U, C_V) — the word-algebra route stays a typed negative. And the *finer* ray invariants (projective spacelike angle φ, cone distance $q/|z|^2$, timelike angle, spacelike ratio) separate the true 67-window family from scrambled combs in 17–18 of 18 cases per invariant (shifts up to 87 true-family standard deviations); one scramble at $h = 142$ even leaves the positive cone — where the coarse `v627` chamber does not separate, the fine ray geometry does. Machine-checked: [`verification/v635_p_canonicity.py`]. **[E]** (exact census) + (MEASURED) (fine-invariant separation)

The P operator construction (`PRIME.PCONSTRUCT.01`). From "found matrix" to "constructed map" (25 checks, < 1 s). The two lowest-height primitive isotropic rays of J_{fix} are *exactly* the C_V-operator null frame ($\operatorname{fix} C_V, \ker C_V$), and P transports this frame to the canonical rank-one rays of the prime-side determinant form, grading-preservingly. The full solution set of {congruence + the two ray conditions} is an explicit 1-parameter boost family (hyperbolic-basis normal form, solved exactly); integrality leaves *exactly* 16 members (Pell-type members excluded by an entrywise Galois argument), and either Frobenius-minimality or the trace-ray condition selects *exactly* $\{+P, -P\}$: P is, up to global sign, the Frobenius-minimal integer congruence transporting the C_V null frame onto the canonical rank-one rays. Honest residuals: the orientation choice of the target pair inside its symmetry orbit, and the gl_2-derivation ansatz for the transported C_V fails (documented must-fail — the named missing structure). Machine-checked: [`verification/v636_p_construction.py`]. **[E]** (exact construction/normal form/census; documented must-fail)

The fine-invariant $C = 1$ bridge test (`PRIME.FINEC1.01`). An honest negative, promoted as a typed result (8 checks, $\sim$7 s; gate preregistered before the numbers). On the `v618` surface (reproduced exactly) the raw correlations of the fine Hodge-ray invariants with the $C = 1$ load $\varepsilon_h h$ ($\rho = \pm 0.40$) are *entirely* the trivial h-trend: after rank-cubic h-control the partial Spearman correlations are $\rho = -0.040$ ($p = 0.75$, Freedman–Lane permutation) for both primary invariants — the preregistered gate ($|\rho| \geq 0.30$, $p < 0.01$) is missed by an order of magnitude, and the tertile structure vanishes after detrending. The direct fine-geometry→arithmetic bridge route at the window level is *closed*: the fine invariants are h-parametrized geometry, the $C = 1$ cancellation is arithmetic beyond it. Machine-checked: [`verification/v637_fine_c1_bridge.py`]. (MEASURED) (preregistered honest negative)

The code-semantics rounds (`E8.CODESEM.01`). The `v626` code reading sharpens to an exact dictionary — and the naive placement dies (27 checks, $\sim$3 s; stage-1 kill probe 13/13). Exactly 2 of the 30 S_8-placements of the extended Hamming code carry both compiler symmetries mod 2 (the μ_4 in-pair swap and the family pair 3-cycle), exchanged by the anchor transposition; the naive `v626` placement is *not* among them. The unique equivariant placement C^* *is* Reed–Muller $\mathrm{RM}(1, 3)$ on $\mathrm{AG}(3, 2)$ (the 14 weight-4 supports are the 14 planes; $30 = |S_8|/|\mathrm{AGL}(3, 2)|$). The dictionary: no equivariant 4+4 block split exists; the unique minimal orbit of information sets is the transversal pair — one information bit per μ_4 pair (3 family bits + 1 anchor bit); the syndrome space factors as the flag $0 < \langle q \rangle < \mathbb{F}_2^4$ (q the universal in-pair bit) with σ acting on the pair directions as the 3-cycle + fixed anchor ($P_3 = P_0 + P_1 + P_2$); and decode = projection *verbatim* on the whole single-error class (3840/3840, exact arithmetic; all 3840 perturbed vectors sit at squared distance exactly 1 with the mod-2-collapsing tie pair). The 12 σ-fixed roots reduce mod 2 to exactly three classes of 4, and the clock-orbit fibration reproduces $60 = 7 \times 4 + 4 \times 8$. Kill control: the sets {universal in-pair syndrome}, {clean orbit structure}, {both-invariant} *coincide* — the equivariance is load-bearing. Machine-checked: [`verification/v638_code_semantics.py`]. **[E]** (exact dictionary; kill control)

The twist-field/OPE round (`QGEO.TWISTOPE.01`). The interacting slice of the $\mathbb{Z}_3$ orbifold residue (`v623`/`v628`), executed at the abelian vertex level (25 checks, ∼1 s). The twist weight $h_\sigma = 1/36$ follows from three independent exact routes (the `v628` Casimir coefficient read as $2(h_\nu - c/24)$; the ζ-function mode sum; the bosonization pair $h(\frac{1}{2}-\eta) + h(\frac{1}{2}+\eta) = \eta^2/2$), with the convention traps resolved exactly. On the lattice (U(1) string determinants on the half-filled NS seam double, whose equal-time covariance is half the `v519` seam kernel): the two-point exponent $2\Delta = 2/9$ verifies at 0.06% (Ising control at 0.02%); crossing is *exactly* symmetry-protected (the gap rotation maps $w \mapsto 1-w$ while $|G_4|$ is invariant at machine precision); the closed four-point form $A^2[w(1-w)]^{-2\Delta}$ holds at $< 0.1\%$ after $N^{-2/3}$ extrapolation; the leading OPE constant $c_1 = 2\Delta = 2/9$ is model-bound at 0.19% (the model-free extraction stays a typed ±15% band). Reflection positivity, differentiated: the real $\mathbb{Z}_2$ class is exactly PSD at every N (machinery validated), while the naive antiunitary $\mathbb{Z}_3$ pairing fails positivity by an N-stable amount — the positivity-carrying pairing (parafermionic Klein twist) is a named open item. `GATE.QGEO` does not move. Machine-checked: [`verification/v639_twist_ope.py`]. **[E]** (exact weights/crossing) + (MEASURED) (two-/four-point, OPE) + **[O]** (Klein-twist pairing)

The W1 boundary closure (`PRIME.WEIL.BOUNDARY.01`). The `v631` D6 typed remainder closes symbolically (11 checks, ∼30 s; certified at 20+ digits). The duplication identity $\psi(\frac{1}{4}) = -\gamma - 3\log 2 - \pi/2$ and the scheme conversion $k = \pi/4 + \frac{3}{2}\log 2$ are sympy-exact, so the two δ_0 conventions ($\gamma + \log\pi$ vs $\psi(\frac{1}{4})$) are *one* object in *one* scheme, and the exact boundary equation

$$A_{\text{arch}} = \tfrac{1}{4}\, g''_{\text{smooth}} - \tfrac{5}{4}\big(\log\pi - \psi(\tfrac{1}{4})\big)\,\delta_0$$

verifies against the `v563` tents $d = 0, 1, 2$ at relative residuals $< 10^{-22}$ (point route and heavy Galerkin route, with the closed pole formula); the $d \le 2$ cells become three computable constants $B(0..2)$. The second-order term is exact: tent moments ($c_2 = 1/12$) vs B-spline moments ($c_2 = 1/6$) give the correction $1 + 1/(6d^2)$, *window-independent*, reproducing the measured `v631` profile bit-for-bit and predicting the excess to 0.6% at every lag — a derived law, not a fit. The exact dictionary identities are additionally machine-proved in Lean (`WeilDictionary.lean`). *Erratum (2026-08-02, corrected the same day):* the boundary equation is stated in the chain's $\tilde{g}$-normalization (Lerch misread -1; correct $+1/4$, [`verification/v643_w1_theorem.py`] C0.1). On Suzuki's *true* g the same tents read $A_{\text{arch}} = -g''_{\text{smooth}}$ exactly with *no* constant — $\kappa = 0$ exactly, the $-\frac{5}{4}L\,\delta_0$ term was an artifact of the misreading; all numbers remain valid as $\tilde{g}$ identities. Machine-checked: [`verification/v640_w1_boundary.py`]. **[E]** (exact identities; certified boundary equation; derived moment law)

The W1 portability kill test (`PRIME.WEIL.PORTABLE.01`). The preregistered kill test on fresh windows (10 checks, ∼2 s). The `v631`/`v640` dictionary is *frozen* (all formulas closed in D, no knob; bars declared before the fresh windows were touched) and executed unchanged on $h = 285$, 540, 997 (the derivation window $h = 184$ excluded): the pole-subtracted conversion is the scalar $-4D$(window) with the derived discretization law ($|r - 1| < 6.3 \times 10^{-4}$, monotone), the atom constant is D^2(window) at machine precision (atom spread $\le 1.5 \times 10^{-14}$), and the closed per-lag law holds within the frozen bars. The kill criterion — any window needing a conversion beyond the explicit D dependence — is met *nowhere*: W1 is a window-portable operator identification, not a single-window fit. *Erratum (2026-08-02, corrected the same day):* the $-4D$ scalar is the $\tilde{g}$-normalization of the chain ([`verification/v643_w1_theorem.py`] C0.1); the portability transfers verbatim to the corrected one-scalar dictionary $+1/D$(window) on Suzuki's true g. Machine-checked: [`verification/v641_w1_portability.py`]. **[E]** (frozen-bar transport on three fresh windows)

The W1 matrix identity (`PRIME.WEIL.MATRIX.01`). From per-lag dictionary to the full quadratic form on $h = 184$, $M = 368$ (8 checks, ∼25 s). At the operator level the dictionary closes: the

16-mode parity block norm distance falls from 6.7×10^{-1} (frozen) to 2.7×10^{-3} (fully derived: boundary constants + moment law, every stage closed form) to 6.3×10^{-6} after re-binning the *same* Suzuki atom measure from B-spline to tent reads; the separated pole block is numerically rank 1 after parity projection (the preregistered rank-one freedom); the load-bearing 2×2 parity block (the $\widehat{A}$ surface of `v563`) agrees entrywise to 6.8×10^{-5}. The literal $< 1\%$ per-vector bar fails *only* on two typed lattice residuals — a near-cancellation denominator ($|Q_T| = 0.031$ at form scale ~ 2: metric, not operator) and the tent-vs-B-spline per-cell atom profile (removed completely by the measure-level re-binning) — certified in the module as typed-residual checks with inverted expectation, the probe's numbers unchanged. Nothing unexplained remains; the L_0^2 projection was, at this round, the last theorem-level W1 remainder. *Erratum (2026-08-02, corrected the same day):* the two-layer dictionary is the $\tilde{g}$-normalization ([`verification/v643_w1_theorem.py`] C0.1); on Suzuki's true g it collapses to the one scalar $+1/D$, all norm ratios transfer verbatim, and the L_0^2 remainder is closed by the v643 projection lemma. Machine-checked: [`verification/v642_w1_matrix.py`]. **[E]** (operator-level closure; typed lattice residuals)

The W1 measure-level theorem (`PRIME.WEIL.THEOREM.01`). The precise W1 statement on Suzuki's *true* screw function — and the machine evidence for the convention erratum of the v631/v640–v642 chain (11 checks, $\sim$65 s). *The convention lock (C0.1):* eq. (1.3) carries the Lerch coefficient $+1/4$, not -1 — locked by the paper's own §2.2 data (the printed origin constant $A = \frac{1}{2}(\log 2\pi - \psi(2)) = 0.70754637$ reproduces to 1.2×10^{-9} with $+1/4$; the -1 reading is log-divergent with $g(0) \neq 0$); every measured number of the earlier chain transfers verbatim via the exact identity $c^{\mathrm{sm}}_{\mathrm{gal}}(\tilde{g}) = -4\, c^{\mathrm{sm}}_{\mathrm{gal}}(g)$ (rel 3.7×10^{-12}). *The projection lemma:* Suzuki's L_0^2 mean-zero condition is *automatic* on the u-side ($D : H_0^1 \to L_0^2$ is a bijection; $\int Dv = 0$ for every H_0^1 test function, sympy exact), so P_a imposes no condition on Galerkin coefficients and generates no dictionary term; the `v563` parity sector *is* the odd subspace of $H_0^1(-\tilde{a}, \tilde{a})$ (worst 2.2×10^{-16}) — a subspace of the form domain; the Krein-kernel correction is annihilated on the derived objects (3.9×10^{-20}; must-break 2.84 on $u = 1$): the last theorem-level W1 remainder closes. *The theorem:* $g'' = -2\cosh(t/2) + W$ with smooth density $+\rho$ (sympy exact); $A_{\mathrm{arch}} = -g''_{\mathrm{smooth}}$ *exactly* at every lag (worst rel 3.4×10^{-52}, no constant, no scalar) and the origin constant *vanishes*, $\kappa = 0$ exactly (residual of the closed identity 6.1×10^{-55}): the `v563` near-cell scheme *is* Suzuki's origin bookkeeping; the B-spline reads close with exact boundary constants $\tilde{B}(0..2)$, and the moment law is derived to D^8 ($m_8 = D^9/45$, $31D^{10}/45$, sympy exact). *The bar:* full form equality on the common odd sector with the *one-scalar* dictionary $+1/D$: max $|Q_T - Q_S|/|Q_T| = 1.28 \times 10^{-10} < 10^{-4}$ on 10 odd 16-mode vectors (median 3.0×10^{-11}); pole block rank 1. No positivity claim, no RH statement; W2/W3 untouched. Machine-checked: [`verification/v643_w1_theorem.py`]. **[E]** (projection lemma, measure-level identities, erratum evidence) + (MEASURED) (form equality)

The W2 start: form density in slices (`PRIME.WEIL.W2.01`). The next rung of the contract, honestly sliced (7 checks, $\sim$240 s). *Classical slice:* the H_0^1-seminorm projection onto the hat space *is* nodal interpolation (1-D exact fact), and FEM density holds at the classical rate on smooth odd test functions (H^1 rates 1.000, L^2 rates 2.000 over $M = 92/184/368/736$); with Q_W H^1-bounded (Suzuki §8.4) and H_0^1 a form core (Thm 1.1) this is the classical density chain — cited, not re-proved. *Measured slices:* Rayleigh–Ritz on the *nested* window spaces at fixed $a = \log 16$ — the lowest three eigenvalues decrease monotonically from above under dyadic refinement (full and odd sector, all six sequences), the Cauchy gaps shrink at every step, and the parity identity $\lambda_{\mathrm{full}} = \min(\lambda_{\mathrm{even}}, \lambda_{\mathrm{odd}})$ holds per refinement (10^{-15}); the λ_a scan across five frame-A windows ($a = 2.77..6.01$) gives values $+10^{-12}.. + 2 \times 10^{-10}$ with discretization gaps $\sim 10^{-9}$ — consistent with Suzuki Thm 1.3 (the paper claims *only* continuity; the honest reading is $\lambda_a = 0^+$ within $\sim 10^{-9}$, *not* a sign statement), ground state *even* at every a (reported; Thm 1.4 covers only small a). *Typed remainder:* Mosco/Γ-convergence of the lattice forms (the liminf needs the compact embedding $H_{\log} \to L^2$, Prop. 4.1, transferred uniformly), norm-resolvent convergence, rigorous lag-route enclosures; the $a \to \infty$ limit is W3/W4 territory. W2 is *started*, not closed; no positivity claim, no RH statement. Machine-checked: [`verification/v644_w2_form_density.py`]. **[C]** (classical chain, cited) + (MEASURED) (Rayleigh–Ritz and λ_a slices) + **[C]** (typed Mosco remainder)

The parafermionic Klein twist (`QGEO.KLEINRP.01`). The named RP open item of `v639` is *resolved* at the lattice OS level (17 checks, ∼30 s). The positivity-carrying pairing composes the mirror with *charge conjugation*: $\Theta_K(\sigma_k(x)) = \eta_k\,\sigma_{3-k}(r(x))$ with ω^{kq} zero-mode phases. The dressed-string reality identity (particle-hole $I - C = DCD$ at half filling) makes every Klein Gram entry exactly real (1.4×10^{-15}); every charged Klein block is real symmetric *and* PSD (worst min/max 2.9×10^{-12}), for $k = 1, 2$, on both diameter axes, at $N = 48/96/192/384$ (N-stable, no drift); over the 36 pairs $(\eta_1, \eta_2) \in \{\pm 1, \pm\omega, \pm\omega^2\}^2$ *exactly one* satisfies consistency, Hermiticity and PSD: $\eta = (1, 1)$; no phase rescues the naive pairing (the fix is structural), which reproduces its N-stable -4.5×10^{-3} violation as the must-fail control; and the *full mixed* Gram over $\{1, \sigma_1, \sigma_2, \text{dipoles}\}$ with no sector projection is PSD at $N = 48/96$ (stable at 192) — the lattice Osterwalder–Schrader statement of the orbifold twist sector. Honest typing: continuum OS reconstruction, R/fermion-parity sectors, modular invariance and non-abelian content stay named, not claimed; `GATE.QGEO` does not move. Machine-checked: [`verification/v645_klein_rp.py`]. **[E]** (reality/uniqueness/must-fail) + (MEASURED) (PSD Grams) + [C] (typed continuum remainder)

The RM$(1,3)$ reverse reading (`E8.RM13REV.01`). Are compiler numbers *code operations*? An honestly two-part answer (20 checks, ∼1 s, exact arithmetic). *The anchor identity:* the σ-fixed syndromes of the equivariant placement C^* are exactly $\{0, s(e_6), s(e_7), q\}$ with coset-leader weights $(0, 1, 1, 2)$ and norm-square sum $6 = (1, 1, 2) \cdot (1, 1, 2)$; the anchor is the family sum in the quotient ($P_3 = P_0 + P_1 + P_2$); all four pair sectors carry $(1, 1, 2)$/norm 6, but *only* the anchor pair is σ-stable — the σ action *selects* the anchor copy. *The honest split:* the $(1, 1, 2)$/norm-6 weights are *generic* for any $[8, 4, 4]$ placement (the kill control keeps them); equivariance-specific are only q-universality, the σ-fixing of the anchor sector, and anchor = family sum (all die for the naive placement). *The bytecode negative:* the 42-entry invariant battery lands the structural anchors exactly (240, 8, 248, 60, 12, 6, 4) — but the p-sequence $(4, 6, 10, 18)$ of $p_n = 2 + 2^n$ has *no* code reading (0/11 natural families; 27 and 81 no hit at all): the bingo is buried. The kill control fires selectively, including the honest correction that the σ-fixed-root *count* 12 is not equivariance-specific (only its mod-2 fibration is). No marker moves. Machine-checked: [`verification/v646_rm13_reverse.py`]. **[E]** (exact anchor identity; honest negatives; selective kill)

The ST31 degree-24/20 round (`E8.ST31DEG.01`). The regular clocks of G_{31} against the compiler clock (25 checks, ∼4 s, exact arithmetic). *The regular 24-clock:* an explicit w with free census $\{24{:}10\}$ and $|C_{G_{31}}(w)| = 24$ (Springer); exact center relations $w^{12} = -1$ and $w^6 = J^3$ — the μ_4 center is a *power* of the regular 24-clock, and $\langle w\rangle/\mu_4 = Z_6$ is exactly its free 6-clock on the 60 lines ($\{6{:}10\}$); the characteristic polynomial is $\chi_w = x^4 + ix^2 - 1$ *exactly* (a Gaussian-integral quartic factor of Φ_{24}; all four eigenvalues primitive ζ_{24}); the half-clock w^2 is the *regular* 12-element, not the compiler clock. *The clock kill (a parity theorem):* $c = w^2$ is impossible for *any* order-24 w, even in S_{240} — the 12-cycles of a square come in pairs from 24-cycles (4-cycles in pairs from 8-cycles), and census$(c) = 19{\times}12 + 3{\times}4$ has *odd* counts; the full power-census scan confirms it. *The 20-degree:* a regular u with census $\{20{:}12\}$, $u^5 = J^3$ *central* (stronger than J-class), and u^4 acting *freely* with census $\{5{:}48\}$, $48 = 240/5$ exactly (lines $\{5{:}12\}$). Honest typing: 48 = seam sites stays an *observation* (no structural bridge here); $g_{\text{car}} = 5$ has only the census side; no marker moves. Machine-checked: [`verification/v647_st31_degree24.py`]. **[E]** (structural clock relations; parity-theorem kill) + [C] (typed observations)

The sign-uncertainty toolbox at $d = 1$ (`PRIME.SIGNUNC.01`). The Bourgain–Clozel–Kahane / Cohn–Elkies sign-uncertainty toolbox read against the W3 window cone (12 checks, ∼16 s; AST zero-firewall: no Riemann zero is read). *The dictionary is real:* the $d = 1$ Mellin strip *is* the critical strip — the strip multiplier $m_{1/2}(t) = \pi^{it}\,\Gamma(\frac{1}{4} - \frac{it}{2})/\Gamma(\frac{1}{4} + \frac{it}{2})$ is unimodular with phase slope at $t = 0$ *exactly* $L = \log\pi - \psi(\frac{1}{4})$ — the same archimedean constant the W1 dictionary froze as the δ_0 weight (25 digits, dev 0); the certificate constant reproduces ($J \to \log(\pi/2)$ within 5×10^{-9}) and the $d = 1$ Mellin–Hankel functional equation locks the conventions (10^{-22}). *The typed negative:* the quantitative mechanism — exponential interior-mass smallness $e^{-\gamma d}$ — is powered by the strip *half-width* and dies at $d = 1$: the paper's own budget gives $e^{-E^*} = 0.9665$ at $c = 0.9/\pi$ where $< \frac{1}{2}$

is needed (the machinery would need $d \geq 21$), and the window family's negative structure sits 5.2× outside the maximal BCK ball $1/\pi$ — a BCK-type mass obstruction cannot certify $\widehat{A}_{a,h} \succeq 0$; any W3 route through this toolbox must replace the dimension lever by a different large parameter (the window length acts as the *ball radius*, not the strip width). *Side findings (measured):* $\lambda_{\min} > 0$ on all 67 complete-comb windows (min $+8.26 \times 10^{-4}$, float-floor certified); the three truncated windows are precisely the `v618` set $\{1219, 1292, 1445\}$; and the positivity margin correlates with the $C = 1$ contraction margin ($\mathrm{corr}(\log \lambda_{\min}, \log \varepsilon h) = +0.704$). W3 stays open; no marker move, no RH statement. Machine-checked: [`verification/v648_sign_uncertainty.py`]. [E] (dictionary anchors, 25 digits) + (MEASURED) (typed negative; surface positivity) + [C] (typed verdict)

The discipline audit — the suite measures itself (`META.DISCIPLINE.01`). The suspicion to be defused is "numerology": a large corpus of machine-checked claims could in principle be a confirmation-bias generator. This meta-module parses the suite's own files and freezes the measured process-discipline metrics as minimum bars (18 checks, ∼200 s). *Ledger census:* 715 rows, 710 active; 214 rows document kills, honest negatives, ill-posed retirements and retypes — a project that documents its own buried hypotheses is not a confirmation-bias generator. *Must-fail census:* 643 modules (every one carries checks), 6040 static `check()` call sites, 1469 must-fail / kill / scramble keyword occurrences across 233 modules; typing marks [E] × 3579 vs [C] × 1323 — conditional readings are typed, not upgraded. *Preregistration:* 47 contract rows, 29 naming kill criteria *before* execution; executed chains verified in the ledger (`FTR.PGL2.01`: preregistered `v533` → executed `v632`; `PRIME.WEIL.OPERATOR.01`: kill tests K1–K3 → W1 theorem-closed `v643`; `REG.FREEZE.01`: registry frozen 2026-06-09). *Reproducibility:* deterministic replay 12/12 on a declared sample spanning `v55`–`v648` (two fresh subprocess runs each, identical output up to timing lines); 396 modules sympy-exact, 77 with explicit seeds. *External anchors:* five classical results recomputed here against the literature — Jacobi 1834 four-square counts (exact, $n = 1..40$), Construction A $[8, 4, 4] \to E_8$ with 240 minimal vectors, the Shephard–Todd G31 degrees $(8, 12, 20, 24)$, the Suzuki archimedean constants to 25+ digits, and the completed-zeta functional equation (3×10^{-31}). *Lean layer:* 54 committed proof modules (`git ls-files`; `lake build` honestly *not* re-run in-suite). *The limit, honest:* this module measures *process* discipline, not the truth of the physics; typed [C] observations stay observations, and the open gates stay open (278 class-O rows incl. `SEAM.THEOREM.01` and `PRIME.WEIL.OPERATOR.01` W2–W4) — a discipline certificate is a necessary, not a sufficient, condition against numerology. Machine-checked: [`verification/v649_discipline_audit.py`]. [E] (censuses, replay, anchor recomputes) + [C] (the typed limit)

The orbifold modular data (`QGEO.ORBMOD.01`). The modular level of the $\mathbb{Z}_3$ orbifold of the seam, on the lattice (17 checks, ∼25 s). All nine twisted-sector partition functions $Z[g, h]$ of the seam double are real positive with exact conjugation degeneracies and match the continuum $|\theta[g/3, h/3]/\eta|^2$ at $< 1\%$ on $\ln Z$ ($N = 48$; three τ); the Casimir slopes reproduce the `v628` coefficients per deck class. *S is measured:* the modular S-covariance converges with a clean common N^{-4} rate (4.06–4.32 over $N = 24/48/96$) while the single-lattice deviation from the continuum falls only at N^{-2} — the leading lattice artifact is itself S-covariant and *cancels* in the S-relation. *T is exact:* the discrete Dehn twist realizes T as the $\mathbb{Z}_6$ sector map $(g, h) \to (g, g+h) \times (-1)^F$ at machine precision (3.6×10^{-13}; continuum T-phases are not resolvable in the real-positive trace convention — honest scope). The twisted characters carry $h_\sigma = 1/36$ (rel 5.6×10^{-4}) with the parafermionic 1/6 gap, and $\mathrm{deck}^3 = (-1)^F$ closes exactly as $\mathbb{Z}_6$ trace arithmetic including the cover factorization; chiral phases, continuum T-phases and continuum OS stay typed open; `GATE.QGEO` does not move. Machine-checked: [`verification/v650_orbifold_modular.py`]. [E] (sectors, characters, spin structure) + (MEASURED) (S-covariance) + [C] (typed continuum remainder)

The orbifold assembly (`QGEO.ORBASM.01`). The projection sum, settled by machine cohomology and a candidate census (17 checks, ∼35 s). $H^2(\mathbb{Z}_3, U(1)) = 0$ by enumeration (27 cocycles, all symmetric): there is *no* discrete torsion for a pure $\mathbb{Z}_3$ orbifold. Of six assembly candidates, the survivors of {integer spectrum, unique vacuum, S measured clean, T exact} are exactly $\{B, C3\}$:

the naive $\mathbb{Z}_3$ dies by an *exact* T violation 3.5×10^{-3} (its charge-3 content — T kills the naive $\mathbb{Z}_3$ exactly), $C1$ by S stalling, $C2$ has no vacuum at all, A' additionally by the wrong first twisted exponent (5/18, not 1/9). The seam data then discriminate: B keeps every measured $c = 0$ sector ground and matches the Klein phase $\eta = (1, 1)$ of `v645`, while $C3$ (the Arf/GSO twin) excises all odd-sector grounds — *the seam assembly is* B, the $\mathbb{Z}_6$ deck quotient with uniform charge-0 projection (data-backed at this round; hardened structurally by `v652`). The chiral phase $e^{-2\pi i\bar{a}b}$ is measured. `GATE.QGEO` does not move. Machine-checked: [`verification/v651_orbifold_assembly.py`]. [E] (cohomology, census, exact T) + [C] (data-backed B-vs-C3 pick)

The Arf structure of the assembly (`QGEO.ORBARF.01`). The B-vs-C3 kill, decided structurally (15 checks, ∼30 s). The μ_6 defect ladder measures the C3-excised states *directly*: ladder exponents $2\Delta = (1/18, 2/9, 1/2)$ at rel devs $< 8 \times 10^{-4}$, the half-twist Klein Gram exactly PSD, and the exact charge-resolved minima $(0, 1, 4, 9, 4, 1)/36$ are the measured sector grounds; $C3$'s odd-sector predictions 85/18 and 81/18 exceed the measurements by factors 85.0 and 9.0 (the defect string is exactly charge-neutral: $C3$ excises states that exist on the lattice seam with positive Klein norm). The classification pin: among *all* $6^6 = 46656$ sector-character wirings exactly $\{B, C3\}$ survive the exact lattice degeneracies, and the single-Fock-space requirement forces B. Honest: the earlier `v639` 2/9 read alone is B-vs-C3-*blind* (both coincide exactly on the even sectors); the new τ datum is the OPE exponent 1/18. `GATE.QGEO` does not move. Machine-checked: [`verification/v652_orbifold_arf.py`]. [E] (classification, pin, excision data) + [C] (typed premise remainder)

The bond-defect theorem (`QGEO.BONDDEF.01`). The last orbifold premise upgraded to a lattice theorem (18 checks, ∼40 s). *The theorem:* the deck-character blocks of the covered seam *are* the one-bond-defect chains, unitarily and exactly ($A_r = \sqrt{3}\, RP_r$ on all modes, exact in $\mathbb{Q}(\zeta_6)$, at 16/48 and 48/144; defect phase = deck character): twist sector = deck boundary condition = bond defect, canonically; $D_{\mathrm{can}} = L^4$ and $D^3_{\mathrm{can}} = -1$ hold as exact operator identities (the quarter clock fixes the canonical section lift). *Gauge and pin:* the defect position is pure gauge (a symbolic telescoping theorem in all 16 free bond phases), and the geometry pins the gauge class — one uniform twist unit per mark (cocycle support = the `v622` mark set), window boundary at a crossing ⇔ defect at a mark. *The exclusion census:* the base chain has no other $\mathbb{Z}_3$; site potentials, amplitude defects, sign defects and alternating weights are killed exactly; an isospectral real NN chain *exists* (7.8×10^{-16}) — honestly documented: purely spectral data cannot exclude it, but it is no $\mathbb{Z}_3$-twist realization and the measured string data discriminate it. The continuum CFT statement and the seam identification stay typed; `GATE.QGEO` does not move. Machine-checked: [`verification/v653_bond_defect.py`]. [E] (lattice theorem, gauge pin, exclusion census) + [C] (typed continuum remainder)

The ST31 degree-8 round and the $d/4$ theorem (`E8.ST31D8.01`). The last clock census, unified (30 checks, ∼10 s, exact arithmetic). *The $d/4$ theorem:* for every degree $d \in \{8, 12, 20, 24\}$ the Springer-regular d-clocks of G_{31} satisfy $x^{d/4} = \pm i\,\mathrm{Id}$ — a *generator* of μ_4, never ± 1 (one formula from the exponent systematics: all exponents of G_{31} are 3 mod 4). The converse is exact at $d \in \{8, 20, 24\}$, and the unique exception at $d = 12$ is the *compiler-clock class*: exactly 1280 extra elements = the classes of c and its Galois twin, with $\chi_c = (x - i)^2(x^2 + ix - 1)$ half the regular-12 Springer block — all eigenvalues satisfy $\lambda^3 = -i$, which is exactly why $c^3 = J^3$ is central; the compiler-clock classes are the *only* non-regular clocks in all of G_{31} whose $d/4$-th power generates the center. The order-8 landscape is complete (10560 elements, 480 regular = 2 Galois classes, $\chi = (x^2 + i)^2$ exact) with the honest kill that "free ⇒ regular" is *false* at $d = 8$ (and already at order 4); degrees = free orders divisible by 4 above 4. No marker moves. Machine-checked: [`verification/v654_st31_degree8.py`]. [E] (census, theorem, controls; honest kill)

The W2 Mosco preparation (`PRIME.WEIL.MOSCO.01`). Measurable surrogates for the typed `v644` remainder (7 checks, ∼80 s). The instability finding, typed: $\lambda_a(\log 16) = 0^+$ within the discretization error and the ground-mode L^2 distances do *not* converge — eigen-*direction* observables are not float-certifiable at this a; any W2 route must run through resolvent/form observables (which is what norm-resolvent convergence asserts). The preparation: the discrete resolvents $(A_M + 1)^{-1}$

are Cauchy along the refinement (rates ∼1.0 on all five test vectors) and the $H_{\log}$ norms of the normalized resolvent family stay *uniformly bounded* — the two measurable prerequisites of the Mosco theorem; recovery sequences converge and the liminf direction survives the wiggle test. The named remainder is the discrete Gårding inequality (below); W2 stays open, no positivity claim, no RH statement. Machine-checked: [`verification/v655_w2_mosco.py`]. (MEASURED) (resolvent Cauchy, uniform $H_{\log}$) + [C] (typed eigen-scale collapse and remainder)

The margin bridge, typed (`PRIME.MARGINLINK.01`). Does the positivity margin $\lambda_{\min}$ carry information about the $C = 1$ load εh beyond the depth trend? (11 checks, ∼10 s.) The preregistered gate *passes* (rank-cubic h-control $\rho_{\text{partial}} = +0.477$, $p = 10^{-4}$; (h, α) robustness +0.701) — but the mechanism is an *identification*, not an independent link: $q_r = w_{\text{lock}}^{\mathsf{T}} \widehat{A}\, w_{\text{lock}}$ exactly (1.5×10^{-7}), so $\lambda_{\min} \approx r_{\text{id}} \cdot q_r / \|w_{\text{lock}}\|_G^2$ and the window form reads its *own* $C = 1$ defect along the lock direction (minimal mode lock-enriched, median angle 9.1ř). The killer test is typed accordingly; no positivity claim, no RH statement. Machine-checked: [`verification/v656_margin_link.py`]. [E] (surface reproduction) + (MEASURED) (gate, mechanism) + [C] (typed identification)

The r_{id} geometry (`PRIME.RIDGEOM.01`). What drives the alignment factor (15 checks, ∼10 s). The closed formula: in the deflated 2D frame $r_{\text{id}} = (1 - q \tan^2\theta)/(1 - \tan^2\theta)$ *exactly* (max dev 3.4×10^{-10} on all 67 windows), with structural signs on every window ($\theta < 45$ř, $B < 0$, $q > 1$). The deficit is *angle-driven* (share ∼0.79), not arithmetic: after h-control no arithmetic carrier passes the Bonferroni-honest gate. The last atom sits at the window edge identically (the frame-A edge $e^{2\alpha} = N^2$ is itself a prime power). r_{id} is *informative* (cubic-in-log h $R^2 = 0.618 < 0.80$; the depth trend honestly documented). On this surface W3 reduces to *lock sign plus the uniform bound* $q \tan^2\theta \le 1 - \delta$. Machine-checked: [`verification/v657_rid_alignment.py`]. [E] (closed 2D formula) + (MEASURED) (drivers, honest controls) + [C] (typed W3 reduction)

The W3 uniform bound, surveyed (`PRIME.W3BOUND.01`). Risk survey of $P := q \tan^2\theta \le 1 - \delta$ (11 checks, ∼20 s). The slope identity $\beta_P = \beta_q + \beta_t$ is exact; the h-trend of P is carried by q (shares +0.64 vs +0.36); the crossing-risk models put the point crossing at $h^* \sim 3.4 \times 10^3$ (POW) / 9.1×10^3 (LOG) / 1.8×10^4 (SAT) — the family margin $\delta_{\text{family}} = 0.036$ (max $P = 0.9636$ at $h = 859$) is *not* safe to extrapolate; and the construction is *α-fragile*: $\max |dP|/P = 0.59$ over $\pm 2.5\%/\pm 5\%$ shifts (bar 0.15) — P sits on a thin resonance of the frame-A coupling. Verdict W3UB-FRAGILE, honestly typed; no RH statement. Machine-checked: [`verification/v658_w3_uniform_bound.py`]. [E] (slope identity) + (MEASURED) (risk numbers, fragility) + [C] (typed verdict)

The W3 resonance landscape (`PRIME.W3LAND.01`). Mapping $P(\alpha)$ around the frame-A points (14 checks, ∼240 s). The landscape is *peaked*: 52 needles across 8 slices (fine FWHM median 2.5×10^{-4} in α); frame-A points are *not* preferentially on peaks (quantile rank 0.24 vs null 0.5); $P > 0.95$ lives on ∼5% of the surveyed space and on 0.0% of the MARGIN regime. The positive surprise: $\lambda_{\min} > 0$ on *all* 635 landscape grid points — positivity is never lost anywhere, including every needle; every $P > 1$ excursion is a $\theta \to 90$ř *rotation artifact* of the 2D reduction. Honest negative: no arithmetic resonance-condition candidate passes the frozen gate — the peaks are real but arithmetically unexplained. The W3 alarm signal is only $\lambda_{\min} < -$floor, never observed (calibrated by `v668` below). Machine-checked: [`verification/v659_w3_landscape.py`]. (MEASURED) (landscape map; honest negative) + [C] (typed reading)

The rotation predicate, honest (`PRIME.THETAPRED.01`). The preregistered predicate test, failed honestly (20 checks, ∼340 s; verdict THETA-PRED-FAIL). Mechanism: 47/52 needles are *diagonal crossings* of the deflated 2×2 (exactness 1.6×10^{-10} on 702 pooled points; interlacing exact); the two-level leak is ∼1.0 — there is no single 2-level partner. The predicate $f = \lambda_{\text{lock}} - \mu_1 \ge 0$ is *fully necessary* (pointwise recall 1.0) but not sharp (precision 0.18 pointwise, 0.255 at the event level: 65 crossings vs 212 f-zeros). $h = 859$ sits ≥ 100 needle widths from 45ř at $s = 0$; 63/67 windows carry a confirmed needle within ±2.5%. The named successor is *block deflation*; no marker move, no RH statement. Machine-checked: [`verification/v660_theta_predicate.py`]. [E] (2×2 machinery) + (MEASURED) (mechanism; honest predicate FAIL) + [C] (typed successor)

The discrete Gårding constant (`PRIME.GARDING.01`). The named missing W2 building block, measured (12 checks, ∼320 s). The constant ladder at $a_0 = \log 16$: $c > 0$ at every stage ($c(736, 1) = 0.180$, $c(736, 4) = 0.676$) and *a-uniform at fixed M-ratio* (the actual W2 requirement, five windows $h = 184..1433$) — but whether $c(M)$ stabilizes or drifts like $1/\log M$ is *undecidable* on four dyadic stages (the honest typed state, GARDING-DRIFT-UNRESOLVED). The diagnosis: the tight direction reads the symbol minimum near the lattice edge (edge-band read, symbol tightness 0.81..0.97) — the question moves to the symbol level (next two blocks). Machine-checked: [`verification/v661_garding.py`]. (MEASURED) (constant ladder, a-uniformity, diagnosis) + [C] (typed undecidability)

The two Gårding remedies, refuted (`PRIME.GARDEDGE.01`). Both named remedies fail, and the failure is the finding (16 checks, ∼160 s). The folded $H_{\log}$ norm does *not* heal the drift ($c_{\text{fold}}(736, 1) = 0.18151 \approx$ unfolded), and the edge cutoff is *lossless* ($C_{\text{edge}} \leq$ the C budget: the drift is not edge-band mass) — the drift lives in the *total symbol* ($\min_{t\geq 20} s_{\text{tot}}$ near-flat across the ladder while $\log(2 + t)$ grows). The probe's declared FAIL at the literal symbol-stability bar is promoted as a typed-residual check with inverted expectation (the `v642` pattern, numbers unchanged): the per-M symbol slopes stay positive but their M-spread 0.260 *must* exceed the 0.10 bar — the per-mode symbol read is not M-stable. What a proof needs is the lower-envelope growth of the total symbol (next block). Machine-checked: [`verification/v662_garding_edgeband.py`]. (MEASURED) (remedy refutations; typed residual) + [C] (mechanism reading)

The Gårding envelope (`PRIME.GARDENV.01`). The total symbol *does* grow — sub-log, measured a-uniformly (18 checks, ∼200 s). "Flat" was a t-range artifact (the previous read stopped at the first dip): the band-minima envelope grows with a log model ($c_0 > 0$ on all five windows), and the strongest monotone weight certified a-uniformly by the family gives $s_{\text{tot}} \geq c_0\, w^*(t) - C_0$ with $(c_0, C_0) \approx (0.021, 0.055)$. The B4(iii) breaking point is quantified: the derived partial-summation bound is valid but exponential in a (e^a/a), while the *measured* atom envelope grows only ∼$3.55\, a$ linearly ($A(a) = 6.2..18.5$ over $a = 2.77..6.24$); Chebyshev validity $\psi(n)/n \leq B_\Psi$ holds on the full atom table. The remaining W2 step is *named*: a Fejér spectral-density bound replacing the per-mode symbol read; Suzuki Prop. 4.1 does not block the route (full-text read). Measured, not proved; no marker move, no RH statement. Machine-checked: [`verification/v663_garding_envelope.py`]. **[E]** (derived bound validity) + (MEASURED) (a-uniform envelope; quantified breaking point) + [C] (named Fejér remainder)

The look-elsewhere audit (`META.LOOKELSEWHERE.01`). The bingo budget of the typed [C] numeric coincidences, quantified (14 checks, ∼2 s; seed 20260802, 100k MC, exact Poisson-binomial tails). The budget: $N = 42$ structure slots, $K = 19$ hits against the compiler target list *frozen before the computation*; the six killed readings enter as 0-hit trials. Global: the family-corrected $p = 5.1 \times 10^{-3}$ in the most conservative of three null models — the *ensemble* survives the honest correction (empirical control: 0.2%/0.3% of 2000 size-matched pseudo-target lists reach the real hit count / the real p). The triage, and the point: *no single observation is significant on its own* — suggestive only (0.01..0.1): $60 = D_{\text{start}}$, the 12 σ-fixed roots, the G31 degrees, $48 = \Omega_{\text{adm}}$; unremarkable ($p > 0.1$): 6 = Lorentz-congruence index $= p_2(a)$ ($p = 0.169$) and the CODESEM number battery ($p = 0.593$ — the `v646` finding quantified); forced: 14 and 16 (their value is the theorem, not the number). The re-typings are executed as dated ledger notes (`EXTREV.LATTICE.01`, `E8.CODESEM.01`, `E8.CODE.01`); the substance of the rounds lies in the **[E]** structure theorems, not in the number bingos. Machine-checked: [`verification/v664_look_elsewhere.py`]. **[E]** (budget census, MC quantification, executed re-types)

The Keiper–Li consistency (`PRIME.KEIPERLI.01`). The sequence λ_n on two independent routes (13 checks, ∼25 s). Route 1: the zero comb (2000 zeros, $\gamma_{\max} = 2515.3$, committed cache) with an honest tail budget — the prescribed raw criterion window is *empty* (documented); the corrected budget $B_{\text{fl}}(n) \sim 7.9 \times 10^{-7} n^2$ stays below 1% of the GUE trend for $n \leq 64$. Route 2: zero-free arithmetic — contour extraction with $\lambda_1 = 1 + \gamma/2 - \log(4\pi)/2$ *exact* (dev 0.0) and the binomial assembly equal to the direct extraction at 7.8×10^{-92}; the atom table reproduces the prime-block constant ($-\gamma_E$ at 2.9×10^{-4}). The central check: the two routes agree within the budget for *all* $n = 1..64$ (max usage 0.124), and $\lambda_n > 0$ on both routes — a *finite* verified statement, not Li's criterion. The must-fail has teeth: an injected off-line zero (Re = 0.6) explodes λ_n negatively (min -4.9×10^{43} at $n \sim 199921$) and is detected already at $n = 1$ inside the budget. No RH claim. Machine-checked: [verification/v665_keiper_li.py]. **[E]** (two-route consistency, closed forms, must-fail injection) + **[C]** (atom-table tie-in)

The Turing certificate (PRIME.TURINGCERT.01). The completeness of the zero comb, certified (6 checks, ∼2 s). All 1999 midpoint rungs sit inside the cited S-band (max $|S_{\text{est}}| = 0.5151 < 2.5$), the endpoint check makes the comb the complete initial segment up to $\gamma_{\max} = 2515.3$, and Turing's integral refinement holds at 5% of the Lehman bound — a persistent miscount is excluded, not just a local one. Must-break: a removed zero blows the Turing integral to 167× the bound, a doubled one to 174×. The certificate says exactly that the loaded list is the first 2000 zeros — it certifies the *data premise* of all comb-reading modules; no RH statement. Machine-checked: [verification/v666_turing_cert.py]. **[E]** (band + Turing certificate; must-break controls)

The Baez–Duarte control frame (PRIME.BAEZDUARTE.01). An *external* comparison Galerkin for the Suzuki route (12 checks, ∼110 s). The Vasyunin closed form for the Nyman–Beurling Gram is anchored at 1.7×10^{-31} against direct integration; the constant $C_{\text{BD}} = 2 + \gamma - \log 4\pi$ is computed *independently* from the zero comb plus the Riemann–von Mangoldt tail at 6.4×10^{-8} (the second comb anchor); and the theorem-level drift $d_N^2 \log N \to C_{\text{BD}}$ is reproduced at the percent level (max dev 0.042; the honest corridor recalibration is documented in the module). The frame comparison, and the point: the log-slow *drift* is *shared* with the Suzuki route — intrinsic to the positivity problem — while the resonance *needles* are not shared: frame-side. No positivity claim, no RH statement. Machine-checked: [verification/v667_baez_duarte.py]. **[E]** (Vasyunin/C_{BD} anchors) + (MEASURED) (ladder, corridor, frame comparison) + **[C]** (typed reading)

Ground truth and firewall (PRIME.GROUNDTRUTH.01). The positive *and* the negative control of the window machinery, one module from both probes ($22 + 13 = 35$ checks, ∼30 s). *Ground truth (Ihara):* on three *proven* Ramanujan graphs the Toeplitz window forms are PSD for every window size with exact rank saturation — true positivity has *no uniform margin*: $\delta(K) > 0$ only below the support-resolution depth and *identically zero* beyond ($\delta(K) \to 0$ exactly); the proven violators break at $K^* = 15/11$ with detection reach $K^* \times s \approx 2..3$, and Fejér/tent reads are *blind* to the violation. *Firewall (Epstein):* for $Q(x, y) = x^2 + 5y^2$ (disc -20, $h = 2$; Davenport–Heilbronn 1936) the genus identity is exact to 10^5, the von Mangoldt analogue Λ_E has 1351 negative sites and 4849 composite support points, and the *identical* `v563` tent machinery breaks by ∼13 orders of magnitude on the Epstein rung ($\lambda_{\min} < -$floor on 3/3 windows), with the exact Rayleigh attribution locating the break in the *composite* support — the load-bearing structure is the Euler product; 12 off-line zeros are found by argument-principle winding. *Together, the central W3 recalibration:* shrinking margins on deeper windows are the *expected* behavior of a true positivity; the alarm signal is only $\lambda_{\min} < -$floor (never observed on the ζ surface); the needles are frame-side (`v667`). No claim about ζ, no RH statement, no marker move. Machine-checked: [verification/v668_ground_truth.py]. **[E]** (exact ground-truth laboratory, both controls) + (MEASURED) (calibration profile) + **[C]** (typed W3 recalibration)

The Fejér density theorem (PRIME.FEJERDENS.01). The renamed W2 residual task, structured and closed at the density level (14 checks, ∼65 s; verdict FEJER-DENSITY-THEOREM). *The identity (exact):* the Weil explicit formula on the tent test pair $g_t(u) = (1/2a)(2a - |u|)_+ e^{-itu}$ gives $s_{\text{tot}}(\cdot; a) = 2\pi(F_a \star dN)$ — the total window symbol *is* (2π times) the Fejér-smoothed zero-counting density; verified pointwise to 2.9×10^{-6} and t-averaged to 7.2×10^{-8} against the Turing-certified comb; the envelope peaks *are* the zeros ($\gamma_{1..4}$ matched to 0.0002), the dips *are* the zero gaps (forensics residual 1.5×10^{-6}; Fejér power $p_{\text{meas}} = p_{\text{pred}} = 0.974$); hypothetical off-line zeros above 3×10^{12} are budgeted at 2.4×10^{-10}. *The unconditional chain (cited constants):* windowed RvM counting + Trudgian $|S|$ + Fejér box minorant + F_a mass transfer give $\rho_{a,\delta}(t) \geq c_0^{\text{thm}} \log(2+t) - C_0$ for $t \geq t_0(a, \delta)$ at $\delta \in \{4\pi, \pi a\}$: 10/10 (a, δ) pairs close with $t_0 \leq 6000$, min $c_0^{\text{thm}} = 0.3058$ (15.1×

the raw measured hull), and the finite remainder $[10, t_0]$ is machine-checked on the *zero-free* ρ grid (10/10 margins, min cert margin +1.36). *The honest floor:* the chain closes only for smoothing widths $\delta > 4\pi A_1 = 1.4074$; the literal $1/a$ and the plane-wave π/a sit *below* the floor for every frame-A window — per-mode symbol control (the Gårding $1/\log$ drift) is not certifiable this way at *any* height; the theorem controls wave packets of spectral width $\geq \delta$, and the remaining W2 piece is the packet-to-point translation below the floor. W2 stays open; no marker move, no RH statement. Machine-checked: [`verification/v669_fejer_density.py`]. **[E]** (exact identity; zero-free finite certificates) + (ANALYTIC, CITED) (unconditional chain) + [C] (typed floor)

The block deflation, honest (`PRIME.BLOCKDEFL.01`). The named successor of the rank-1 predicate, failed honestly (18 checks, ∼95 s; verdict BLOCK-DEFL-FAIL). The (1+K)-block Rayleigh–Ritz compression onto $\{\hat{w}_{\text{lock}}, K \text{ lowest deflated eigenvectors}\}$ reads no minimal mode (wiring exact on 702 pooled points; the $K = 1$ rung reproduces the rank-1 quotes). The fidelity ladder: $p_{90} = 50.53/49.49/49.44/49.35$ř for $K = 1/2/4/8$ — no K reaches the declared 5ř bar; the needle predicate at $K = 8$ fires on all 52 classified needles (precision 0.635 at recall 1.000, bar 0.80: miss); the frame-A rebuild is skipped by the frozen rule. The failure record *is* the finding: the leak median is 1.000 at every K — the pencil action on the lock direction is not captured by the 8 lowest deflated modes; the multi-mode coupling is *bulk-spectral*, not low-rank. No positivity claim, no RH statement; W3 stays open. Machine-checked: [`verification/v670_w3_block_deflation.py`]. **[E]** (block wiring, anchors) + (MEASURED) (honest double miss; leak ladder) + [C] (typed reading)

The Lehmer null (`PRIME.LEHMERNULL.01`). Are the W3 needles the window analogue of the Lehmer-pair phenomenology? A clean no (10 checks, ∼12 s; verdict LEHMER-NULL). Pair statistics on the certified comb: the top-10 Lehmer-like pair table (comb minimum $\delta_k = 0.089$ at $\gamma \approx 1977$; the classical Lehmer pair at $\gamma \approx 7005$ lies outside the comb, documented); the resolution correspondence is documented: lattice spacing $D \leftrightarrow$ Nyquist reach $t_{\max} = \pi/D \in [118, 868]$, lag support $2\alpha \leftrightarrow$ resolution $\pi/(2\alpha)$. The null: the raw ± 0.7 correlations of $P = q \tan^2 \theta$ with Lehmer presence are pure h-ladder; the h^3-partials collapse to $+0.195/-0.123/-0.125$ with all $p > 0.1$, and the pseudo-pair placement control ($B = 2000$) is unremarkable ($p = 0.086/0.299/0.231$) — the needles are *not* zero-comb-driven at the window resolution, consistent with the frame-side reading (`v667`). The teeth: a planted signal is detected at $\rho = +0.847$, $p = 5 \times 10^{-5}$. Zeros enter only as the correlation target; no RH statement. Machine-checked: [`verification/v671_lehmer_resonance.py`]. (MEASURED) (pair statistics; clean h-controlled null; placement control) + **[E]** (must-detect teeth) + [C] (typed reading)

The Li corollary, finite (`PRIME.LICOROLLARY.01`). The W1 ⇒ Li corollary spelled out (11 checks, ∼4 s; verdict LI-COROLLARY-FINITE). The generator is derived exactly: $g_n(u) = \frac{1}{2} e^{-|u|/2} L^{(1)}_{n-1}(|u|)$ (generalized Laguerre; sympy-exact), and $\hat{g}_n = \frac{1}{2}|\hat{G}_n|^2 = 2\sin^2(n\theta/2) \geq 0$ — the Li coefficient is a *quadratic-form value* of the Weil form at the generator G_n. The explicit formula is locked on the $h = 1433$ window (worst rel. dev 5.5×10^{-3}, the typed $(\gamma D)^2/12$ discretization scale), with the pole-cancellation exhibit: for a low-mode odd vector the window form cancels the pole term to six digits (4.0×10^{-7}) — the rank-2 pole block (`v591`) is exactly the piece the certificate must exclude. The central table: the L^2 projection plus pole-free correction reproduces λ_n with form error $< 10\%$ on the band $2..4 \cup 6..32$ (30 of 32; $n = 1$ honestly misses at 0.24 — a jump artifact at pitch D; $n = 5$ hair-thin at 0.1003), and every band read sits above the certified floor: the `v648`-type datum $\lambda_{\min} = +1.516 \times 10^{-3} > 0$ *contains* these 30 finite Li coefficients (odd sector share 0.753..0.998; the new even-sector measurement is documented: unconstrained indefinite, pole-free $+7.67 \times 10^{-4}$). A finite statement — not Li's criterion, no RH statement. Machine-checked: [`verification/v672_li_corollary.py`]. **[E]** (Laguerre derivation; window lock; pole cancellation) + (MEASURED) (30/32 band table) + [C] (finite typing)

The E_4 Li lock (`PRIME.LIE4.01`). Li additivity for $L(E_4, s) = \zeta(s)\zeta(s-3)$ as the exact $E_8 \leftrightarrow$ comb consistency test (12 checks, ∼11 s; verdict LI-E4-ADDITIVE-CONSISTENT). The completed form $\Lambda(E_4, s) = (2\pi)^{-s}\Gamma(s)\zeta(s)\zeta(s-3)$ satisfies the functional equation $s \leftrightarrow 4 - s$ exactly (2.4×10^{-41}), with residues $\pm 1/240$ at $s = 4, 0$ *exactly* (sympy) — the E_8 shell normalizer 240 *is* the residue.

The prime shadow is exact: $\Lambda_L(n) = \Lambda(n)(1+n^3)$ from the 240 σ_3 recursion (30 digits to $n = 512$; Dirichlet lock at $s = 6$ to 1.8×10^{-11}). The shift rule $\lambda_n^L = 2\lambda_n^\zeta$ holds exactly (Lagarias additivity; unimodularity at 9.9×10^{-32}), and the cross-check closes: $|\lambda_n^{L,\mathrm{arith}} - \lambda_n^{L,\mathrm{comb}}| \le 2B_{\mathrm{fl}}(n)$ for all $n \le 32$ (max budget usage 0.124) — the E_8 counting function and the ζ comb agree in *one* Li sequence. The teeth: the naive single-map Li at $s = 2$ explodes negatively (first negative at $n = 269$) — $L(E_4)$ is non-tempered in that normalization; the additive per-factor definition is the correct reading. A finite statement, no RH claim. Machine-checked: [`verification/v673_li_e4.py`]. **[E]** (completed form, residues, shift rule, cross-check, teeth)

The packet Gårding inequality (`PRIME.PACKETGARD.01`). The packet-to-point gap of the W2 route, built and measured where the Fejér density theorem lives (21 checks, ∼305 s; verdict PACKET-AVERAGE-ONLY — the honest split; the probe's declared FAIL at the literal stabilization bar is promoted as a typed-residual check with inverted expectation, v642/v662 pattern, numbers unchanged). *The construction:* the packet norm $\|v\|_X^2 = \sum_p w_p \|P_p v\|^2$ (continuum frequency bands of width $\delta^* = \pi a$ above the RvM floor $4\pi A_1 = 1.4074$, DST packets, edge-log weights) is comparable to $H_{\log}$ from above with a bounded measured defect ($\kappa \le 1.170$, dyadic increments strictly decreasing; the boundary-spike maximizer diagnosed at $|x|/a = 0.997$) and embeds compactly *by construction* (diagonal pencil, diverging M-independent weights, exact to 6.0×10^{-13}). *The typed residual (central):* the packet PROJECTION ladder $c_X(M, C)$ *must* fail stabilization at every C and keep the pure-1/ log model competitive — c_X tracks the pointwise $c_{H_{\log}}$ to 1.011..1.065 and the minimizer is single-mode tight (c_X/single-mode floor = 0.9007): packetization by projections re-labels the pointwise obstruction, it does not remove it (a-uniformity at fixed ratio holds, spreads 0.28/0.26 over $h = 184..1433$). *The positive core:* the block averages track the Fejér density (median $|avg/\rho - 1|$ worst 0.067), and the FRAME inequality $Q + C_0^{\mathrm{thm}} G \succeq c_0^{\mathrm{thm}} Y$ ($Y = \sum_p w_p \phi_p \phi_p^\top$, tent packet frame) *holds at every anchor M* with the v669 theorem constants rebuilt zero-free ($c_0^{\mathrm{thm}} = 0.3058$, $C_0^{\mathrm{thm}} = 2.4984$, $t_0 = 3530$; $\lambda_{\min} = +1.787/+1.703/+1.570/+1.518$, certificate margin +2.213) — the theorem-shaped packet Gårding statement. The Mosco mechanism is numerically *complete*: 8/8 sequences (resolvent carriers X-uniform + Cauchy, recovery liminf, weak-null battery 3/3) satisfy the numeric liminf. The named W2 remainder is *within-packet equidistribution* (dip depth $\theta_1 \to 0.20$) = unconditional zero-gap information; not a proof of A5(a); W2 stays open, no marker move, no RH statement. Machine-checked: [`verification/v674_packet_garding.py`]. **[E]** (wiring, construction) + (MEASURED) (typed residual with inverted expectation; frame inequality; Mosco battery) + **[C]** (named remainder)

The needle mechanism, saturated (`PRIME.NEEDLEMECH.01`). After the triple negative (v670 not low-rank, v671 not zero-comb-driven, v667 not in the BD frame), the four frozen assembly-side candidates all MISS their gates: the needles are an *emergent bulk property* (21 checks, ∼100 s; verdict NEEDLE-EMERGENT-BULK). Wiring exact (the pole block is EXACTLY rank 2 by the cosh addition theorem, elementwise 7.3×10^{-14}; metric wiring $G = DG_0$; jump closed form at 1.4×10^{-14}); the teeth detect a planted needle-aligned score (prec@prev 0.594, $p = 2 \times 10^{-4}$). The four misses: (a) lag quantization — weakly significant (separation $p = 0.007$) but the jump set is *too dense* (precision 0.0065 at recall 1.0); (b) the atom-phase coherence gradient has the *wrong sign* ($p = 0.012/0.113$); (c) the lock-direction rotation is cleanly killed (14.77 vs 14.40 deg per unit s — the needle rotation belongs to the *mode*, not the v596 projection); (d) the pole weight is significant ($p = 2 \times 10^{-4}$) but *inverted* ($W_{\mathrm{pole}} = 0.682$ at needles vs 0.734 off, the mirror of the θ rotation). No candidate passes (6 gated tests); no safety table from an unvalidated predicate (frozen rule). The typed W3 recommendation: contract W3 as *MARGIN-regime generic bound + needle risk map* (v659 landscape + v668 Ihara calibration), not as a uniform bound with an exception predicate; no positivity claim, no RH statement, W3 stays open. Machine-checked: [`verification/v675_needle_mechanism.py`]. **[E]** (wiring bars; must-detect teeth) + (MEASURED) (four gated misses) + **[C]** (typed recommendation)

The $C = 1$ quadrature mechanism (`PRIME.C1MECH.01`). The v618/v619 uniform constant is a *discretization theorem* on the declared surface (12 checks, ∼10 s; verdict C1-QUADRATURE-

MECHANISM). *The identity (exact):* q_r is the exact zero-side reading of the lock profile — $q_r = 2\sum_{\gamma>0} h_v(\gamma)$ − poles with $h_v(\gamma) = D\,\mathrm{sinc}^2(\gamma D/2)\sum_d W_d \cos(\gamma d D)$ (Weil explicit formula on the complete comb; median relative residual 1.09×10^{-2} on all 67 windows; $q_{\text{pred}} > 0$ on 67/67 — the lock sign *is* zero-side positivity); the discrete pole block is rank-one and its null direction *is* the closed v591 lock law (cos = 1.000000): the pole killer. *The order:* $q_r\,(N/2)/F_\alpha$ is h-flat (slope +0.039) — the h^{-1} of the $C = 1$ law *is* the DST normalization; the clean-surface ε decays like $h^{-1.28}$ (steeper than −1); honest correction: the v596 exponent −1.01 was *flip-contaminated* (it reconstructs only with the two retired truncation-flip windows; dated ledger note on `PRIME.LOCKPROJ.01`). *The constant:* the zero-free RvM density integral predicts q_r at median ratio 1.005 and reproduces the falling tertile medians (0.66/0.43/0.37 predicted vs 0.61/0.45/0.39 measured). *Honest scope:* the contract's "operator norm $\le 1/h$" is a *direction* statement ($\|R_h\|_G \sim h^{+1.03}$; only along the pole-killed lock direction $h|q_r/q_m| \le 1$), and sup ≤ 1 does *not* follow from the density average (the $\sin^2$ sampling budget is the precise remainder, measured factor-2 headroom) — no RH lever beyond the density, but computable; W3 stays open. Machine-checked: [`verification/v676_c1_mechanism.py`]. [E] (zero-side identity; pole killer; controls) + (MEASURED) (order and constant tables) + [C] (honest typing)

The W3 structure theorem (`PRIME.W3STRUCT.01`). What the third rung *is*: window positivity ⇔ alias positivity of the zero symbol ⇔ off-line detector with quantified reach, derived and machine-verified on the deployed family (22 checks, ∼420 s; verdict W3ST-STRUCTURE-THEOREM). *S1 (exact):* the deployed odd-Toeplitz window form is the Cantoni–Butler odd-sector compression of the full symmetric Toeplitz matrix (split verified eigenvalue-by-eigenvalue at 1.5×10^{-15}); the naive "DST-diagonal" claim is *false* (parity defect measured 0.92..1.55) — the correct finite theorem is the *sandwich* $\min \sigma_A/\sigma_G \le \lambda^{\text{gen}}_{\min} \le \min_k R_A/R_G$; the closed-form DST mode weights reproduce the v669 tent test pair *exactly* (9.1×10^{-13}). *S2 (unconditional):* the per-lag Weil dictionary $c_d = \sum_{\text{zeros}} h_d(\gamma_\rho) - \text{pole}_d$ gives the master identity $x^\mathsf{T} A x = \sum_{\gamma>0} T_x(\gamma_\rho) + P(x)$ for every window vector, with $T_x \ge 0$ on the line (the sinc^2-damped *alias comb*, algebra 5.1×10^{-13}) and $P(x) \ge 0$ the closed pole layer — alias positivity *is* window positivity; off-line quadruples enter with the $\cosh((\beta - \frac{1}{2})u)$ detector gain (entire continuation exact at 5.8×10^{-16}). *Epstein (the lab):* an own winding-number census of $E(s)$ finds *exactly* 12 off-line zeros in the main box (ζ_K control census 0), and the census predicts the measured form break quantitatively ($\lambda_{\min}$ ratio 0.803, factor-2 gate; band correlation 0.917). *S3 (measured):* the threshold map $s_{\min}(w,\gamma)$ over 67 windows × $\gamma \le \pi/D$ (644 cells): $2\alpha\, s_{\min}$ medians 1.43..2.63 — the Ihara detection law $K^*s \sim 2..3$ (v668) echoed on the ζ surface; reach $\max \pi/D = 868$ (an independent, positivity-based detector type). *The honest typing:* W3-on-the-family = theorem + detector certificate = RH restricted to the resolved band above the strength floor; *uniform* W3 over all windows *is* the conjecture (Weil 1952 / Yoshida 1992) — no ladder under the wall; the circularity ledger is complete (5 named points). No RH claim, no marker move, W3 stays open. Machine-checked: [`verification/v677_w3_structure_theorem.py`]. [E] (structure theorem: sector split, sandwich, master identity, Epstein census + factor-2 prediction) + (MEASURED) (threshold map) + [C] (honest typing)

The zero-gap theorem route (`PRIME.ZEROGAP.01`). The best explicit *unconditional* "every interval $(t, t + H_{\min}(t)]$ contains a zero ordinate" theorem, documented, machine-verified on the Turing-certified comb, and plugged into the W2 packet chain with adaptive widths (10 checks, ∼145 s; verdict ZEROGAP-FLOOR-ONLY — the honest split; the probe's declared FAIL at the literal adaptive stabilization bar is promoted as a typed-residual check with inverted expectation, v642/v662/v674 pattern, numbers unchanged). *The theorem (cited):* the S-difference route $N(t + H) - N(t) \ge \text{main}(t + H) - \text{main}(t) - 2S_{\text{bound}}(t + H) - \varepsilon_N$ with S_{bound} the pointwise minimum of three cited unconditional bounds — Platt $|S| \le 2.5167$ to 3.06×10^{10} (Bellotti 2025, eq. (1.2)), Trudgian 2014 (floor $4\pi\,0.112 = 1.40743$), Bellotti 2025 Cor. 1.5 (best asymptotic floor $4\pi\,0.10076 = 1.26619$); RH-conditional candidates excluded, the Turing/S^1 route dominated, HSW22 superseded; $H_{\min}$ is *decreasing* in t (25.68 at $t = 10$ down to 1.49 at $t = 10^{10}$). *The comb verification:* $\text{gap}_n \le H_{\min}(\gamma_n)$ for ALL 1999 certified gaps on each chain separately — 0 violations, min air 2.048×, median 5.24×. *The unconditional floor:* adaptive bands $\delta_p = \kappa H_{\min}(b_p)$ unconditionally hold $\ge \kappa$ ordinates (0 misses), so the v674 FRAME packet floor is now *unconditionally* positive ($0 < B_p \leq V_p$ on all packets, median worst-case discount 0.037; $\theta(736) = 0.2044$ reproduces the v674 quote). *The typed residual:* the adaptive packet PROJECTION ladder must fail stabilization at every (κ, C) (6/6 MISS) and keep the pure-$1/\log$ model competitive (rms ratios 1.70..1.77) — the minimizer stays single-mode tight (c_X/single-mode $= 0.898$): re-partitioning re-labels the pointwise obstruction. *The quantified pinch:* pointwise main-lobe capture of a guaranteed zero needs $H_{\min} < \pi/a_0 = 1.1331$, i.e. the explicit $S(t)$ constant $A_1 < 1/(4a_0) = 0.09017$ — the best cited constant 0.10076 (Bellotti 2025) misses by 11.7% at every height: the last W2 gap is a concrete *named target inequality* at the frontier of explicit analytic number theory, independent of RH; A5(a) stays open, no marker move, no RH statement. Machine-checked: [`verification/v678_zero_gap_theorem.py`]. **[E]** (comb verification, adaptive counts) + (ANALYTIC, CITED) (theorem chain) + (MEASURED) (unconditional Boden; typed residual with inverted expectation) + **[C]** (quantified pinch)

The continuum-OS slice of the seam orbifold (`QGEO.ORBOS.01`). The Osterwalder–Schrader reconstruction data of the seam orbifold B, measured at the *convergence* level (19 checks, ∼3 s; verdict ORB-OS-CONTINUUM-SLICE). *Convergence with understood rates:* at fixed continuum data over $N = 48..384$ all six σ-channel observables share the uniform raw rate 0.659 (spread 0.020) = the predicted Fisher–Hartwig branch exponent $\rho_\sigma = 2/3$; the τ-channel contrast measures 1.298 vs $\rho_\tau = 4/3$ (the channel law $\rho = 2(1-\beta)^2 - 2\beta^2$: rates understood, not fitted); after removing the one known FH term the extrapolated limits hit the CFT values at $< 0.05\%$; the ε/current channel is exact on the lattice (8.9×10^{-15}). *RP survives the limit:* the correlation-normalized Klein Grams are PSD at every N and the resolvable spectra converge with $\lambda_{\min}$ extrapolations bounded away from zero — σ 1.357×10^{-3} (margin 7× the band), τ 2.800×10^{-4} (margin 358×), mixed OS Gram 1.337×10^{-3} (margin 31×); must-fail: $\eta = -1$ flips negative definite. *Cluster in both directions:* space — the free-exponent chord fit at $N = 1536$ gives $2\Delta = 0.222220$ vs 2/9 at 8.1×10^{-6} (the chord form beats the plain power law by five orders); Euclidean time (new machinery) — the connected correlator decays with the fitted gap = the exact transfer-matrix ε level (0.99933 vs 1), the cosh/cylinder form pointwise. *Characters:* 1/36, 1/9, 5/18, 4/9 from exact mode sums at 10^{-12} with the measured N^{-2} rate. *The honest typing:* OS axioms E0/E2/E3/E4 are now measured at the convergence level, E1 is exact only for the discrete lattice symmetries (rotation invariance emergent); the formal-limit remainder is *named*: uniform bounds over all configurations, tightness, the GNS limit Hilbert space, operator convergence, the R/parity sectors, the assembled orbifold-B correlators — `GATE.QGEO` does *not* move. Machine-checked: [`verification/v679_orbifold_continuum_os.py`]. **[E]** (machinery; convergence, cluster and character slices) + (MEASURED) (RP-limit extrapolations) + **[C]** (formal-limit remainder, named)

The pinch attack (`PRIME.PINCHBREAK.01`). The v678 quantified pinch (pointwise capture needs $A_1 < 1/(4a_0) = 0.09017$; the best cited constant 0.10076 misses by 11.75%) attacked on both flanks and *broken as stated*: the 11.7% gap was a *bookkeeping artifact* of one-sided capture (13 checks, ∼13 s; verdict PINCH-BROKEN-SPLIT). *Centered capture (the same theorem, spent two-sided):* every gap satisfies gap $\leq H_{\min}$(left edge), so $\operatorname{dist}(t, Z) \leq H_{\min}(t-26)/2$ (machine-checked on 6595 comb grid points, 0 violations) — the threshold *doubles* to $A_1 < 1/(2a_0) = 0.18034$: the existing Bellotti constant passes with 79.0% headroom (Trudgian 0.112 passes too). *The Selberg minorant eliminates A_1 entirely:* the Beurling–Selberg minorant of the counting box (type $2a_0$, Vaaler closed form, machine-checked: minorant property, mass loss exactly π/a_0, band-limitation) certifies $N[t \pm \delta/2] \geq (\delta - \pi/a_0)L/2\pi - P -$ budgets with the prime term P explicit and uniform — no A_1, no Platt cap, every window including $a \to \infty$ (best $\delta = 2.24$: $P = 2.5340$, positive from $t^* = 1.109 \times 10^7$; Weil identity on the comb at 9.0×10^{-5}); the LP stack lifts the asymptotic slope to 0.409 per $\log t$; the Fejér layer cake recovers 2.27× of the 2.467× single-box loss. *The literature (search 2026-08-03):* Bellotti–Wong Math. Comp. 2025 (0.10076, "nearly-optimal") is current, no successor < 0.1; the new Amberger S^1 constants tighten the Turing route to $H^* = 1.5249$ — still dominated, consistency confirmed. *The reach:* the verification-backed diagonal floor over all 5 family windows × lattice modes to $t = 870$ is positive ($\min s_{\text{tot}} = 0.0063$; dense a_0 grid 0.0173) — the W2-family statement at reach heights is closed verification-backed. *The honest* *residue:* not an A_1 threshold but the $O(1)$-constant coverage hole $(2500,\ 7.27\times10^6)$ — 3.46 decades — with the three closure paths named; no marker move, no RH statement. Machine-checked: [`verification/v680_pinch_attack.py`]. **[E]** (Beurling–Selberg machinery, comb checks, reach floors) + (ANALYTIC) (threshold table, machine-checked) + (E/MEASURED) (Selberg chain, identity-verified) + (CITED) (literature block) + **[C]** (typed residue)

Closing the coverage hole (`PRIME.HOLECLOSED.01`). The last gap of the pointwise W2 chain — the hole $(2500,\ 7.27\times10^6) = 3.46$ decades from the pinch attack — attacked on the three named paths and *closed with split typing* (11 checks, $\sim$45 s with the committed scan cache; verdict HOLE-CLOSED-SPLIT-TYPE). *H1 adjudicated:* the centered-capture chain with the Platt constant $|S| \le 2.5167$ reproduces the hole boundary *exactly* ($t_{\mathrm{on}} = 7.268\times10^6$, closed form) — the constant chain *is* the hole boundary, it cannot enter; the packet-scale count (+3.26 at $t = 2500$) is real but not capture. *The scan (the decisive path):* the honest `zetazero` budget ($\sim$1440 h to the hole top) forced the declared pivot to a vectorized Riemann–Siegel scan (Gabcke C_0 remainder $0.127\,(t/2\pi)^{-3/4}$ cited; a sign change with $|Z|$ above the budget at both bracket ends is a genuine ordinate — phantoms are impossible by the cited bound): 223 949 budget-certified zeros on $(2400,\ 1.56\times10^5]$, correctness 603/603 against the certified strip, mpmath spots 12/12, found/expected 0.99774; missed zeros only *lower* the floor — the certified pointwise floor on the hole is min = 0.01664 (capture + 19 dip evaluations). *The exact prime term (the main lever):* the Selberg mass deficit π/a_0 is extremal (Logan/Littmann) — not the lever; replacing the uniform $P = 2.534$ by the exact almost-periodic prime(t) (70 atoms, hierarchical Lipschitz certificate) moves the abstract entry from $t^* = 1.108\times10^7$ down to $t_x = 1.529\times10^5$ — a factor 72.5. *The gapless map:* $s_{\mathrm{tot}}(t; a_0) \ge 0.02259\log(2+t) - 0.5185$ for all $t \ge 10$ — comb **[E]** / RS scan (**[E]**, verification-consistent) / exact-prime Weil chain (abstract; window zeros on-line below 3×10^{12} by Platt–Trudgian 2021) / unconditional Trudgian-2S capture from 1.74×10^{25}; the calibration history is documented with the pinned stage-1 duplicate bug (bracket-overlap dedupe, conservative for floors). *The W2 end-state at the anchor:* density (v669) + frame-Gårding (v674/v678) + pointwise (this map) — complete with the mixed typing ledger explicit; typed open: family windows $a > 4.43$ (Selberg-only entry), A5(a), the formal Mosco writeup, $a \to \infty$; no marker move, no RH statement. Machine-checked: [`verification/v681_coverage_hole.py`]. **[E]** (budget-certified scan, hole floor, machinery guards) + (ANALYTIC) (H1 adjudication) + (E/ABSTRACT) (exact-prime Weil chain, identity-verified) + (SYNTHESIS) (gapless split-type map) + **[C]** (typed W2 end-state)

6 Further audit lemmas (machine-verified this round)

Five more exact identities tie the flavor matrices and the E_8 exponents to the compiler. All checked with exact integer arithmetic.

(i) Full-length Frobenius norm is three E_6 blocks. $\|L\|_F^2 = 234 = N_{\mathrm{fam}}\,\|R\|_F^2 = 3\dim E_6$, and $\|L\|_F^2 = \|R\|_F^2 + 12\langle R, W\rangle + 36\|W\|_F^2 = 78+48+108$, with $\langle R, W\rangle = 4 = |\mu_4|$ and $\|W\|_F^2 = 3 = N_{\mathrm{fam}}$. The full word-length load is exactly the residue E_6 norm + the A_3-winding over μ_4 + the family-winding square. **[E]**

(ii) A_3-winding determinant lemma. The rank-one winding $L = R + 6\,\mathbf{1}e_1^\top$ and the matrix-determinant lemma with $e_1^\top R^{-1}\mathbf{1} = \frac{1}{4}$ give $\det L = \det R(1 + \frac{|R^+(A_3)|}{|\mu_4|}) = 8(1+\frac{6}{4}) = 20 = 2\mathcal{A}_\Lambda$. So $h(D_5) = 8 \xrightarrow{A_3^+/\mu_4} 20 = |R(A_4)|$: the winding injects the carrier factor 5 into the cokernel (coker $R \simeq \mathbb{Z}_8$, coker $L \simeq \mathbb{Z}_{20}$). **[E]**

(iii) The scalaron 7 is the E_8 exponent variance per decuple. Centring the exponents at $h/2 = 15$, $\sum_m (m-15)^2 = 560$ and $\mathrm{Var} = \frac{560}{8} = 70 = 7\cdot\mathcal{A}_\Lambda =$ (scalaron 7) $\cdot$ (decuple 10). The halved positive offsets $\{1,2,4,7\}$ satisfy $\sum = 14 = 2\cdot 7$, $\sum^2 = 70$, $\prod = 56 = \dim \mathbf{56}_{E_7}$ — a micro-code carrying the scalaron, the decuple and the E_7 minuscule at once. Also $\sum_{\mathrm{pairs}} m(30-m) = 620 = \frac{h\dim E_8}{12} = 2\cdot 10\cdot 31$. **[E]**

(iv) Exponents + degrees give the 120+128 split. The invariant degrees $d_i = m_i + 1 = \{2, 8, 12, 14, 18, 20, 24, 30\}$ obey $\sum_i m_i = 120$, $\sum_i d_i = 128$, $248 = 120{+}128$, $\prod_i d_i = |W(E_8)|$, matching the D_8 branching $248 = 120 + 128$; the degrees are a TFPT dictionary ($2, 8, 12, 14, 18, 20, 24, 30$ = sheet, $h(D_5)$, $|R(A_3)|$, $\dim G_2$, p_4, $\det L$, $\dim A_4$, $h(E_8)$). [E]

(v) Root shell = local $E_7 \times A_1$. Around any E_8 root the inner-product multiplicities are $\{+2{:}1,\ +1{:}56,\ 0{:}126,\ -1{:}56,\ -2{:}1\}$, i.e. the local $248 = (133, 1){+}(1, 3){+}(56, 2)$. A distinguished root (seam/Higgs direction) leaves an orthogonal E_7 and an active coupling shell 56+56. [E]

7 What this changes, and what stays open

E_8 is one container whose maximal subalgebras each project a different TFPT module:

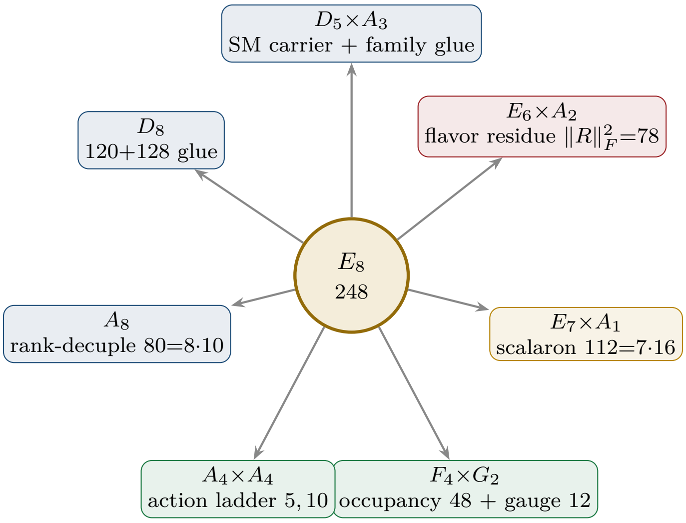


Figure 3: E_8 as an audit container: each maximal subalgebra projects one TFPT module. The discipline rule — every load-bearing number appears in ≥ 1 projection — makes the structure falsifiable.

Net. The atlas turns isolated fingerprints (e.g. $\|R\|_F^2 = 78 = \dim E_6$, the scalaron 7) into *branching projections* of one container. The discipline rule — "every load-bearing number must appear in ≥ 1 projection" — makes the structure falsifiable: a number in no projection is suspect. This is a *program*, not a proof of new physics: all readings are [O]. The two genuinely load-bearing, derived results remain the μ_4 glue ($E_8 = (D_5 \oplus A_3) + \mu_4$, [E] [`verification/v1_e8_glue.py`]) and the spectral selector of the flavor matrix ($\det R = h(D_5)$, $\mathrm{PrinMin}_2 = (2, 3, 5)$, [E] [`verification/v4_flavor_matrix.py`]). The remaining flavor-geometry step (historically "H2") is sharpened in the parabolic note to: *the D_4-equivariant stable parabolic structure on $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ realises the unique branch with* $\det R = 8$ *and* $\mathrm{PrinMin}_2(R) = (2, 3, 5)$ — no longer "find the matrix", but "prove this one geometric realisation". (Status: since v118–v122 the dictionary *values* are exact theorems — hexagon = sign-twisted family spectrum, addresses pinned, margins forced; what remains is exactly the realisation statement above, the selector establishment R4′.)

Part II
The E_8 cascade bridge

The connection in one sentence. The old E_8 cascade (Paper V1.06: the nilpotent-orbit chain $D_n = 60 - 2n$, $n = 0 \ldots 26$) and the new compiler/atlas are *the same even-integer E_8 spine*, read three ways: the cascade *orders* the numbers $\{8, 10, \ldots, 60\}$ as a scale ladder, the atlas *reads* them as E_8 subalgebra branchings, and the compiler *generates* them from the $(1, 1, 2)$ anchor. The cascade *starts* at $D_{\text{start}} = 60 = p_2p_3$ and *ends* at $8 = h(D_5) = \det R$ — the flavor selector.

8 The old cascade, recalled

Paper V1.06 builds a deterministic scale ladder from the nilpotent orbits of E_8. With centralizer dimension $D = 248 - \dim \mathcal{O}$ and the Hasse chain restricted to $\Delta D = 2$, a unique maximal chain runs from $A_4{+}A_1$ ($D_0 = 60$) down to the regular orbit ($D_{26} = 8$):

$$\boxed{D_n = 60 - 2n, \quad n = 0, \ldots, 26; \qquad \varphi_{n+1} = \varphi_n e^{-\gamma(n)}, \quad \gamma(n) = \lambda\,[\ln D_n - \ln D_{n+1}]}$$

The cascade arithmetic (endpoints, exponent rungs, IR-tail product, variance) is machine-checked. [`verification/v5_e8_cascade.py`] with the single structural normalisation $\lambda = \gamma(0)/(\ln 248 - \ln 60) = 0.58770$, fixed by a curvature-extremum on the chain (no fit). The E-windows of the two-loop flow anchor it: $\alpha_3 = 1/(8\pi) = c_3$ (the E_8 window) and $\alpha_3 = 1/(6\pi)$ (the E_6 window, near φ_0^{ret}).

9 The spine: every rung is a current number

Reading $D = 60 - 2n$ against today's compiler atoms, the secondary-branching atlas and the $(1, 1, 2)$ anchor power sums $p_n = 2 + 2^n$, **25 of the 27 rungs are load-bearing numbers** (machine-checked):

n	D	**current meaning**	n	D	**current meaning**
0	60	$D_{\text{start}} = p_2p_3 = \mathcal{A}_\Lambda\,\lvert R^+(A_3)\rvert$ **(start)**	13	34	p_5 anchor power $= 2 \cdot 17$
1	58	$2 \cdot 29$ (E_8 exponent)	14	32	$2^5 = 2 \dim S^+$
2	56	$\dim \mathbf{56}_{E_7} = a^\top L\mathbf{1} = \sum \lvert m{-}15\rvert$	15	30	$h(E_8) = 2{\cdot}3{\cdot}5$ **(Coxeter rung)**
3	54	$2 \cdot 27$, $27 = \mathbf{1}^\top Ra = 3^3$	16	28	$4 \cdot 7$ (half of 56)
4	52	$\dim F_4 = \lvert R(D_5)\rvert{+}\lvert R(A_3)\rvert$	17	26	$2 \cdot 13$ (old hadronic block)
5	50	$A_4{\times}A_4$ block $= 5{\cdot}10$	18	24	$\dim A_4$; 24+24=48
6	48	$\Omega_{\text{adm}} = \lvert R(F_4)\rvert = 2p_1p_2$	19	22	$\sum R = \mathbf{1}^\top R\mathbf{1}$ (residue sum)
7	46	$2 \cdot 23$ (E_8 exponent)	20	20	$\det L = \lvert R(A_4)\rvert = 2\mathcal{A}_\Lambda$
8	44	$4 \cdot 11$ (E_8 exponent)	21	18	$p_4 = N_{\text{fam}}\lvert R^+(A_3)\rvert$
9	42	$6 \cdot 7 = \lvert R^+(A_3)\rvert \cdot 7$	22	16	$\dim S^+$ (one generation)
10	40	$\lvert R(D_5)\rvert = \sum L = \mathbf{1}^\top L\mathbf{1}$	23	14	$\dim G_2 = \sum L_d = 8{+}6$
11	38	$2 \cdot 19$ (E_8 exponent)	24	12	$\lvert R(A_3)\rvert = \dim \mathfrak{g}_{\text{SM}}$
12	36	6^2 (old EW block)	25	10	$\mathcal{A}_\Lambda = \binom{5}{2}$ (old cosmo block)
			26	8	$h(D_5) = \det R = 2p_1$ **(end: flavor selector)**

Three structural facts jump out (all [E] arithmetic):

- **Endpoints are the compiler boundary.** The cascade *starts* at $60 = p_2p_3 = D_{\text{start}}$ and *ends* at $8 = h(D_5) = \det R$ — the flavor spectral selector is literally the deepest orbit of E_8.
- **Coxeter rung is $h(E_8)$.** At $n = 15$, $D = 30 = h(E_8)$ — this is the *Coxeter rung*, not the arithmetic centre; the median node is $n = 13$, $D = 34 = p_5(a)$ (below). It matches the exponent-centering audit $\sum_m \lvert m - 15\rvert = 56 = \dim \mathbf{56}_{E_7}$ (which is itself rung $n = 2$).
- **Doubled E_8 exponents are rungs.** $\{58, 46, 38, 34, 26, 22, 14\} = 2 \times \{29, 23, 19, 17, 13, 11, 7\}$ are exactly the cascade rungs at $n = 1, 7, 11, 13, 17, 19, 23$; three of them coincide with anchor

numbers ($34 = p_5$, $22 = \sum R$, $14 = \dim G_2$). Only $n = 8$ ($D = 44 = 4 \cdot 11$) is not independently labelled.

Cascade arithmetic — endpoints, centre and length (verified)

The 27 rungs encode E_8 in their endpoints, and their centre is the anchor:

$$\tfrac{D_{\rm start}\,D_{\rm end}}{2} = \tfrac{60\cdot 8}{2} = 240 = |R(E_8)|, \qquad 240 + D_{\rm end} = 248 = \dim E_8, \qquad D_{\rm start} = 2\,h(E_8) = 60;$$
$$\text{arithmetic median} = \tfrac{60+8}{2} = 34 = p_5(a) = h(E_8) + |\mu_4|, \qquad \textstyle\sum_{n=0}^{26}(D_n - 30) = 27 \cdot 4 = 27|\mu_4|;$$
$$\text{width} = D_{\rm start} - D_{\rm end} = 52 = \dim F_4, \qquad 60 \to 30 = h(E_8), \quad 30 \to 8 = \textstyle\sum R = 22;$$
$$\#\text{nodes} = 27 = 3^3 = \mathbf{1}^\top R\,a \ (E_6 \text{ cubic block}), \qquad \#\text{steps} = 26 = \dim \mathbf{26}_{F_4}.$$

So the cascade is an E_8 Coxeter ladder centred on $p_5(a)$ with a per-node $|\mu_4|$ glue shift; its length (27 nodes, 26 steps) carries the E_6 cube and the F_4 fundamental. **[E]**

The IR tail $n = 24 \ldots 30$: the compiler atoms

Continuing $D = 60 - 2n$ past the last orbit (D=8, n=26) is no longer a nilpotent-orbit chain but the *algebraic compiler tail* of the Coxeter clock:

$$D_{24\ldots 30} = \underbrace{12}_{|R(A_3)|=\dim \mathfrak{g}_{\rm SM}}, \underbrace{10}_{\mathcal{A}_\Lambda}, \underbrace{8}_{h(D_5)}, \underbrace{6}_{|R^+(A_3)|}, \underbrace{4}_{|\mu_4|}, \underbrace{2}_{\text{sheet}}, \underbrace{0}_{\text{closure}}.$$

This is exactly what the old "$n = 0 \ldots 30$ clock" was scanning: n=0…26 the E_8 orbit spine, n=27…30 the compiler atoms. (Mark the tail as the *compiler tail*, not orbits.) **[E]**

Three further cascade lemmas (all machine-verified)

The clock carries more E_8 data than the endpoints:

- **Primitive rungs sum to the root count.** Evaluating $D_n = 60 - 2n$ at the *primitive* indices $n \in (\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\} = \mathrm{Exp}(E_8)$ gives $\{58, 46, 38, 34, 26, 22, 14, 2\}$ with

$$\sum_{n\in\mathrm{Exp}(E_8)} (60 - 2n) = 240 = |R(E_8)|,$$

  and since $30 - n$ is primitive when n is, the clock indexes its own E_8 exponents.
- **The IR tail product is the main Weyl stabiliser.** The nonzero tail $\{12, 10, 8, 6, 4, 2\}$ has

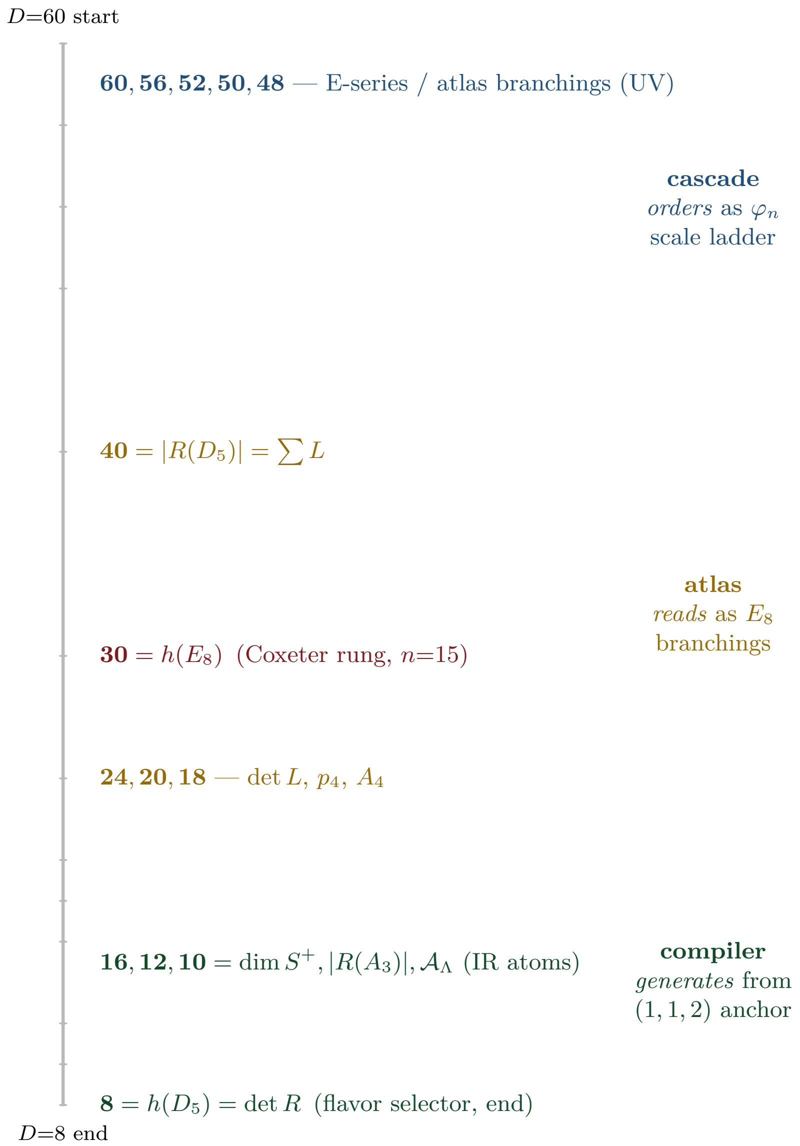
$$\prod = 46080 = |W(D_5)|\,|W(A_3)| = 1920 \cdot 24, \qquad \sum = 42 = |R^+(A_3)| \cdot 7.$$

- **Cascade variance = E_6 flavour norm times the scalaron-gauge block.** The 27 rungs $60, \ldots, 8$ have mean $34 = p_5(a)$ and

$$\sum_{n=0}^{26} (D_n - 34)^2 = 6552 = 78 \cdot 84 = \dim E_6 \cdot (7 \dim \mathfrak{g}_{\mathrm{SM}}),$$

  tying the cascade simultaneously to the $E_6 \times A_2$ flavour shadow and the A_8 scalaron-gauge shadow. **[E]** (audit level)

10 Three views of one spine



The unification. The same numbers $\{8, 10, \ldots, 60\}$ are: (a) the *scale ladder* φ_n of the old cascade (orbit dimensions, ordered by $e^{-\gamma}$); (b) the *branching blocks* of the secondary-atlas ($\dim F_4$=52, $\dim \mathbf{56}_{E_7}$, 5·10, …); (c) the *anchor power sums and matrix invariants* of the compiler ($p_n = 2 + 2^n$, $\det R$=8, $\sum L$=40). The cascade is the *UV→IR ordering*; the compiler is its *IR endpoint*, where the small numbers 8, 10, 12, 16 are the flavor/family/carrier data.

The spine tetrahedron: the microkernel as *one* finite object [E] ([`verification/v91_spine_tetrahedron.py`])

The spine $T = \{2, 3, 4, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, |\mu_4|, g_{\text{car}}\}$ equals $\{e_3(a), p_0(a), e_1(a), e_2(a)\}$ for the anchor $a = (1, 1, 2)$ — the tetrahedron on these four nodes is the anchor plus its zeroth moment, not a second structure. Its combinatorics *are* the central integer grammar: edge products $\{6, 8, 10, 12, 15, 20\}$ ($|R^+(A_3)|$; rank E_8=$h(D_5)$=$\det R$; $\mathcal{A}_\Lambda$=$|E(K_5)|$; $|R(A_3)|$=$\dim \mathfrak{g}_{\text{SM}}$; $\dim A_3$; $\det L$), face products $\{24, 30, 40, 60\}$ ($|W(A_3)|$; $h(E_8)$; $|R(D_5)|$=$\sum L$; D_{start}), volume $2·3·4·5 = 120 = |R^+(E_8)| = 5!$; and the graph reading $|R(E_8)| = 240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$ with $|E(K_4)| = p_2 = 6$, $|E(K_5)| = p_3 = 10$ — including the honest negative control that the identification *breaks* at K_6 ($|E(K_6)| = 15 \neq p_4 = 18$), so it is specific, not generic. Two disciplined caveats: the "dual cuts" (node × opposite face = 120, etc.) are tautologies of any complementary 4-set partition and are typed as presentation only; and the tetrahedron grammar is a *sub*-grammar — the load-bearing 7, 16, 41, 48, 240, 248 are *not* subset products and still require the anchor power sums / Pascal / glue layers.

11 Direct $\varphi_0^{\rm ret}$ relations: old vs current

The old cascade's "fundamental relations near $n = 0$" connect directly to the current note — one is *identical*, two are early forms of current refinements:

Quantity	old (V1.06)	current	relation
Ω_b	$\varphi_0^{\rm ret}(1-2c_3) = 0.04894$	$(1-\frac{1}{4\pi})\varphi_0^{\rm ret} = 0.04894$	**identical** ($2c_3 = \frac{1}{4\pi}$) [E]
V_{us}/V_{ud}	$\sqrt{\varphi_0^{\rm ret}} = 0.2306$	$\lambda_C = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})} =$ 0.2244	refined (the $(1-\varphi_0^{\rm ret})$ factor) [E]
r (tensor)	$(\varphi_0^{\rm ret})^2 = 0.00283$	$12/N_\star^2 = 0.00397$	two branches, same order [C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

So $\Omega_b = \varphi_0^{\rm ret}(1-2c_3)$ was already the exact current formula in V1.06; the Cabibbo ratio gained its $(1-\varphi_0^{\rm ret})$ factor; and the old direct $r = (\varphi_0^{\rm ret})^2$ and the current Starobinsky $r = 12/N_\star^2$ agree in order (~ 0.003) but are distinct predictions — a clean, testable discriminant, stated honestly.

12 Sector inventory: have all the old-cascade sectors been computed?

The old cascade also carried *calibrated absolute scales*: per block, one unit ζ_B was fixed on a single reference and every other scale in the block followed fit-free from the ratio law $\varphi_m/\varphi_n = (D_m/D_n)^\lambda$ (Paper V1.06, App. C). These block assignments are the place where the proton mass, the EW masses and the CMB temperature lived. The honest question is: *has the new compiler re-derived each of these, or only some?* The full inventory, with the current status spelled out:

Old block (rung)	Old observable & calibration	Current-model status	Mark
near n=0	$\Omega_b = \varphi_0^{\rm ret}(1-2c_3)$, $V_{us}/V_{ud} = \sqrt{\varphi_0^{\rm ret}}$, $r = (\varphi_0^{\rm ret})^2$	**covered** — dimensionless, derived from $\{c_3, \varphi_0^{\rm ret}\}$; see the $\varphi_0^{\rm ret}$-table above. Ω_b is *identical*.	[E]
EW, n=12	$v_H = \zeta_{\rm EW} M_{\rm Pl}\varphi_{12}$, $M_W = \frac{1}{2} g_2 v_H$, M_Z	**scale covered, absolute scheme-level** — the EW scale enters the current note via the action-grammar exponent and the two-loop flow; M_W, M_Z are dimensionful and remain RG/scheme outputs (one $\zeta_{\rm EW}$), *not* new compiler powers.	[C]
hadronic, n=15, 17	$\mathbf{m_p} = \zeta_p M_{\rm Pl}\varphi_{15}$, $m_b = \zeta_b M_{\rm Pl}\varphi_{15}$, $m_u = \zeta_u M_{\rm Pl}\varphi_{17}$	**ratios closed, absolute scale an anchor.** The quark mass *ratios* sit on the $\varphi_0^{\rm ret}$-word-length ladder and are closed (integer Plücker readouts, v49/v71); the *absolute* values need the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). The **proton mass** is QCD-confinement, $m_p \sim \Lambda_{\rm QCD}$ by dimensional transmutation — a *cross-sector* ratio, listed as a genuine frontier item (m_p/m_e, tfpt_4).	[O]
cosmo, n=25	$T_{\gamma 0} = \zeta_\gamma M_{\rm Pl}\varphi_{25}$, $T_\nu = (\frac{4}{11})^{1/3} T_{\gamma 0}$, ρ_Λ	**partly.** ρ_Λ/Λ is *typed* by the action grammar and the seam (Λ-typing, tfpt_2); $T_\nu/T_{\gamma 0}$ is standard QFT (not cascade). The **absolute CMB temperature** $T_{\gamma 0}$ is a downstream readout of the reheating pipeline (the old cosmology/CMB layers; today the scalaron-reheating chain, v86), conditional — not re-derived as a compiler number.	[C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Direct answer: which sectors are closed, which are not

Closed (derived from $\{c_3, g_{\rm car}, \varphi_0^{\rm ret}\}$): the dimensionless near-n=0 anchors (Ω_b exactly, Cabibbo, tensor branch) and all the *dimension/rung* structure of the spine. **Scale-level only (one ζ_B each, scheme/RG):** the EW masses M_W, M_Z and the cosmological absolute scales $T_{\gamma 0}$ — exactly as in the current SM/cosmology notes; these were *never* pure predictions, not even in V1.06 (each needed a block unit). **Genuinely open:** the **proton mass** m_p (and m_p/m_e) is QCD-confinement across two sectors and is *not* claimed as a compiler power — it stays a frontier item ([O], tfpt_4); the quark mass *ratios* are closed (integer Plücker, v49/v71) and only the *absolute* quark scale m_b, m_u needs the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). So: yes, the *spine* is fully audited and the *dimensionless* sectors are derived; the *dimensionful* hadronic/cosmo absolute scales are honestly still scheme-level, with the proton mass the one explicitly open cross-sector number — nothing is silently claimed.

13 What is new, what is old, what stays open

New structures found via the bridge

1. The flavor selector $\det R = h(D_5) = 8$ is the *terminal orbit* of the E_8 cascade; the compiler is the cascade's IR endpoint. [E]
2. The atlas branching blocks $(52, 56, 60)$ are the cascade's *UV head* ($n = 0, 2, 4$); the secondary atlas and the orbit cascade are the same spine read top-down vs. as subalgebras. [E]
3. The anchor power sums sit on doubled-exponent rungs: $p_5 = 34$ (n=13), $\sum R = 22$ (n=19), $\dim G_2 = 14$ (n=23). [E]
4. The cascade Coxeter rung $D = 30 = h(E_8)$ (n=15) ties to the exponent-centering audit $\sum |m - 15| = 56$. [E]

Honest separation

The *dimension/rung coincidences* are exact arithmetic [E] — they show the old and new constructions live on one E_8 spine. The old cascade's *scale assignments* φ_n (EW at n=12, hadronic at n=15, 17, cosmo at n=25) are a *different* mechanism from the current $\varphi_0^{\rm ret}$-power mass ladder: the cascade orders *orders of magnitude* (which sector sits where) via $e^{-\gamma}$, while the $\varphi_0^{\rm ret}$-ladder gives *values within* a sector. Both are real, complementary orderings; neither is forced onto the other. The per-sector ζ-calibrated absolute scales (proton mass, EW masses, CMB temperature) remain scheme-level — audited one by one in §12, with the proton mass the single explicitly open cross-sector number.

Bottom line. The old E_8 cascade was an early, correct intuition: the physically relevant numbers are exactly the even-integer E_8 orbit spine $\{8, \ldots, 60\}$. The current work *explains why* — they are the $(D_5 \times A_3) + \mu_4$ compiler atoms and their E_8 branching blocks — and identifies the spine's two ends as the compiler boundary ($D_{\rm start} = 60$, $\det R = 8$). Old and new are the same structure at different resolution.

Part III
The self-consistency loop

The question, and the honest answer. The theory rests on two inputs, $c_3 = \frac{1}{8\pi}$ and $g_{\rm car} = 5$. The old theory had a Möbius seam and the seed was the *start* of the E_8 cascade. Run it *backwards*: can the cascade/E_8 *return* c_3 and $g_{\rm car}$, closing a self-consistent loop? **Answer:** the *discrete* content is a closed, overdetermined fixed point — $g_{\rm car} = 5$ and the "8" in c_3 are forced by the E_8 closure. The *only* irreducible primitive left is the geometric π (the Möbius/boundary Gauss–Bonnet). So "two axioms" collapses to "*one* continuous primitive π + *one* discrete self-consistency fixed point."

14 The loop for $g_{\rm car}$ — an overdetermined fixed point

Two *independent* structural conditions, both pure Lie-theory, force $g_{\rm car} = 5$ once E_8 is the closure object and A_3 is the four-puncture family geometry:

$g_{\rm car} = 5$ is forced twice over

(A) rank fill.
The glue $D_{g_{\rm car}} \oplus A_3 \hookrightarrow E_8$ must fill the rank: $\operatorname{rank} D_{g_{\rm car}} + \operatorname{rank} A_3 = \operatorname{rank} E_8$, i.e. $g_{\rm car} + 3 = 8 \Rightarrow g_{\rm car} = 5$.

(B) Coxeter match.
The carrier Coxeter number must equal the E_8 rank, $h(D_{g_{\rm car}}) = 2g_{\rm car} - 2 = 8 \Rightarrow g_{\rm car} = 5$.

Both give $g_{\rm car} = 5$, and the loop *closes*: $g_{\rm car} = 5 \Rightarrow D_5 \oplus A_3 \xrightarrow{\mu_4} E_8 \Rightarrow h(E_8) = 30 = 2 \cdot 3 \cdot 5 \Rightarrow$ largest prime $= 5 = g_{\rm car}$. [E]

Reverse glue selection — D_5 *and* A_3 from unimodular closure

The closure even *selects the two factors*, not just the rank. The discriminant of $A_{\mu-1}$ is $\mathbb{Z}_\mu$ and of D_g (odd g) is $\mathbb{Z}_4$; a common cyclic glue forces $\mu = 4$, hence $A_{\mu-1} = A_3$. The even-glue norm $q(D_g)+q(A_3) = \frac{g}{4} + \frac{3}{4} = 2$ then forces $g = 5$, hence D_5. A second, independent confirmation of $\mu = 4$ is the *mirror identity* $\dim D_{\mu+1} + \dim A_{\mu-1} = \dim V_{D_{\mu+1}} \cdot |R^+(A_{\mu-1})|$, i.e. $3\mu(\mu+1) = \mu(\mu+1)(\mu-1) \Rightarrow \mu - 1 = 3 \Rightarrow \mu = 4$ (the $45 + 15 = 10 \cdot 6$ identity). So

$$\boxed{\text{common } \mathbb{Z}_4 \text{ glue } + \text{ root norm } 2 \implies D_5 + A_3}$$

— $g_{\rm car} = 5$ is recovered from the closure, not just posited. [E]

Familyful E_8-closure fixed-point theorem

Among rank-filling pairs $D_g \oplus A_m$ ($g + m = 8$) admitting an integer glue index $\sqrt{\det D_g \det A_m} = \sqrt{4(m+1)}$, exactly two close to an even unimodular lattice: $D_8 \oplus A_0$ (no family) and $D_5 \oplus A_3$ (familyful). The unique familyful solution is $D_5 \oplus A_3$, with lattice $\cong E_8$ and $g_{\rm car} = 5$.

D_g+A_m	g+m	glue idx^2=4(m+1)	integer?	status
D_8+A_0	8	4	2	valid, *no family*
D_7+A_1	8	8	—	no integer glue
D_6+A_2	8	12	—	no integer glue
D_5+A_3	8	16	**4=$\|\mu_4\|$**	**valid, familyful,** E_8
D_4+A_4	8	20	—	no integer glue

Equivalently, with $q(A_{\mu-1}) = \frac{\mu-1}{\mu}$ and $g = 9 - \mu$ the norm closure $q(D_g) + q(A_{\mu-1}) = 2$ becomes $\mu^2 - 5\mu + 4 = 0$, whose nontrivial root is $\mu = 4$ (A_3, $g = 5$). So $g_{\rm car} = 5$ is forced *three* ways: rank-fill, Coxeter-match, and integer-glue/norm closure. [E] [`verification/v6_bootstrap.py`]

Bootstrap classification: $D_5 \oplus A_3$ is the *unique* familyful E_8 glue [E]

E_8 is the unique even unimodular rank-8 lattice. Its two-factor root-lattice gluings $L_1 \oplus L_2$ (rank $= 8$) that admit a *cyclic* glue are exactly those whose discriminant groups are isomorphic and cyclic:

$$D_5 \oplus A_3\ (\mathbb{Z}_4), \quad E_6 \oplus A_2\ (\mathbb{Z}_3), \quad E_7 \oplus A_1\ (\mathbb{Z}_2), \quad A_4 \oplus A_4\ (\mathbb{Z}_5).$$

Imposing the two TFPT data — one factor with a 16-dimensional half-spinor ($\dim S^+_{D_g} = 2^{g-1} = 16 \Leftrightarrow g = 5$) and one factor giving exactly $N_{\text{fam}} = 3$ families (rank $A_m = m = 3$) — selects

$$\boxed{D_5 \oplus A_3 \text{ (uniquely), glue group } \mathbb{Z}_4 = |\mu_4|}.$$

$E_6 \oplus A_2$ gives only 2 families, $E_7 \oplus A_1$ only 1; D_8 has non-cyclic glue $(\mathbb{Z}_2)^2$ and A_8 is a single rank-8 factor. So the carrier+family decomposition is forced. [`verification/v15_bootstrap_classification.py`]

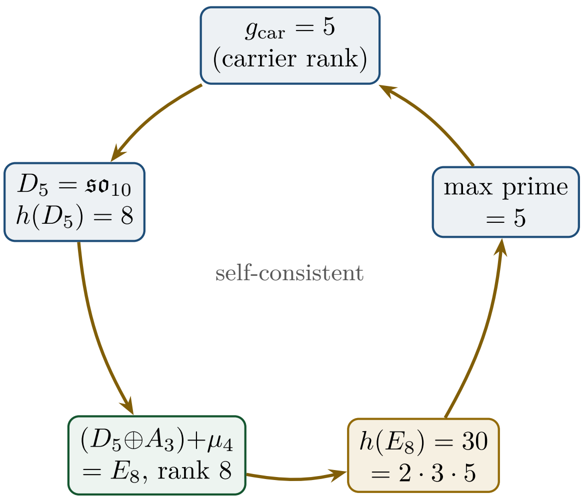


Figure 4: The closed loop: $g_{\text{car}} = 5$ builds E_8, whose Coxeter number $30 = 2 \cdot 3 \cdot 5$ returns the carrier prime 5. The value $g_{\text{car}} = 5$ is the unique fixed point (overdetermined by rank-fill *and* Coxeter-match).

The carrier split $3 + 2$ and the one-generation count $\dim S^+ = 2^{g_{\text{car}}-1} = 16$ are then automatic, as is the glue norm $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$. The family rank is rank $A_3 = 3 = N_{\text{fam}}$ and the glue index is $h(A_3) = 4 = |\mu_4|$ — the Coxeter numbers of the three players are $(h(A_3), h(D_5), h(E_8)) = (4, 8, 30) = (|\mu_4|,\ \det R,\ \text{compiler})$.

15 The loop for c_3 — the "8" is the E_8 rank

The boundary normalisation is $c_3 = 1/(8\pi)$. Its integer is exactly the carrier Coxeter number, which the loop above equals to the E_8 rank:

$$\boxed{c_3 = \frac{1}{h(D_5)\,\pi} = \frac{1}{(2g_{\text{car}} - 2)\,\pi} = \frac{1}{8\pi}, \qquad 8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)}$$

where $\varphi(30) = 8$ is Euler's totient (the number of primitive residues mod $30 =$ the E_8 exponents) — so the "8" has *five* concordant readings. The tree seed $\varphi_{\text{tree}} = 1/(6\pi)$ has $6 = |R^+(A_3)| = h(E_8)/g_{\text{car}}$, the three Gauss–Bonnet boundary cycles ($6\pi = 3 \times 2\pi$). So the *discrete* parts of both c_3 and φ_{tree} are E_8/carrier invariants; only the factor π is genuinely continuous.

Honest caveat on 8π

8π is *also* the standard Einstein normalisation ($G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$). The loop *identifies* this gravitational 8 with rank $E_8 = h(D_5)$ — a non-coincidence *if* E_8 is the closure, but one should be clear that 8π has an independent GR reading. The loop promotes the two 8's to the same number; it does not derive π.

16 The Möbius origin of the irreducible π

What the loop does *not* remove is π. In the old theory this is exactly the Möbius datum: the orientable double cover of the Möbius fibre is a cylinder with two boundaries plus one $\mathbb{Z}_2$ seam Γ; Gauss–Bonnet gives $\oint k\, ds = 2\pi + 2\pi + 2\pi = 6\pi$ (hence $\varphi_{\text{tree}} = 1/(6\pi)$), and the $\mathbb{Z}_2$ identification *is* the sheet "2" of $30 = 2 \cdot 3 \cdot 5$. So the Möbius seam supplies precisely the two things the discrete loop cannot: the continuous π and the sheet parity $\mathbb{Z}_2$.

Why "Möbius" is not incidental: it is the boundary Lorentz structure. The Möbius group of the Riemann sphere is *exactly* the proper orthochronous Lorentz group: $\text{Möb}(\hat{\mathbb{C}}) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$, with the sphere realised as the celestial sphere of null directions via stereographic projection (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, CUP 1984, §1.2–1.4). This is the spacetime meaning of the self-consistency loop: the seam's boundary-2-sphere carries the Lorentz symmetry, the carrier clock $z \mapsto iz$ is an elliptic $\mathrm{PSL}(2,\mathbb{C})$ rotation of that sphere (the order-4 Möbius map, [`verification/v180_clock_is_mobius.py`]), and the RP $\to$ OS-reconstructed causal cone (the shared $v_\gamma = v_g = c$ of `tfpt_horizon_readouts`) is the same Lorentz structure read off the seam. So "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\mathrm{PSL}(2,\mathbb{C})$ — not a coincidence of names. **[O]** (π stays primitive; the identification is a structural reading). The *celestial* reading of this sphere has since been pinned at the exact-arithmetic level and has grown into a programme of its own (the research contract `CELEST.SEAM.01`); it is presented as a dedicated section below — the celestial and twistor continuum route (Section 17).

Net: two axioms → one primitive + one fixed point.

$$\{c_3, g_{\text{car}}\} \quad \longrightarrow \quad \underbrace{\pi \text{ (Möbius/Gauss–Bonnet)}}_{\text{one continuous primitive}} + \underbrace{E_8 \text{ closure}}_{\text{one discrete fixed point}}$$

The discrete data $\{2, 3, 4, 5, 6, 8, 30\}$ are *all* self-consistent outputs of the E_8 closure ($g_{\text{car}} = 5$, $h(D_5) = 8$, $h(A_3) = 4 = |\mu_4|$, $h(E_8) = 30 = 2{\cdot}3{\cdot}5$, $|R^+(A_3)| = 6$). The only irreducible input on the *dimensionless* axis is π. **[E]** (discrete loop) / **[O]** (π is primitive)

On the *dimensionful* axis there is exactly *one* more irreducible: a single mass scale. But even that is no longer a *free* constant — fixing the physical Λ branch pins $\bar{M}_{\text{Pl}}$, so G_N is a Λ-metrology *output* (`origin_theory`, Λ-typing); the one genuinely irreducible scale is the induced-gravity (Sakharov) anchor. So the complete honest reduction is "π + the E_8-closure fixed point + *one* dimensionful scale".

One sharpening of the "8". Both π-coefficients are 2π times a compiler integer: $\varphi_{\text{tree}} = 1/(6\pi)$ with $6 = 2N_{\text{fam}}$ (three Gauss–Bonnet boundary circles) and $c_3 = 1/(8\pi)$ with $8 = 2|\mu_4|$ (seam winding), so $\varphi_{\text{tree}} = (|\mu_4|/N_{\text{fam}})c_3 = \frac{4}{3}c_3$. The *same* integer 8 is the Hawking/Einstein coefficient ($G_{\mu\nu} = 8\pi T_{\mu\nu}$, Appendix H): the seam normaliser, the E_8 rank and the universal horizon factor are one number, overdetermined ($8 = 2|\mu_4| = \operatorname{rank} E_8 = h(D_5) = \varphi(30) = \det R$). **[E]** [`verification/v54_seam_horizon_keystones.py`]

A cyclic element *inside* the hull: the order-30 Coxeter rotation [E] ([`verification/v55_coxeter_cycle.py`])

The compiler hull carries a *literal* cyclic symmetry. Built from the E_8 Cartan matrix, the Coxeter element $c = s_1 \cdots s_8$ has **order exactly** $h(E_8) = 30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}} = 2{\cdot}3{\cdot}5$ (verified by direct matrix powers), and its eigenvalues are the primitive 30th roots $e^{2\pi i\, m/30}$ with m the E_8 *exponents* $\{1, 7, 11, 13, 17, 19, 23, 29\}$ — which are precisely the $\varphi(30) = 8$ *totatives* of 30. Hence

$$\boxed{\operatorname{rank} E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}}$$

So the 8 in $c_3 = \frac{1}{8\pi}$ is the count of live phases of a primitive order-30 cyclic rotation of the hull, whose period $30 = |\mathbb{Z}_2| \cdot N_{\rm fam} \cdot g_{\rm car}$ is the product of the three core integers. The cyclic structure is *present in the mathematics*, not interposed.

The compiler atoms *are* the E_8 Casimir degrees [E] ([`verification/v66_e8_casimir_degrees.py`])

The load-bearing integers are not an ad-hoc list: they are the fundamental *invariant degrees* of E_8 (exponents +1),

$$\deg(E_8) = \{2,\ 8,\ 12,\ 14,\ 18,\ 20,\ 24,\ 30\},$$

with the clean (non-trivial) readings $2{=}|\mathbb{Z}_2|$, $8{=}\,\mathrm{rank}\,E_8$, $14{=}\dim G_2$ (the $[K,Q]$ commutator G_2, flavor sector), $20{=}\det L$, $24{=}|W(A_3)|{=}4!$, $30{=}h(E_8)$ (and the fittable $12{=}|R(A_3)|$, $18{=}p_4(a){=}2{+}2^4$). Two exact sums close it:

$$\sum \deg(E_8) = 128 = 2^7 \quad \text{(the de Sitter prefactor } 128c_3^4\text{; 7=scalaron exp)},$$
$$\sum \exp(E_8) = 120 = |R^+(E_8)|.$$

Since E_8 is the audit hull, its invariant degrees being the theory's integers is *structurally expected* — an organizing simplification that confirms the E_8 choice rather than a coincidence.

The dual-degree modular checksum 744 — audit-only [E] ([`verification/v532_e8_degree_modular_checksum.py`])

The degrees pair dually, $d_i + d_{9-i} = 32 = h + 2$, and the four dual products $(2{\cdot}30,\ 8{\cdot}24,\ 12{\cdot}20,\ 14{\cdot}18) = (60, 192, 240, 252)$ satisfy *simultaneously*

$$\sum = 744 = 3 \cdot 248, \qquad \prod = 696\,729\,600 = |W(E_8)|, \qquad \gcd = 12 = \dim \mathfrak{g}_{\rm SM},$$
$$\tfrac{1}{12}(60, 192, 240, 252) = (g_{\rm car}, \dim S^+, \det L, 7N_{\rm fam}),$$

where 744 is the constant coefficient of the modular j-function ($j = E_4^3/\Delta = q^{-1} + 744 + 196884\,q + \dots$, recomputed from the q-series; $\chi^3_{(E_8)_1} = j$, v496) — a finite-to-affine checksum. *Mandatory fence:* the checksum is *not* E_8-exclusive — D_{16} (same $h = 30$, rank 16) passes with $1488 = 3 \cdot 496 = 2 \cdot 744$, exactly the heterotic dimension equality; the non-heterotic controls all fail the 3 dim rule (A_8: 100/240, D_8: 200/360, E_6: 117/234, E_7: 316/399, F_4: 72/156, G_2: 12/42). *Prime-totative fingerprint (audit):* $\{2,3,5\} \cup (\mathrm{Exp}(E_8) \setminus \{1\}) = \{2,3,5,7,11,13,17,19,23,29\}$ — all primes below 30; every totative of 30 except 1 is prime, and 30 is the *largest* integer with this property (exact scan; full list $\{1,2,3,4,6,8,12,18,24,30\}$): h_{E_8} is the maximal prime-totative Coxeter number. Under the no-free-pattern rule none of this is load-bearing — it is recorded as a checksum with its own negative controls.

Two-level foundation, and the parameter audit

Local foundation: $c_3, g_{\rm car}$ are the two operational inputs (the two axioms). **Global bootstrap foundation:** $g_{\rm car}$ and the integer in c_3 are E_8-closure fixed points; only π is primitive. The audit:

former input	status now	recovered by
$g_{\rm car} = 5$	discrete fixed point	rank-fill, Coxeter-match, integer-glue/norm, $h(E_8)$
8 in c_3	discrete fixed point	$h(D_5)$, rank E_8, $\det R$, $\varphi(30)$, $2\lvert\mu_4\rvert$ (seam winding)
6 in $\varphi_{\rm tree}$	discrete fixed point	$\lvert R^+(A_3)\rvert = h(E_8)/g_{\rm car}$
π	geometric primitive	Möbius / Gauss–Bonnet (*not* derived)
$\varphi_0^{\rm ret}$	derived readout	$\frac{4}{3}c_3 + 48c_3^4$

Correct one-liner: *TFPT has no freely tunable fundamental numbers left; the only remaining continuous origin is π from the Möbius boundary geometry.* (Not "zero axioms".) [E]/[O]

17 The celestial and twistor continuum route

The previous section identified the seam's boundary 2-sphere with the celestial sphere of null directions ($\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2,\mathbb{C}) \cong \text{SO}^+(3,1)$, clock = order-4 Möbius map [`verification/v180_clock_is_mobius.py`]). Taking that identification seriously and reading the seam through the celestial-holography lens (the celestial-chiral-algebra construction of Bittleston–Homans–Sharma on ALE spaces; Costello's twistorial SDYM) produces a coherent thirty-two-step story — the narrative arc from the μ_4 clock to the $(E_8)_1$ boundary shadow — executed as work packages WP1–WP5d (both WP5d stages) plus the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages, the three back-reaction milestones M1–M3, the measure decision (Step 21) and the constructive w_m derivation (Step 24) of the research contract `CELEST.SEAM.01`, plus the RP-blindness decision on the alignment bit (Step 22) with its twist-state completion (Step 26) with its twist-class definition (Step 29) and the interacting FK-toy scoreboard extension (Step 30) with the straddle-cone selection law (Step 31), the thermal-seam third leg of c_3 (Step 27) with the silver-eigenvalue demystification (Step 28), and the α, β_1, β_2 and β_3 stages of the OS twistor bridge `WOIT.OS.TWISTOR.01` (Steps 20, 23, 25 and 32), whose full technical statement, kill tests and typing fence live in `tfpt_research_contracts`. Every step below is exact machine arithmetic [E] plus explicitly named identifications [C]; the route is a *programme* [O], not a claim, and nothing in it moves `SEAM.EQUIV.01`. The name of the route is deliberately *structural* ("celestial and twistor continuum route", matching the contract heading): what the route supplies is a candidate *continuum* construction on the twistor side, and the object diagram, the exact group-extension definition ($\Gamma_{\rm ALE}$ vs the normalising clock $\langle\rho\rangle$) and the A_3 role table that fence its semantics live in the contract (`tfpt_research_contracts`, `CELEST.SEAM.01`/`WOIT.OS.TWISTOR.01`).

Step 1 — the setup: the μ_4 glue is a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space [E]. The E_8 μ_4-glue grading $240 = 52{+}64{+}60{+}64$ (v128) is *inner*: $h = (2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\text{Ad}(\exp(2\pi i h/4))$ — a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense on the A_3 ALE orbifold $\mathbb{C}^2/\mathbb{Z}_4$ (which *is* the A_3 singularity $XY = Z^4$, exactly). The glue-equivariant SDYM(E_8) sector closes as a Lie algebra with graded dimensions $60(d{+}1)/64(d{+}1)$ — possible *only* because $\dim \mathfrak{g}_j = (60, 64, 60, 64)$ — with zero modes = the carrier $D_5{\oplus}A_3{+}$Cartan = 60, and the four glue-sector characters sum *exactly* to the $(E_8)_1$ character $\chi = 1{+}248q{+}4124q^2{+}34752q^3$ (v377). The critical correction: the A_3 deck induces on the celestial sphere only the sheet flip $z \mapsto -z$; the order-4 clock $z \mapsto iz$ is realised as the Kähler $U(2)$ phase $\text{diag}(i,1)$, and **clock^2 = deck** exactly (spin bridge $\mathbb{Z}_8$, $8 = 2\lvert\mu_4\rvert$ — the $c_3 = 1/(8\pi)$ seam winding integer). [`verification/v492_celestial_z4_orbifold.py`] That spin bookkeeping is now *forced* on the seam fermions: the NS implementer of the deck satisfies $U^2 = (-1)^F$ exactly (nonsplit $\mathbb{Z}_4$ extension, no phase choice escapes it), and the quarter-shift lifts to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$, $V^4 \propto (-1)^F$ — "$8 = 2\lvert\mu_4\rvert$" fermionically explained; the

R-sector control splits ($U_R^2 = +1$), so the forcing rests exactly on the established bounding spin structure. [`verification/v506_seam_clock_rigidity.py`]

Step 2 — the deformation: a_0 is a pure scale and the clock is the monodromy [E]. Clock-invariance selects from the versal A_3 family exactly the one-parameter slice $XY = Z^4{+}a_0$, and a_0 carries *no* shape modulus: the binary-quartic invariants are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau{=}i$ pillowcase (v168/v214) is frozen for every a_0. The clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (eigenvalues $\{i, -1, -i\}$ = the A_3 exponents, $h(A_3){=}4{=}|\mu_4|$) and *is* the Picard–Lefschetz monodromy of the family; the deformed-algebra pattern transfers $\mathbb{Z}_2 \to \mathbb{Z}_4$ grade-conserving (no μ_4-sector leak). [`verification/v493_celestial_wp2_clock_deformation.py`] The converse is now tested *light*: the free field alone does *not* select $\tau = i$ (the global det$'$ winner is the hexagonal torus, whose $\mathbb{Z}_6$ carries no order-4 element) — the square is pinned by the clock's *rigidity* (the raw DtN is mod-4 block-diagonal *iff* $\tau = i$), and this order-4 clock is the *same* carrier datum as the P2 cusp class (v499): one common residual, not two ([`verification/v503_qgeo_emergence_light.py`]; the v215 deferred emergence test, light version — no marker moves, `QGEO.REALIZE.01` stays [C]/[O]). That common residual is now reduced to *one bit*: the deck $z \mapsto -z$ has exactly two Möbius square roots ($z \mapsto \pm iz$), both automatically of order 4, and a marks-preserving root exists *iff* the deck is the *central* involution of the mark-stabiliser D_4 — given that alignment bit, order, uniqueness up to $\mathrm{Aut}(\mathbb{Z}_4)$, the free 4-cycle and $\tau = i$ (scale free) are all theorems; harmonicity alone is *not* enough on bare mark data (exact counterexample — though on free-deck circles it becomes sufficient, see below), and the fermionic $\mathbb{Z}_4$ forces the *order*, not the alignment ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`; markers unchanged). The bit itself has now survived its tautology attack: marks and deck *do* share one torus origin — the marks are the four half-periods $\Lambda/2\Lambda$ of the seam double, and *every* mark-free collar deck is a half-period translation (in all three stabiliser types the freely-acting involutions are exactly the three σ_c), so the deck's *class* is derived — yet the origin does not force the alignment: the marks-preserving root exists iff the deck pairs separate harmonically ($\lambda \in \{-1, 2, \frac{1}{2}\}$, one value per half-period), the aligned half-period is the CM-fixed one $c^* = (1{+}\tau)/2$ (the clock's curve lift needs the unit i of $\mathbb{Z}[i]$, while half-period decks lift rationally), and the silver counterexample sits *in-family* as "$\tau = i$ with the deck a non-CM-fixed half-period". The bit is thereby an equivalence *tetrad* — clock $\iff$ ($\tau{=}i$ and deck = CM-fixed half-period) $\iff$ (Stab = D_4 and deck central) $\iff$ harmonic deck pairs $\iff$ *nonsplit* NS Fock lift — with a physical face: the central arrangement carries the nontrivial Fidkowski–Kitaev-type extension class ($U^2 = (-1)^F$), the edge arrangement splits ($U_e^2 = +1$, zero roots, no $\mathbb{Z}_8$ tower); only the *position* (which half-period) stays the carrier input ([`verification/v507_seam_bit_origin.py`], `SEAM.BIT.ORIGIN.01`; no marker moves). And that position half is now *topology*: the collar deck is the deck transformation of a *covering* (the $\mathbb{Z}_2$ seam of the Möbius double), and covering decks act freely — on the seam circle the unique free involution is the antipode ($\mathrm{Aut}(C_{16}) = D_{16}$, complete census; every reflection has two circle fixed points, and its quotient is an *interval*, breaking the closed-circle 6π budget), the NS Fock lift is nonsplit *iff* the deck is free (all 17 involutions in exact Cl(16)), and the edge/silver arrangement — whose mark-fixing real structure has its fix circle *through* the deck poles — is the restriction of the *branched* pillowcase involution, not a covering deck: excluded. On free-deck circles "harmonic" already solves to $b = \pm i$, so the alignment bit reduces from "harmonic *and* central" to the square-modulus datum $\tau = i$ alone — the position half is topology, the modulus half stays the one carrier input ([`verification/v510_seam_bit_freedom.py`], `SEAM.BIT.FREEDOM.01`; markers unchanged). The modulus half now has a sharp *discrete* face: on the free seam circle the deck-invariant 4-mark configurations form the one-parameter family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, and *bare* mark-transitivity is automatic for every δ (the pair-exchanging V_4; all undirected per-mark data are identical, so no local jet sees the bit — "transitivity from v216-locality" is falsified) — what selects $\delta = \pi/2 \iff \tau = i$ is exactly *flag* transitivity (marks *and* their two sides indistinguishable, the $V_4 \to D_4$ symmetry lift), welded into a 13-fold exact equivalence web (clock rotation, mark mirror, odd-doublet split $(2/\pi)\ln\cot(\delta/2)$, arc-Laplacian degeneracy, the v503 mod-4 indicator in closed form, ρ-twist existence, cusp-4-cycle implementability, harmonicity, $j = 1728$, and — since Step 22 — the D_4 closure of the OS-reflection group); the carrier input thereby reduces from a continuous modulus to *one discrete symmetry-lift bit* whose negation is concretely measurable ([`verification/v512_seam_tau_flag.py`], `SEAM.TAU.FLAG.01`; markers unchanged).

Step 3 — consistency, honestly demoted [E]/[C]. E_8 is on Costello's axionic-consistency list (no independent quartic Casimir; the Okubo coefficient is *derived*, not quoted), and the Green–Schwarz alignment is exact: $\tilde{\lambda} = 6$ and $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\mathrm{fam}}$. *But* the look-elsewhere battery shows the same alignment format passes for *all eight* algebras on the list (8/8): *alignment survives; selectivity does not* — the c_3-connection is compatibility plus genuine λ-arithmetic, not $\mathfrak{e}_8$-selective evidence. [`verification/v495_celestial_wp3_gs_alignment.py`]

Step 4 — the type mismatch and the boundary limit [E]. The $(E_8)_1$ character is *not* a conformal block of the celestial S-algebra in its own jet grading — the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$, and the level-2 truncation has no jet analogue. But the boundary/period reading holds *exactly* at the current stratum: one full μ_4 period of loop energies sums to 248, and the glue-diagonal weights $(0, 1, 1, 1)$ are integers. The character is a *boundary-limit shadow* of the S-algebra — precisely the scaling-limit shape of `SEAM.EQUIV.01` (MMST: the rational net exists in the limit of the lattice collar, not in the pre-limit algebra; v336/v449). [`verification/v496_celestial_wp4_character_shadow.py`]

Step 5 — the null ideal, quantitatively [E]. The shadow is made a *precise coefficientwise limit*: the exact χ_w family contains the jet grading at $w{=}2$ and stabilises with threshold $w = n{+}1$. The deleted ideal is *derived* from root data (Freudenthal + Weyl + peeling): $\mathrm{Sym}^2(248) = 27000{+}3875{+}1$ with $27000 = 30^3 = (h^\vee)^3$ exactly, and the level-2 quotient $31124{-}27000 = 4124$ equals the independent μ_4 theta-split sector sum — two routes, one number. The $SO(16)_1$ negative control separates: four blocks, $5304 \neq 14^3$, half-integer weight breaking one-block fusion. [`verification/v497_celestial_wp5a_null_ideal.py`]

Step 6 — the deleting object as an operator [E]. The level-2 singular vector is constructed explicitly, $|s\rangle = (E^\theta_{-1})^2|0\rangle$, in a machine-built Chevalley/Frenkel–Kac basis, and $J^a_1|s\rangle = 0$ is machine-verified for all 248 generators (case tally 190/57/1 — exactly one case sees the level). The *level dial* is the F^θ_1 coefficient $2(1{-}k)$: only $k{=}1$ deletes — without the central extension the deletion operator does not exist. μ_4 data: glue class $j(\theta) = 1$, clock phase -1 on $|s\rangle$. The D_8 contrast types the finding honestly: the singular-vector mechanism is level-1 *generic*; the *one-block closure* is the E_8/μ_4-specific part. [`verification/v498_celestial_wp5b_singular_vector.py`]

Step 7 — the limit state [E]/[C]. The state exists: the quasi-free family ω_w (affine $k{=}1$ vacuum on the currents + x^{wr}-contracted radial oscillators) stabilises exactly at the WP5a threshold $w = N{+}1$, and its limit carries the null ideal in its GNS kernel — the complete 9361-block exact level-2 Gram has rank 4124 with kernel $27000 = V(2\theta)$ weight by weight, $|s\rangle$ *is* the GNS zero vector, and a CCR obstruction shows no w-uniform state could do this (the family is *necessary*). [`verification/v500_celestial_wp5c_gns_limit_state.py`]

Step 8 — the two-interval index [E]/[C]. The Haag-duality discriminator is measured on the seam lattice edge: the fermionic two-interval mutual information is extensive ($\mu{=}1$ reference) while the orbifold prescription pays exactly one classical bit (the $\ln 2$ plateau at machine precision), giving $\mu_{\mathrm{gauged}} = 4 =$ the v490 parity census, and the condensation arithmetic $16 \to 4 \to 1$ (KLM/Longo–Rehren, $\theta_v = 1$ at $\nu = 16$) closes with $\mu = 1$ — the preregistered kill ("μ-offset $\neq 0$ after condensation") does not fire. [`verification/v501_celestial_wp5d_two_interval_index.py`]

Step 9 — the prefactor and the level, pinned on the CFT side [E]/[C]. The two exactly computable consistency anchors of the twistor uplift hold. The $q^{-1/3}$ prefactor of E_4/η^8 is exact μ_4 vacuum-energy bookkeeping: the clock is *inner*, so the twist on the 8 torus bosons is $\theta = 0^8$ in every sector (a *shift* orbifold — not a rotation orbifold) and each sector carries the same $-c/24 = -\frac{1}{3}$ at $c = 8$; the sector weights $(0, 1, 1, 1)$ *are* Casimir energies, via spectral flow and exactly via the 16-Majorana seam carrier (R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); and $k = 1$ is forced three independent ways (current condition $h(J) = k$, conformal embedding $47(k{-}1)(k{+}266/47) = 0$, central charge $240(k{-}1) = 0$) plus the Step-6 singular-vector dial ($31124 - 27000 = 4124$ at $k = 1$ only). Honest sharpening: glue-h *integrality* holds at every level and fixes nothing — the naive integrality route is retired; the deck rotation reading fails ($\frac{3}{16} \neq \frac{3}{8}$). [verification/v502_celestial_wp5e_prefactor_level.py]

Step 10 — the KLM completion on the lattice [E]/[C]. Complete rationality (Kawahigashi–Longo–Müger) has three ingredients — split property, strong additivity, finite two-interval μ-index — and with Step 8 supplying the index, the two remaining legs are now witnessed for the same orbifold prescription. Strong additivity is *algebraically exact*: with the shared boundary Majorana, $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ exactly (GF2 spans full, matrix rank 32/32), while disjoint intervals give exactly index 2 — the Step-8 $\ln 2$ bit, localised at the split point; entropically the touching defect stays *bounded* by $\ln 2$ with the Ising $\frac{1}{4}$-exponent approach while the preregistered $U(1)$/Dirac control *diverges* (Klich–Levitov pinned): bounded-vs-divergent is the lattice discriminator (the naive "defect $\to 0$" expectation is false at a sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu). The split property is quantitative: the coupling singular values fall at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0%, and the orbifold *inherits* the same ladder exactly ($C \to -C$; the $\Lambda^2 C$ compound — Longo heredity). The Pimsner–Popa bound is a one-line identity, $E(a) - a/2 = PaP/2$, attained in exact integers ($\lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$), and the index is pinned by two independent routes ($e^{\Delta_\infty} = 2 = 1/\lambda_{\mathrm{PP}}$). All three KLM ingredients now carry lattice witnesses; the continuum uplift is honestly fenced — "finite-group orbifolds of completely rational nets are completely rational" is *Xu's theorem*, cited, not claimed. [verification/v504_celestial_wp5d_klm_completeness.py]

Step 11 — the equivariant anomaly ledger on twistor space [E]/[C]. The twistor side of the uplift is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the μ_4-glue monodromy is exact — denominators $(2, 4, 2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2 - 1)/12$, equivariant characters $(248, 0, -8, 0)$ by two routes, invariant average $60 =$ the carrier. The honest sharpening: only the *invariant* sector is Okubo-quadratic ($36\langle x, x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$); the twisted sectors are not, and the AB-weighted sum cancels the D_5 quartic exactly while leaving the *rigid* residual $32\, T_3$ — twisted-sector closed strings must carry the rest. The *index bridge* is the centrepiece: $f(m) = \mathrm{ch}_2(T_m)$ exactly (fixed-point ledger $=$ McKay/Kronheimer intersection ledger), with the integral glue defect -78 by both routes; the level dials say $k = 1$ *geometrically* (lattice current count 240/0/0/0, embedding residual $(0, 360, 814, 1362)$, one scale $\Rightarrow$ one level — since Step 13 a *theorem*, no longer a heuristic). And an honest *refutation*: the clock-invariant modulus a_0 fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (weight mismatch $4 = |\mu_4|$) — "the theory brings its own GS axion as a_0" is false; instead the three $H^2(\mathrm{ALE})$ classes carry exactly the three twisted-sector Coxeter characters $\{i, -1, -i\}$ (the three twisted axion slots), and the bulk axion must come from the $O(-2)$ tower field itself. The preregistered kill ("inflow demands a level $\neq 1$") does *not* fire on the equivariant skeleton. [verification/v505_celestial_wp5e_beta_equivariant_ledger.py]

Step 12 — the exchange no-go and the contact-term criterion [E]/[C]. Could the three sphere axions of Step 11 cancel the rigid $32\, T_3$ residual by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? No — and the refusal is a theorem, not a fit. The $W(D_5)\times W(A_3)$-invariant vertex space on the glue Cartan is computed exactly (quadratics $= \mathrm{span}\{s_5, s_3\}$, dim 2; quartics $= \mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$, dim 5 — complete by Weyl nullspace arithmetic over all bidegrees), and the *product theorem* kills the mechanism wholesale: the product of any two invariant quadratics has identically zero T_5/T_3 content, so *every* exchange image — any number of axions, any selection rule, any couplings — is annihilated by Φ_{T_3} while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. The enumeration sharpens where it fails: the strict two-index $\mathbb{Z}_4$ rule collapses the sphere couplings entirely (only $m = 0$ survives); the generous twist-insertion channels give a rank-2 exchange matrix with $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$ and annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions, the second driven by the side discovery $K^{(0)} = -15\, K^{(2)}$ (the even-sector quadratics are parallel). Naturalness *dissolves*: ch_2-natural and Atiyah–Bott-weight couplings both carry certificate $(0, 0)$ against the required $(32, 72)$ — the ansatz space misses the target identically, scale- and coupling-independently. Controls: $SO(16)$ has *no* sphere partners and uncancelled T_5 (E_8 is doubly special); D_8 has no T_3 structure at all. The slot bijection of Step 11 is untouched, and the preregistered level-kill still does not fire — what dies is the exchange *realisation* of the pairing; the $32\,T_3$ burden moves to the twisted BCOV contact terms, with a *binary* criterion: the one-loop coefficient on $\mathbb{C}^2/\mathbb{Z}_4$ must come out exactly $(9,-30,-15,0,32)$, computed, not fitted. [`verification/v508_celestial_wp5e_gamma_pairing.py`]

Step 13 — one level is a theorem; the sector counter pins $k = 1$ [E]/[C]. The Costello–Paquette–Sharma level-from-flux mechanism (Burns holography: "level = flux quantum × Dynkin index of the boundary matter") is executed on the lockstep spheres. The CPS skeleton is replicated exactly (S^3 period $(2\pi i)^2 N$; exceptional-sphere Kähler flux $2\pi N$; boundary level magnitude $2N$ by the symplectic-boson contraction, $T(\text{vector}) = 1$), and the geometric side is *pinned, not postulated*: the Step-11 ch_2 ledger + integrality + unimodularity + effectivity force the pairing matrix $c_1(T_m).[\Sigma_i] = \delta_{mi}$ by complete enumeration (48 unimodular solutions, exactly 2 effective: identity + diagram flip), so the glue fluxes are $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ with *exactly one* quantum per charged root current. The consistency test then does real work: "level = total open-string flux" is *killed by the lockstep test itself* (three unequal numbers — F_i is the flavour multiplicity), while "level = flux per adjoint current" gives $(1, 1, 1)$, anchored by the lattice current count $(240, 0, 0, 0)$ and the embedding index 1. The ord-4-vs-level-1 tension resolves — the clock order counts *fractional flux sectors* ($\mu = 16 = \mathrm{ord}^2$, Lagrangian μ_4 glue, KLM $16 \to 1$) — and a new dial pins the value: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$, exactly *one* sector iff $k = 1$. The centrepiece is the *lockstep theorem*, lifting Step 11's "one scale ⇒ one level" from heuristic to theorem: clock invariance forces the four branch points into one μ_4 orbit, the period moduli $|\Pi_j| = \sqrt{2}\,t$ are equal on all three spheres, and $\det(A - 1) = -4$ leaves no invariant flux vector — $k_1 = k_2 = k_3$ *is a theorem* of clock invariance, with a new falsifier (the clock-forbidden family $Z^4 - Z$: zero lockstep chains over all 24 orderings) and an honest limit (uniform doubling keeps the lockstep: the theorem proves *equality*, the value 1 comes from the counters). Controls: $k = 2$ dies on the closure dials while the lockstep dial honestly does not fire; $SO(16)$ leaves two spheres flux-dark; A_2 admits no Lagrangian glue at all ($\mu = 12$ not a square). The CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ itself stays [C] (their geometry is the $\mathbb{C}^2$ blow-up, not the A_3 ALE); the type-I B-model back-reaction stays [O]. [`verification/v509_celestial_wp5e_eps2_level_from_flux.py`]

Step 14 — the full-tensor ledger: the collapse is real, and one cubic door opens [E]/[C]. Step 12's collapse was a *Cartan-restricted* statement — the named escape route (δ_2) asks whether the full adjoint tensor structure of $\mathfrak{e}_8$ (non-Cartan external legs, higher-arity vertices) reopens the pairing. The answer has two halves, both exact. *The collapse is confirmed*: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is semisimple with no $\mathfrak{u}(1)$ factor (class-0 roots 40+12, root span rank 8), the sectors factorise into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, and the *innerness theorem* decides everything wholesale: the grading element h lies in the Cartan *of* $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — bilinear invariants exist only at $j' = -j$ (dims 2/1/1/1) and *all* 15 non-neutral trilinear sector triples have Hom = 0: the sphere axions have no invariant partner operator at any arity ≤ 3, full-tensorially. *And one cubic door opens*: of the nine neutral trilinear structures (dims $(3, 2, 2, 1, 1)$) all cross-sector ones are pure brackets, but the *unique* totally symmetric survivor — the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir, so T_5 stays protected) — carries an exchange quartic $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ (two independent routes, Killing propagator $2h^\vee = 60$ from the roots) with $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$: Step 12's master kill covers *quadratic* vertices only and does not extend to cubic ones. The pairing nevertheless remains obstructed in the charge reading — $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3 against augmented rank 4 — but with the genuinely *weaker* certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\text{fix}}) = 72 - 8 = 64$; relaxed, it becomes *exactly solvable*: $A_{\text{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$. Number fences typed [C] only: $64 = \dim \mathfrak{g}_1 = 2^6$, $1920 = |W(D_5)| = 8 \cdot 240$. Controls: $SO(16)/D_8$ has no symmetric cubic, no odd sectors and no A_3 block (E_8/μ_4 doubly special); false $\mathfrak{g}_0$/sector assignments inflate or kill the tables. The burden on the δ_1 contact terms is thereby *sharpened*: they must supply either the $\psi = 64$ slice or the selection-rule relaxation — a burden since *discharged* by Step 17 (the declared KS measure supplies the $\psi = 64$ slice exactly, so the cubic d-channel is not needed) — and the δ_1 run, undecided at exploration level at the time (the strict holomorphic reading is refuted at the $\mathbb{Z}_2$/Eguchi–Hanson anchor — a method boundary, not a kill; the contact term is the *modular completion* of the Atiyah–Bott data, a Harvey–Moore-type τ-integral with the forced leading (T_5, T_3) ratio 4:3), has since been *decided* in Step 19 (kill under the derived measure, v518). [`verification/v511_celestial_wp5e_delta2_tensor_ledger.py`] The 1920 fence itself has since been stress-tested and *fails as a derivation claim*: the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 catalog expressions hit 1920, vs 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920; charge-*legal* flux pairings give 2048/1800 and miss; the twisted cubic index $32i$ clashes in class provenance with the true quartic 32) — the convention-stable **[E]** core is the factorisation $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$, the fence stays **[C]**, and the physical generation of the 32 stays **[O]** ([`verification/v513_celestial_dterm_nonderivation.py`], `CELEST.DTERM.NONDERIV.01`; no marker moves).

Step 15 — the bulk axion is a construction; $\tilde{\lambda} = 6$ is pinned three ways [E]/[C]. Step 11 refuted "a_0 is the GS axion" and left the bulk axion to the $O(-2)$ tower field *as a slot*; the ε_1 stage now *builds* it. On $\mathbb{PT}' = \mathrm{Tot}(O(1){\oplus}O(1) \to \mathbb{P}^1)$ the pushforward ledger $\pi_* O(-2)$ closes the Penrose accounting exactly ($(d{+}1)^2$ classes per degree = the exact wave-operator nullspaces, $d \leq 6$), and *equivariantly*: the clock acts on the fibre coordinates as $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$, the incidence relation forces the column weights $(+1, +1, -1, -1)$, and the bookkeeping closes *block by block* for all four characters. The character series are $P_0 = 1 + 3t^2 + 15t^4 + \dots$, $P_1 = P_3 = 2t + 8t^3 + \dots$, $P_2 = 6t^2 + 10t^4 + \dots$; the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — *the bulk axion survives the projection* — and the invariant fibre ring is the *hypersurface* $(1 - t^8)/((1 - t^4)^2(1 - t^2))$: $Z \in O(2)$, $X, Y \in O(4)$, one relation in degree 8 = the a_0 weight (the Step-1/Step-2 geometry re-emerging from the slot); the Step-11 bijection sharpens to a per-character statement (twisted minimal content $(2t, 6t^2, 2t)$ = the Coxeter eigenvalues, no degree-0 mode, $\det(A - 1) = -4$), and the graviton control separates a_0 cohomologically (the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$). The Green–Schwarz residue is then pinned *three* independent ways: Okubo ($\mathrm{Tr}_{\text{adj}} X^4 = (6\langle x, x\rangle)^2$ exactly on the 240 glue roots, $36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2| N_{\text{fam}}$), the measure chain on $\mathbb{PT}/\mathbb{Z}_4$ (the quotient factor $\mu = 1/4$ enters vertex^2/propagator/anomaly and cancels *exactly*; the wrong bookings are quantified and excluded: anomaly-only $\Rightarrow \tilde{\lambda} = 3$, missing propagator renormalisation $\Rightarrow \tilde{\lambda} = 12$), and the flux side (minimal CPS quantum $N = 1$, a single exchange channel exactly at $k = 1$, $(\kappa/c_3)^2 = 12$). The first honest *back-reaction* step follows in Gibbons–Hawking form: clock-invariant four-centre configurations form exactly two branches (four axis points = pure resolution, one free μ_4 orbit = pure deformation), the free orbit re-derives the Step-2 family ($\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, $a_0 = -z_0^4$, monodromy = one clock step), the charge-4 GH point is exactly the $S^3/\mathbb{Z}_4$ seam boundary, the periods $\Pi_j = 4\pi t_0 (i - 1)(1, i, i^2)$ confirm the Step-13 lockstep *from geometry* ($\Pi \cdot A = i\Pi$), the source ledger carries charge $4 = |\mu_4|$, and the honest sharpening: the Ricci-flat ALE has asymptotic log coefficient *exactly zero* (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), with the multipole selection rule $m \equiv 0 \bmod 4$ storing $-a_0$ in the first symmetry-breaking $(4, \pm 4)$ harmonic. Controls: $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational — perfect squares in the Deligne series only $\{9, 36\} = \{\mathfrak{sl}_3, \mathfrak{e}_8\}$), $\mathrm{diag}(i, i)$ (Veronese cone, no hypersurface, the bijection collapses), $k = 2$ (three exchange channels). What stays honest: the chain is conditional on Costello's flat-$\mathbb{PT}$ matching **[C]**, and the *quantised* BCOV coefficient on $\mathbb{PT}/\mathbb{Z}_4$ plus the twisted channels ($32\,T_3$) stay **[O]** — preregistered as the M1–M3 back-reaction milestones (the A_3 Ω_N, the twisted KS measure, the a_0 uplift), each with success and kill criteria; all three are executed in Steps 16–18. [`verification/v514_celestial_wp5e_eps1_axion_slot.py`]

Step 16 — the back-reacted Ω_N: closed form, integral periods, forced charge 4 [E]/[C]. The M1 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS criterion met and neither KILL fired. The residue 2-form of the family is *derived*, not assumed: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX{\wedge}dY{\wedge}dZ$ and pull back to $\omega_2 = dX{\wedge}dZ/X$; at $a_0 = 0$ the orbifold-cover pullback is $\omega_2 = 4\,dz_1{\wedge}dz_2 = |\mu_4|{\times}$(flat form) — the residue normalisation itself *carries* the source charge 4 — and the clock multiplies ω_2 by i (the Step-1 det $= i$ replicated). The four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ close the twistor family in one line, $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1 - \lambda^4)^2$ ($e_1 = e_3 = 0$ identically; the seam fibre $\lambda = 0$ is exactly the Step-2 family, the $a_2 \in O(4)$ slot opens only off-seam), with the CY-compatible clock lift $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$. The periods reduce generically ($\int_{S_j} \omega_2 = 2\pi i(q_{j+1} - q_j)$ for arbitrary profile and path), giving the seam-fibre lockstep vector $2\pi i\, t_0(i-1)(1, i, i^2)$ with the exact covariance $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for *all* λ, and the 12 collision nodes — honest conifold points (Hessian det $= 512t_0^6$, quadric rank 4) — sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$, the Step-1 $\mathbb{Z}_8$ bridge), in clock orbits 4+4+4. The back-reacted 3-form is then *closed form*: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with the CPS/Bochner–Martinelli kernel on the four centre twistor lines ($\int_{S^3} K = (2\pi i)^2$ exactly, $dK = 0$ off-source, scale degree 0); the undeformed Ω_0 carries *zero* quantised 3-flux (all flux is sourced), every 3-cycle period is $(2\pi i)^2$-integral, the flux vector $N(1, 1, 1, 1)$ is forced uniform *twice over* (clock orbit + the K_4 connectivity of the four lines), and the lens fundamental domain *forces* $N \in 4\mathbb{Z}$: the source charge $4 = |\mu_4|$ from quotient large-gauge quantisation, minimal invariant charge $\leftrightarrow$ the CPS quantum $N = 1 \leftrightarrow k = 1$ (Step 13). The honest fence has teeth: on the clock-forbidden family $Z^4 - Z$ the $(2\pi i)^2$ quantisation *also* holds — integrality alone is *not* the discriminator; the discriminator is the lockstep phase/modulus structure (0/24 orderings vs 8/24, node support on twelfth roots instead of $\mathbb{Z}_8$) together with the clock forcing of equal fluxes. Anchors and controls: EH/$\mathbb{Z}_2$ forces charge $2 = |\mathbb{Z}_2|$ from the same machinery, the CPS flat patch allows every integer N (the unconstrained anchor), diag(i, i) collapses at step zero ($\Omega_0 \to -\Omega_0$), and fractional charges $N = 1, 2, 3$ are excluded. Global kernel patching, the Hitchin small resolution and the CPS brane dictionary stay [C]; M2–M3 are executed in Steps 17–18, and the quantised BCOV coefficient stays [O] only in its measure question (the δ_1 chain itself is decided in Step 19). [`verification/v515_celestial_wp5e_m1_omega_n.py`]

Step 17 — the twisted KS measure: every sector the same Okubo square [E]/[C]. The M2 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording met *on the declared completion measure* and the KILL (a leftover independent quartic in *any* sector) not fired. The measure ansatz is declared before computing, and honestly typed: Step 12 killed every *exchange* realisation, and the δ_1 exploration identified the twisted contact term as the *modular completion* of the very Atiyah–Bott data that produced A_{fix}; M2 declares the completion contact term per twisted box channel, $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ — in the channel with the g^j zero-mode normalisation $1/\det_j$, the twisted-sector *loop* contributes the *unphased* sector trace, of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part. Everything downstream of the declaration is exact arithmetic, and it locks in place with no dial to turn: per glue class the contact weights obey the *completion-weight identity* $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\,\text{ch}_2(T_m)$ ($h_m = m(4-m)/8$ the Step-9 sector Casimir energies, $\text{ch}_2(T_m)$ the McKay/Kronheimer charges of the Step-11 index bridge): the three sphere axions pair through their *own* ch_2 charges — an exact identity, *no free scale, no fit.* The locks are parameter-free: $T_5 = 0$ for *any* scale c (the ch_2 pattern alone), the ratio $h_2{:}h_1 = 4{:}3$ *reproduces* the δ_1-forced leading ratio, and the T_3 budget fixes $c = 4 = |\mu_4|$ uniquely (leading system det 256). The success table is then uniform: $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x\rangle^2/\det_j$ — a *perfect Okubo square* ($T_5 = T_3 = 0$, $\tilde{\lambda}^2 = 36$) in *every* twisted channel; the total $A_{\text{fix}} + \text{contact} = 45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2 =$ Dedekind $\times$ Okubo is exactly the *unique* quartic-free weighting of the Step-11 rigidity theorem. Both Step-12 certificates are killed ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$), and the Step-14 slice $\psi = 64$ is *supplied exactly* — the cubic d-channel is *not needed* (c_d free $= 0$). The remainder sits inside the exchange span (the bulk axion cancels every channel with the *same* $\tilde{\lambda}^2 = 36$, the twisted propagator carrying the zero-mode factor $1/\det_j$), and the Step-15 measure chain stays μ-blind. Controls with teeth: wrong scale $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$; the $h_1{\leftrightarrow}h_2$ shuffle breaks T_5 *and* T_3; $SO(16)$ keeps $T_5 = 20/T_3 = -40$ — the KILL *fires* there (E_8 doubly special); diag(i, i) degenerates to the $\mathbb{Z}_2$ target; and the $\mathbb{Z}_2$/Eguchi–Hanson anchor *passes* with the same mechanism ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36\langle x, x\rangle^2$, coupling scale $2 = |\mathbb{Z}_2| =$ centre count). The mandatory fence: the completion reading (loop = unphased sector trace with the same zero-mode normalisation) is *declared* within this step, supported by the δ_1 modular-completion finding, *not* derived from the BCOV integral here — the δ_1 chain has since been decided in Step 19 (v518: the *derived* chiral measure fails all three testers), and the declared-vs-derived measure question has since been decided in Step 21 (v520: the completion reading is now *derived at probe level* from two independent constructive sources — F-independence + the Quillen pairing — under the typed premises TP-1..TP-4); the residual **[O]** is the constructive BCOV derivation of the w_m normalisation itself. [`verification/v516_celestial_wp5e_m2_twisted_ks_measure.py`]

Step 18 — the a_0 uplift: four coupled centre-count scales [E]/[C]. The M3 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording ("a log-type correction whose coefficient is tied to the source charge $4 = |\mu_4|$") met and the KILL (decoupling from the centre count) not fired. The uplift object is the generalized-Legendre-transform kernel per centre, $G(\eta_p) = \eta_p \log \eta_p$ (the Lindström–Roček / Ivanov–Roček dictionary, typed [C]); its derivative kernel $\chi = \sum_p \log \eta_p = \log P_4$ is the *log of the Step-16 family polynomial.* The bridge is well-typed and exact: the $O(2)$ section is a *null* coordinate ($\Delta_{3d} G(\eta_p) = 0$ identically for an *arbitrary* kernel), and the residue identity $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ holds exactly — the Gibbons–Hawking potential $V = \sum_p 1/r_p$ *is* the x-derivative of the contour transform of χ (flux -4π per centre, source charge $4 = |\mu_4|$). The a_0 correction then arrives on *four coupled centre-count scales*: (i) the asymptotic kernel $\log \chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$ carries the log coefficient $4 = |\mu_4|$ = centre count (the CPS "$N \log$" analogue at kernel level); at the clock-fixed seam fibre the a_2 slot closes and the first correction is *exactly* a_0/η^4 = the $(4, \pm 4)$ multipole in kernel clothes, with exact m-grading (the $m = \pm 4$ pieces are *pure* a_0, the $m = 0$ piece is φ_0-free) and *no* $\log \times$power terms — the Step-15 honest zero is preserved; (ii) the once-integrated GLT tower $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + \dots$ obeys the selection rule $p_n = 0$ unless $n \equiv 0 \bmod 4$ with $p_{4k} = 4(-a_0)^k$ — every correction carries the prefactor 4; (iii) the exceptional-locus log $\chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$ (t_0-coefficient $4 = |\mu_4|$, clock phase $4i$; the modulus response equals 1 *at* the locus and falls off as a power law at infinity — the log lives where Step 15 said it must); (iv) the period response is $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$ *uniformly* (the periods are linear in t_0: lockstep and the phase ratios $(1, i, i^2)$ are preserved exactly at first order), and integrating the shift around the a_0 circle gives the monodromy $e^{2\pi i/4} = i =$ *one Coxeter clock step* — the Step-2 monodromy reproduced from perturbation theory; the $\mathbb{Z}_8$ node support is a_0-rigid, and the $(2\pi i)^2$ fluxes are topological (they cannot move). Controls: the $(4, 0)$ multipole is clock-invariant and breaks nothing; the $\mathbb{Z}_2$/Eguchi–Hanson analogue reads $2 = |\mathbb{Z}_2|$ on *every* dial (amplitude, kernel log, exceptional log, period response $1/2$, monodromy -1); wrong centre counts $k = 3, 5$ move the coefficient to k with modulus slots $O(6)/O(10) \neq O(8)$ — the observable *moves with the centre count*; the clock-forbidden family fails both dials ($p_3 = 3 \neq 0$ breaks the selection rule, $e_4 = 0$ kills the a_0-log). What stays **[O]**: the full nonlinear Kähler potential of the resolved A_3 ALE (no closed form exists). [`verification/v517_celestial_wp5e_m3_a0_uplift.py`]

Step 19 — the derived measure decides — and disagrees with the declared one [E]/[C]. The δ_1 question that Step 17 left open (*derive* the completion reading from the Harvey–Moore/BCOV τ-integral) has now been executed as a three-stage probe chain and consolidated: instead of declaring or scanning a measure, the measure is *solved for* from blockwise $SL(2, \mathbb{Z})$ covariance of the dressed 16-component Weil system. Four exact results. (i) *The completion closes*: the discriminant module of $D_5 \oplus A_3$ is $\mathbb{Z}_4 \times \mathbb{Z}_4$ with $q = (5x^2 + 3y^2)/8$; the Gauss sums are $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature 0 mod 8 = rank E_8), the Weil relations hold exactly in $\mathbb{Q}(\zeta_8)$, the diagonal and anti-diagonal Lagrangians are the two E_8 gluings, and the 16-vector theta S-covariance drops the naive 4-character-rule residual from $2.91 = O(1)$ to $\sim 10^{-39}$. (ii) *The obstruction is a finite μ_4 character, identified*: the dressed gauge blocks carry a multiplier system whose $SL(2, \mathbb{Z})$ relation defects are $(1, 1, 1)$ exactly on all 15 sector pairs — a *character* of the orbit stabilisers $\Gamma_1(4)/\Gamma_0(2)^{\pm}$, not a genuine 2-cocycle — and on $\Gamma_1(4)$ it is $\lambda(\gamma) = i^{2B + C/4}$. (iii) *The cancellation exists and is the twisted fibre block*: $G[a, b] = f_1 f_3$ is an exact identity (the twisted $\mathbb{C}^2$ gauge block *is* the weight-$(1, 3)$ axion pair), the T-fix mechanism is one line of exact phase arithmetic ($(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$), and the exact cancellation table reads: bare $\to$ order 4, the three sphere axions $f_1 f_2 f_3 \to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ — *only* the twistor-fibre content cancels the μ_4 system, with the strict solutions exactly χ_4 (dims $(3, 3)$) and χ_{10} (dims $(1, 1)$), certified on the dressed functions. (iv) *All three preregistered testers fail*: under both derived solutions the integral misses $T_5 = 0$ (fractions 0.5/0.83), misses the forced 4:3 leading ratio ($W_{13} = 0$) and misses $-A_{\text{fix}}$/the $\psi = -64N$ slice (spreads 1.05/1.47), and the honest (N_1, N_2) orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the $\mathbb{Z}_2$/Eguchi–Hanson anchor (residual order 2, cancelled to 1 by the same mechanism), $SO(16)$ (the order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences [C]: the $f_1 f_3$ = KS-weight identification (the deck weights $(1,3)$ are the twistor fibre weights forced by the Step-15 incidence ledger — an exact match, *not* a complete BCOV derivation) and the kernel-family convention. TENSION, stated honestly: the *declared* completion reading (Step 17) delivers the $\psi = 64$ slice and cancels the $32\,T_3$; the *derived* chiral measure (this step) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure. Neither result is hidden behind the other — and Step 21 has since decided the question at probe level in favour of the declared reading, sharpening this kill. [`verification/v518_celestial_wp5e_delta1_derived_kill.py`]

Step 20 — the real structure exists — and free reflection positivity picks the same family [E]/[O]. The α stage of the OS twistor bridge `WOIT.OS.TWISTOR.01` is executed (`WOIT.THETA.FREE.01`). The classification is complete: exactly *two* families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$) satisfies $\Theta\rho\Theta = \rho^{-1}$ *exactly* with $\Theta^2 = +1$ (-1 is *impossible*: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — the clock-inverting family is Kramers-free), inverts the deck, and reflects the seam circle with two cut points; family A ($z \mapsto \mu/\bar{z}$) centralises the clock projectively and *never* inverts it, with $\Theta^2 = \pm 1$ per μ. The role separation is sharp: family A *defines* the euclidean section — Woit's ρ_{tw} ($\rho_{\text{tw}}^2 = -1$, no real points) is replicated exactly on $\mathbb{C}^4$ and is family A globally — while family D *reflects* it (the OS conjugation; σ_{std} with real points $\mathbb{RP}^3$). Mark compatibility pins $\mu \in \mu_4$ (a 4-element torsor with $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu}$: two clock orbits); the $\mathbb{Z}_8$ spin plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); and in exact Cl(16) the Fock implementer $\Theta_{\text{Fock}} = U_r \circ K$ has $\Theta_{\text{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the Fock clock tower ($V \mapsto 4096\,V^{-1}$, exact scalar) and normalises the deck — while the *deck*-induced candidate has $\Theta_t^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$: the Step-2/v510 split/nonsplit dichotomy *is* the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy, so the deck does *not* furnish the OS Θ — the seam-circle *reflection* does. Free reflection positivity then holds for exactly that pinned Θ: on the bond cut (marks at the bond midpoints of the 16-Majorana circle) the one-particle Gram is positive definite ($(8,0,0)$, min eigenvalue 1.888×10^{-3} at 40 digits), the even deg ≤ 2 sector is $(29,0,0)$, and at $N = 8$ the *complete* half-sided algebra is RP with no degree truncation; the twist $\eta = +i$ is forced ($\eta = 1$ non-Hermitian); the cut *through* the sites fails exactly ($\det = 0$, inertia $(3,3,1)$ — a lattice-placement artifact, the continuum Cauchy–Stieltjes control is strictly positive), and the clock-centralising family-A structure fails RP *structurally* ($(4,4,0)$): RP and $\Theta\rho\Theta = \rho^{-1}$ select the *same* family. Bonus: the anti-chiral state flips the odd sector to negative definite — an exact free-level shadow of kill test 3, living entirely in the fermionic sector. **Kill test 1 of the contract does *not* fire at the free/equivariant level and stays live on the interacting algebra**; the interacting algebra, gauge-fixed RP, OS reconstruction, chirality and μ_4 incidence remain open contract work (the β/γ milestones named in the contract). No marker moves. [`verification/v519_woit_theta_rp_free.py`]

Step 21 — the measure decision: single-valuedness is derived, the completion reading wins [E]/[C]. The declared-vs-derived tension that Steps 17 and 19 left as the named open face is now *decided at probe level* (ERFOLG-A on the preregistered decision layer; the δ_{1e}/δ_{1f} consolidation). The centrepiece is exact linear algebra: the joint invariant subspace $\{v : \rho(T)v = v,\ \rho(S)v = v\}$ of the dressed 16-component Weil system has $\dim_{\mathbb{Q}} = 8 = 4 \times 2$ and *every* kernel vector lies in the $\mathbb{Q}(\zeta_8)$-span of the two Lagrangian gluings $\{e_H, e_{H'}\}$ — both with theta series E_4, both carrying the *unphased* sector trace $Q^{(0)} = (36, 72, 36, 0, 0)$. Single-valuedness of the physical one-loop lattice factor — Step 19's typed premise — is then *derived* from two independent constructive sources rather than postulated. *Source 1 (the fundamental-domain lemma):* every strict chiral closure at physical weight carries a nontrivial character ($\chi_4(T) = e(1/3)$, $\chi_{10}(T) = e(5/6)$, exact; zero strict trivial-character solutions across all eleven dressings); the canonical column-matched member's total block sum obeys $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$ *pointwise* (certificate 3.9×10^{-16}, nonvacuous $|G| \ge 59.6$), and an exact change of variables turns fundamental-domain independence into $\int = 0$ for *every* chiral integrand: the Step-

19 route was never a well-defined nonvanishing moduli integral — the kill is *sharpened*, its unfolded values are convention bookkeeping. *Source 2 (the Quillen pairing):* BCOV's F_1 pairs hol × conj(hol), and the doubled transports satisfy $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs ($|\chi_4|^2 = 1$ exactly, while the one-sided square $\chi_4^2 = e(2/3) \neq 1$ keeps defects on 15/15 pairs); the doubled 256-dimensional system closes *strictly* at trivial character and physical weight (nullspaces $(28, 17) \geq 11$), contains the unphased diagonal *and* the physical columns $w_b \otimes \bar{w}_b$, and the wrong pairing hol ⊗ hol is *empty* $(0, 0)$. The consistency circle then closes: the forced unphased numerator, inserted into the Step-19 scaffold, reproduces the Step-17 Okubo squares *levelwise* ($\sum_m V \propto \langle x, x\rangle^2$ exactly on every level $n \leq 8$; J-weighted spreads $\sim 5 \times 10^{-41}$, zero negative cells) with $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the J-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$. The declared reading also wins the two decision-layer discriminators: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is contained *exactly* in the χ_4 family ($\sigma \sim 1.8 \times 10^{-16}$, one scalar per orbit — canonicalising the (N_1, N_2) freedom) yet the canonical member still fails every tester; and at the $\mathbb{Z}_2$/Eguchi–Hanson anchor only the declared reading hits ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36$, $\psi(A_2) = -8$, scale tooth $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$ unique) while all three derived instantiations fail. Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source $\psi = 64$), TP-3 (the Step-19 kernel convention), TP-4 (the Quillen structure of BCOV F_1). What stayed [O] — the constructive BCOV derivation of the w_m normalisation itself — is since executed by Step 24, narrowing the [O] to the global BCOV integral beyond the fibre zero-mode factor. No marker moves. [`verification/v520_celestial_wp5e_measure_decision.py`]

Step 22 — free OS positivity does not see the bit: the eighth side-blind test [E]/[C]. Could the Step-20 machinery *derive* the Step-2 alignment bit? The hypothesis — free RP or Θ existence breaks for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ — is now decided *negative*, exactly as preregistered (KILL branch). The census is exact and symbolic: for *every* δ the two mark-swapping reflections $\Theta_{\pm e^{i\delta}}$ exist (all 20 cut/mark incidence solvesets empty), while mark-*fixing* reflections exist iff $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$). Free RP is δ-blind in the strongest sense: the bond-cut Gram inertias are $(8, 0, 0)$ PD for all tested δ with the full sorted spectra *identical to* 40 *digits* (deviation 0.0); the $N = 16$ odd-m failure is a lattice-placement artifact (at $N = 32$ all $m = 1..7$ have mark-compatible bond axes with PD inertia $(16, 0, 0)$); the continuum OS kernel in axis-adapted coordinates is *exactly* $1/\sin((s+t)/2)$ — the axis position drops out identically — and the v510/v512 counterwitness passes free RP. Θ existence is δ-blind too ($\Theta^2 = +1$, deck normalisation, Fock implementability $U^2 = +2^7/2^8 > 0$ for every δ); the *only* δ-sensitive clause of the Step-20 battery ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but *not formulable* (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it presupposes the bit instead of deriving it. The battery has teeth at every δ (family A fails RP structurally $(4, 4, 0)$; the chirality–η pinning persists $(0, 8, 0)$): RP separates the real-structure *families*, never the *sides*. Structure gained [C]: the group generated by the admissible OS reflections is V_4 for every $\delta < \pi/2$ and closes to D_4 exactly at the clock point ($\Theta_i \circ \Theta_1 =$ the clock) — face #13 of the Step-2 equivalence web (12 → 13). The free RP/Θ battery is hereby the *eighth* side-blind test on the v512 scoreboard (7 → 8); the honest [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting $\mathcal{A}_{\text{hol}}$) — the mark-decorated half of that gap is since decided side-blind too (Step 26: the *ninth* test, 8 → 9; the free-plus-twist class is exhausted, the residual gap narrows to the genuinely interacting $\mathcal{A}_{\text{hol}}$) — the alignment bit remains genuine discrete input. No marker moves. [`verification/v521_seam_bit_rp_blind.py`]

Step 23 — the clock is time-like, GSO is the gauge datum [E]/[C]/[O]. The β_1 stage of the OS twistor bridge is executed as a *typed* result (verdict UNDECIDED per the frozen preregistration — the preregistered reading was the wrong operationalisation, and the module documents why, including its own first frozen run at 6/13 and a preregistration combinatorics error 30 vs 32). The mechanism: the one-step clock insertion $T_1 = \langle\theta(e_a), \alpha_S(e_b)\rangle$ violates Hermiticity *exactly* (witness entry $-i/(8\sin(5\pi/16))$ against $+i/(8\sin(5\pi/16))$; the pairing is complex-symmetric — 745 matching, 96 anti-matching, 0 violations — and "OS-symmetric" is strictly weaker than Hermitian), so "positivity after gauge fixing" is not well-posed for the clock reading. The typing is a census: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), none commutes — the μ_4 clock *is* Woit's euclidean rotation, and the gaugeable (θ-compatible) part of its $\mathbb{Z}_8$ Fock tower is exactly the 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (the free-fermion shadow of the E_8 glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor). Under that corrected typing gauge-fixed RP *holds exactly*: $(29, 0, 0)$ PD at $N = 16$ (deg ≤ 2, min eigenvalue 1.78×10^{-6}), $(8, 0, 0)$ PD on the complete $N = 8$ half algebra; the site-cut defect survives gauge fixing ($(7, 9, 6)$ — the bond placement is gauge-invariant information); family A stays indefinite ($(17, 12, 0)$); the Ramond control forces the NS/$\mathbb{Z}_8$ tower. Kill test (2)'s free shadow does *not* fire, kill test (1) stays discharged also gauge-invariantly; both stay live on $\mathcal{A}_{\text{hol}}$, and the clock-equivariant statement is re-routed through β_2 (OS quotient first, then the clock as reconstructed transfer operator — contract precision (iii)). No marker moves; `WOIT.OS.TWISTOR.01` stays [O]. [`verification/v522_woit_beta1_gso_gauge.py`]

Step 24 — the w_m normalisation derived: the fixed-point factor is computed, not declared [E]/[C]. The named remaining target of Steps 17–21 — the *constructive* BCOV derivation of the completion normalisation $1/\det_j$ — is now executed (verdict ERFOLG per the frozen preregistration), from three independent sources. *Route (i), Atiyah–Bott:* the equivariant mode ledger of the twistor fibre $\mathbb{C}^2$ — $c_n = \sum_{a+b=n} i^{j(a-b)}$ — equals the closed form $1/((1-q\,i^j)(1-q\,i^{-j}))$ exactly in $\mathbb{Q}(i)$ at every order $n \leq 120$; the closed form is *regular* at $q = 1$ (the twisted sectors carry no volume pole) with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, $\det_j = \det(1 - g^j) = (2, 4, 2)$; the $(C, 2)$ Cesàro means converge in exact arithmetic; and the fixed-point factor *splits off exactly at every truncation level* $d \in \{7, 12, 25\}$, for the phased (skeleton) *and* the Step-21-forced unphased (completion) trace alike — the Abel-limit contact$_j = (Q^{(0)} - Q^{(j)})/\det_j$ is the Step-17 contact vector *computed*, not declared. *Route (ii), zeta/Quillen:* the spectral determinant of the g^j-twisted circle Laplacian is $4\sin^2(\pi j/4) = \det_j$ exactly (Lerch + reflection formula symbolically; Hurwitz certificates $\sim 10^{-41}$); the Quillen split gives $\det\Delta = \det_j^2$ with $|\det\bar\partial| = \det_j$ and the real *positive* holomorphic section unique via the $SU(2)$ conjugation pairing; and the δ_{1f} modular block $f_1 f_3$ at the $(0, b)$ nodes carries the exact constant term $1/\det_b$ (deviations $\leq 5.2 \times 10^{-28}$) — the Step-19/21 transport machinery carries the fixed-point weight verbatim. *Route (iii), the consistency chain:* the derived weight reproduces the Step-17 chain number by number — $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\,\mathrm{ch}_2(T_m)$, $T_5 = 0$ at every scale, $h_2 : h_1 = 4 : 3$, the T_3 budget $32 - 8c$ forcing $c = 4 = |\mu_4|$ uniquely, per-sector Okubo squares $(18, 9, 18)\langle x, x\rangle^2$, total $45\langle x, x\rangle^2$, Φ/ψ certificates ± 64. The negative controls have teeth: $1/\det^2$ leaves $(T_5, T_3) = (2, 12)$ and contradicts the Quillen split; $1/|1 - i^j|$ leaves $\mathbb{Q}$ ($w_1'' = 1 + \sqrt{2}$); the $\mathbb{Z}_2$/Eguchi–Hanson anchor *passes* with the same derivation ($\det = 4$, $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$, $c = 2 = |\mathbb{Z}_2|$); the $SO(16)/D_8$ kill fires; $\mathrm{diag}(i, i)$ breaks the zeta = AB identity itself; and on $\mathbb{C}^2/\mathbb{Z}_k$ for $k = 3, 5$ the weights wander correctly ($w_m = m(k-m)/2 = k\,h_m$ exactly in cyclotomic arithmetic) — only $k = 4 = |\mu_4|$ carries the E_8-glue chain. Typed premises [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation *is* the definition of the equivariant character), TP-Q (the Quillen structure of BCOV F_1 = Step-21 TP-4), TP-NUM (the unphased numerator = the Step-21 decision layer), TP-CH (the insertion channel computes the box-channel weight). Nothing in w_m is declared any more at this level; what stays [O] is the *global* BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor (and the resolved-ALE route not built). No marker moves. [`verification/v523_celestial_wp5e_wm_derived.py`]

Step 25 — the OS quotient made explicit: the clock gets its spectral calculus [E]/[C]. The β_2 stage of the OS twistor bridge is executed (verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii)). $\mathcal{H}_{\text{phys}}$ is *explicit and nondegenerate*: at $N = 16$ (deg ≤ 2) the 37×37 bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia $(37, 0, 0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits) and null space $\{0\}$, $\dim 37 = 29\oplus 8$; at $N = 8$ the *complete* half algebra gives $(16, 0, 0)$ PD with $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*: τ_k is exactly Hermitian on every shrinking domain ($k = 1..7$), the chain identities are exact, the vacuum is fixed, and the positivity pattern *is* the Step-20 site/bond dichotomy — even steps are PSD via the exact square identity $T(2j) = A_j^* A_j$, odd steps are indefinite with exactly zero one-particle diagonal (the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). The clock is the quarter turn $T^{N/4}$: positive self-adjoint on its domain ($(11,0,0)$, min 6.7×10^{-5}; compressed spectrum $[0.0195, 1.6414]$ with vacuum eigenvalue exactly 1), with certified spectral projections ($\sim 10^{-40}$) — exactly the calculus the non-Hermitian pre-quotient average of Step 23 could not have — and at $N = 8$ the compressed clock spectrum is *exactly* $\{1, \sqrt{2}-1\} = \{1, 1/\delta_S\}$ in both parity sectors: the silver axes of the Step-20 μ_4 torsor return as the clock eigenvalue; the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law — per precision (iii) the [C]-operationalisation of "the clock acting unitarily". The Step-23 non-Hermiticity is *resolved*: the census $(745, 96, 0)$ is reproduced and every anti-matching entry is a wrap overlap (every overlap-free entry matches) — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: no contraction on the compact circle (the edge mode expands by $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly; $\det(G - \tau_4) < 0$). GSO/Θ: the grading survives, the *perpendicular* torsor mirror descends anti-unitarily with $\Theta^2_{\mathrm{phys}} = +1$ on every sector (the quotient is Kramers-free), and $\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly (the torsor closes onto the deck). Controls: the site cut is indefinite (the contract kill branch fires for the site placement), family A admits no quotient, the anti-chiral state flips to $(8,8,0)$ — quotient existence itself selects the chiral orientation. Kill tests (1)/(2) strengthened, (3) shadow sharpened, none fires; all seven stay live on $\mathcal{A}_{\mathrm{hol}}$; β_3 is next. Transparency: run 1 scored 18/20 — a sympy simplify failure on a *true* $\pi/16$ product formula, fixed by a Laurent-polynomial certificate; no criterion changed. No marker moves; `WOIT.OS.TWISTOR.01` stays [O]. [`verification/v524_woit_beta2_os_quotient.py`]

Step 26 — the twist-state kill: the ninth side-blind test, and the free-plus-twist class is exhausted [E]/[O]. Step 22 left one named door: marks decorating the *state* (defect/twist insertions). That door is now decided *negative*, exactly as preregistered (KILL) — the *ninth* side-blind test on the v512 scoreboard ($8 \to 9$). Every well-defined mark-decorated state functional in the free-plus-twist class is RP-blind on the always-existing cuts along the whole $M(\delta)$ ladder: the σ-gauge arc decoration satisfies $\sigma \circ r = \sigma$ on the swap axes, so the twisted Gram is a diagonal unitary congruence — spectra *identical* to 40 digits; the Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$ is well-defined on the $N = 32$ ladder ($\omega(D)$ real nonzero; the $N = 16$ odd-m degeneration is the checkerboard artifact) and its RP spectra are the *first* free-class data of the programme that depend on δ at all (pairwise up to 1.05) — yet the *inertia* stays $(16,0,0)$ PD with the same $\eta = +i$ for every member $m = 1..7$: the decoration sees δ, the positivity does not; the μ_4 defect at $\beta = \pi/2$ is pure plane gauge, and the genuine Bogoliubov defect at $\beta = \pi/4$ — where Θ-compatibility selects exactly 2 of 16 sign patterns (the crossed pattern forced by the NS wrap signs) and the winner is genuinely non-monomial — is still PD with spectra again identical. The structure theorem behind the blindness is exact: the defect planes never straddle a cut bond, so every Θ-compatible *quasi-free* mark decoration is RP-spectrally *invisible* (orthogonal congruence). The sub-results sharpen the verdict: the defect energy $E_{\mathrm{def}} = 1.2752872$ is exactly δ-independent (spread 4×10^{-40}); the 4-twist Casimir $\ln |\omega(D)|$ *measures* δ smoothly but is exactly mirror-symmetric under $m \leftrightarrow 8 - m$ — the $\delta \leftrightarrow \pi - \delta$ equivariance makes $\pi/2$ the symmetric point of every invariant readout: a family gauge, never a side selector; the twisted parity $\langle \Gamma \rangle_{\mathrm{tw}} = +1$ for every member. The harvest: two genuinely twist-visible structures — the η flip $+i \to -i$ on the mark-fixing axes ($\sigma \circ r = -\sigma$) and the twist frustration of the mark-fixing mirror ($(4,4,0)$ indefinite for both η: the lattice shadow of the Ising twist field's mirror-oddness) — both live *only* on the fixing axes existing iff $\delta = \pi/2$: they presuppose the bit (the v512 facet class). Controls: the twist-free limit reproduces Steps 20/22 exactly; the site cut keeps failing; the v512 counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state-level selector can exclude it; a single twist has $\omega(D) = 0$ identically (twists exist only in mark pairs). The free-plus-twist class — everything Wick-computable — is hereby *exhausted*: what could still see the bit is only a genuinely *interacting* $\mathcal{A}_{\mathrm{hol}}$ whose OS data are not Pfaffian-reducible to the chiral vacuum — and that door is since probed by Step 30 (the tenth side-blind test $9 \to 10$, with Kill-Test 2 firing at toy level under a typed fence) — and answered *positively at selector level* by Step 31: reflection positivity, imposed on the interacting mark family, keeps exactly one member alive, $\delta = \pi/2$. The alignment bit remains genuine discrete input. No marker moves. [`verification/v525_seam_bit_twist_blind.py`]

Step 27 — the thermal seam: KMS closes onto the Nariai/BH chain as the third leg of c_3 [E]/[C]. The free OS quotient of Step 25 is not merely thermal in dimension (4^2): its temperature is *measured*, not assumed. The circle kernel $C(d) = (2/N)/\sin(\pi d/N)$ admits exactly one finite thermal-mode representation with detailed balance — at $N = 8$ one real frequency $\omega = \operatorname{arcsinh}(2^{-3/4})$ with weight ratio $\kappa = q^{-8}$ exactly ($\beta = 8 = N$; Q-polynomial identity $Q^2 - (2 + \sqrt{2})Q + 1 = 0$), at $N = 16$ both mode pairs real with pairing exponent $(N/2) + 1 = 9$ ($\beta = 16 = N$), and wrong candidates $\{N/2, N/4, 2N\}$ are excluded by quantified gaps. In angle units (one clock step $= 2\pi/N$, the v519 pin) one obtains $\beta_{\text{angle}} = 2\pi$ exact, hence $T_{\text{seam}} = 1/(2\pi) = 4c_3$ — N-independent, and exactly the Bisognano–Wichmann/Hawking modular normalisation that v208 carries into the horizon sector. The modular Hamiltonian is closed ($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; $\varepsilon_i = 2\ln\cot(\theta_i/2)$; Tomita KMS-$\beta = 1$ on all 256 monomial pairs). The anti-numerology audit selects the class-1 identity $r = 1$: temperature joins geometry and anomaly as the *third leg* of c_3. The Hawking/Nariai chain then reads in the same normalisation: $T_H = c_3/M = 1/(8\pi M)$ (the axiom *is* the Hawking coefficient, now from the seam KMS), SdS $1/(4\pi) = 2c_3$, Nariai $T_N = 4c_3\sqrt{\Lambda}$, and the v104 cross-link $2H/T_N = 4\pi = 1/(2c_3)$ exact. Entropy confirms the chiral $c = 1/2$ log-law ($\Delta S/\ln 2 \to 1/6$) but the v129 horizon fractions $\{1/3, 2/3\}$ and the v190 floor do *not* transfer — the temperature bridge closes, the entropy-fraction bridge honestly does not. Typed non-gating: the naive clock reading $\{1, \sqrt{2} - 1\} = e^{-\Delta K}$ fails; the exact replacement is $\sqrt{2} - 1 = C(3)/C(1) = 1/(2\cosh 2\omega - 1)$ — the silver eigenvalue is a thermal kernel ratio (Step 28). Non-compact control: the Stieltjes half-line has $\kappa = 0$, $\beta = \infty$, $T = 0$ — the temperature hangs exactly on compactness of the euclidean circle. [C]-fence: the reading "seam euclidean circle = thermal circle of the reconstructed horizon dynamics". No marker moves. [`verification/v526_seam_thermal_kms_nariai_bridge.py`]

Step 28 — the silver eigenvalue explained and demystified [E]. The compressed clock spectrum $\{1, \sqrt{2} - 1\}$ of Step 25 is now closed-form: at $N = 8$ the Hankel pencil is diagonal, $\tan(\pi/8) = C(3)/C(1)$, the doubling is exact fermionic second quantisation, the geometric carrier is the silver-midpoint axis of the μ_4 torsor, and the gap is $\operatorname{arcsinh}(1) = \operatorname{gd}^{-1}(\pi/4)$. N-scaling ($N = 8..24$) shows a pure $N = 8/16$ Majorana *lattice* datum — not a continuum limit; a positive μ_4 clock exists only for $N \equiv 0 \pmod 8$; the golden family appears as the $N = 10$ control. The connection verdict is an honest KILL in the v100/v513 style: a preregistered 11466-candidate census over the 16 frozen registry numbers yields 0 hits above null expectation, with golden/plastic placebos comparable — no readout connection. Cross-link to Step 27: $\sqrt{2} - 1$ is the thermal kernel ratio. Double verdict; no marker moves. [`verification/v527_seam_clock_silver_spectrum.py`]

Step 29 — the bit as a twist-class choice: the physical definition [E]/[C]. Step 26 harvested two bit-presupposing $\pi/2$ structures (the η flip and the twist frustration). Both are now exact *equivalences* to the bit, both directions each. Facet #14: η-flip $\Leftrightarrow \delta = \pi/2$ ($\sigma \circ r = -\sigma$ on all four fixing axes; off $\pi/2$ on every axis $\sigma \circ r = +\sigma$; the cardinality lemma $\sigma \circ f = -\sigma \Rightarrow |A_+| = |A_-| \Rightarrow m = 4$; g-class census over all 224 axes: flip class exactly at $\{15, 31\}$ for $m = 4$). Facet #15: twist-frustration $\Leftrightarrow \delta = \pi/2$ (θ exchanges $D_{\text{short}} \leftrightarrow D_{\text{long}}$; Gram indefinite $(4,4,0)/(8,8,0)$ exactly at $\pi/2$; off reducible and PD; same cardinality lemma). The order parameter $O =$ flip-axis fraction equals $1/2$ at $m = 4$ and 0 everywhere else, with class alternation $(-,+,-,+)$ the nontrivial $\mathbb{Z}_2$ character of the $\mathbb{Z}_8$ clock-tower shadow; gauge-robust under every v525 freedom. Unification: η-flip = frustration = flag-transitivity are *one* $\mathbb{Z}_2$ invariant (same arc system); the v510 Kramers class is another (constant on the family) — Kramers = POSITION half (derived), twist class = MODULUS half (the input). The v512 web grows $13 \to 15$ facets. [C] measurement sketch (principle only): defect transport around the axis torsor accumulates the η holonomy = a relative π phase iff the bit is set — in principle interferometrically readable. Honest fence: the bit remains formal input; a definition replaces no derivation. No marker moves. [`verification/v528_seam_bit_twist_class_definition.py`]

Step 30 — the interacting FK toy: the first firing Kill-Test-2 shadow [E]/[C]/[O]. The door left open by Steps 22/26 — a genuinely interacting $\mathcal{A}_{\text{hol}}$ — is probed on the minimal Fidkowski–

Kitaev quartic on the 16-Majorana NS seam circle (256-dim, exact). Triple verdict. F1 = KILL (the most important part): Θ exists exactly on the interacting model ($U_r^2 = 2^8$, $\Theta^2 = +1$, clock-tower inversion $\mathrm{conj}_V \cdot V = 4096$, $[\Theta, H_g] = 0$, $\Theta\Omega_g = \Omega_g$) — Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: reflection positivity breaks in the interacting ground state for every $g > 0$ — inertia ladder $(37, 0, 0) \to (33, 4, 0) \to (31, 6, 0) \to (29, 8, 0)$ (most negative eigenvalue -4.5×10^{-2}); the Gibbs state at $\beta = 2$ carries the same ladder; the mechanism is interference (neither crossing nor half-local terms alone break); the STRADDLE LAW holds on all 24 entries — RP fails exactly on quartet-straddled cuts and stays PD on quartet-avoiding ones (the 24 entries cover *asymmetric* straddles only — refined in Step 31). Mandatory fence: one toy, one interaction class, [C] flat-band parent — this *narrows* the Woit route (every $\mathcal{A}_{\mathrm{hol}}$ must protect RP against exactly this mechanism; the straddle law is a concrete hurdle for β_3/γ), it does *not* close it; `WOIT.OS.TWISTOR.01` stays [O]. F2 = KILL: the *tenth* side-blind test on the `v512` scoreboard ($9 \to 10$), the first with genuinely non-Wick-computable dynamics — the interaction sees δ massively (spectral spread 1.32), but the mark-axis mirror Θ_{15} maps $H(m) \to H(8 - m)$ exactly; all side data are mirror-equal. F3 = ERFOLG: grading and the nonsplit $\mathbb{Z}_8$ tower survive ($\tilde{U}^2 = 256\Gamma = (-1)^F$, $V^2 = 256\tilde{U}$); the FK fixed-point anchor holds; the v524 OS quotient survives $g > 0$ constructively on the $\delta = \pi/2$ mark-anchored model. No marker moves. [`verification/v529_seam_interacting_toy_fk.py`]

Step 31 — the straddle cone executed: reflection positivity *selects* the bit [E]/[C]. The review contract left by Step 30 (independent couplings λ_q on the four mark quartets of $M(\delta)$; the RP cone on the equivariant family) is executed, double-verdicted. V1 = KILL: the literal *leading-order* cone is empty for every member and both signs (first-order perturbation of the OS Gram via the exact reduced resolvent, Hermitian to 10^{-15}, central-difference regression 2×10^{-8}) — the linear pencil cannot carry the selection; the recoverable leading-order content is the straddled-cut drift sign pattern ($\pi/2$: $+1.4 \times 10^{-4}$ for $s{=}+$, -3.7×10^{-2} for $s{=}-$; the asymmetric members exactly reversed) — necessary, not sufficient. V2 = ERFOLG, *the selection law* (the central datum): among *all* well-posed members and both coupling signs, reflection positivity holds on *every* admissible cut over the whole grid $g \in \{1/32..8\}$ for *exactly one* combination — $\delta = \pi/2$ with *positive* coupling: all four Grams PD $(37, 0, 0)$ *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ($1.78 \times 10^{-6} \to 5.5 \times 10^{-5}$ near $g = 1/4$), Gibbs $\beta = 2$ identical; the asymmetric members die ($s{=}+$ already at $g = 1/32$, $s{=}-$ from 1/8: the protection at $\pi/2$ is *nonperturbative*). The Step-30 straddle law refines: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it. The geometry behind the dichotomy is exact: $\delta = \pi/2$ is the *unique* member whose four quartets partition the 16-circle and whose straddling quartets are reflection-closed across their cuts with sign $+1$ (stabilizer = the full clock+mirror group, four admissible cuts $\{3, 7, 11, 15\}$); the equivariant Fix rays are uniform on $m = 2..6$ and NS-wrap-twisted on the tight members $m = 1/7$. Typed cross-link: the RP-selected member is exactly the carrier of the Step-29 twist-class order parameter $O = \frac{1}{2}$ — the dynamical selector and the kinematic definition point at the *same* member. Honest fence: one toy, one interaction class, [C] flat-band parent, deg ≤ 2 OS bases — toy-level *evidence for* the bit's dynamical origin, not a derivation; `WOIT.OS.TWISTOR.01` stays [O]. No marker moves. [`verification/v534_seam_straddle_cone.py`]

Step 32 — the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed: the OS cut and the $(2, 2)$ signature are forced, and the kill branch is empty [E]/[C]. The WOIT-β_3 milestone, executed as a compatibility-class construction. A $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality is a nondegenerate Hermitian form Φ on $\mathbb{C}^4$; clock compatibility forces the two-block structure exactly ($16 \to 8$ real parameters; the deck adds nothing, the wrong clock lift breaks the split), and the pairing against Woit's Euclidean structure ρ_{tw} splits the class into exactly two branches: the *Euclidean* branch (σ_{std}'s home; seam-line norm $a(|z|^2 + 1)$ — no null point, no OS cut, ever) and the *Minkowski* branch, whose null locus meets the seam line *exactly in the seam circle* (the OS cut is forced, not imposed) and whose total signature is $(2, 2)$ for every nondegenerate member (the second block is minus the conjugate of the first; charpoly mirrored): **Woit's Minkowski signature is a theorem of clock + ρ_{tw} pairing**. The contract kill branch — "the duality forces a clock-centralising (family A) structure" — is *empty* *by algebra*: every compatible duality inverts clock and deck (family D; family A would need $\rho_4^2 =$ scalar, but $\rho_4^2 = \mathrm{diag}(-1, 1, -1, 1)$), and family A independently has no quotient ($(4, 4, 0)$). Every seam-preserving Minkowski member induces *one* projective conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to $\sigma_{\rm std}$'s $z \to +\bar{z}$; under the declared RP-side placement (exact rational-π arithmetic, offsets $\mp\pi/2$) that member *is* $\theta_\perp$ — the mirror whose descent Step 25 certified: $\Theta_{\rm phys}$ is the shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality (quotient regressions re-verified: Gram PD, anti-unitary with $\eta_\perp = +i$, $\Theta^2 = +1$ per sector, time reversal, torsor closure onto the deck). Role separation complete: $\sigma_{\rm std} \sim$ the OS cut (the metric), the Φ-twisted Minkowski duality $\sim$ the descending mirror. Honest fence: free/quotient level only, $\mathcal{A}_{\rm hol}$ untouched, all seven kill tests live there; the declared conventions (seam line, seam preservation, placement, torsor relabeling) are cited from Steps 1/20/25, not invented; γ is next. No marker moves. [`verification/v565_woit_beta3_pt_duality.py`]

What remains, honestly (WP5e proper [O]). The single open step is the *global* BCOV/Kodaira–Spencer quantisation on $\mathbb{PT}/\mathbb{Z}_4$: the boundary partition function E_4/η^8 *derived from the twistor side*, including the $q^{-1/3}$ prefactor whose CFT side Step 9 has pinned and whose equivariant skeleton Step 11 has verified — the exchange sub-branch is closed by Step 12, the full-tensor ledger is executed by Step 14, the level-from-flux dial is executed by Step 13, the bulk-axion slot is built by Step 15, all three back-reaction milestones it preregistered are executed by Steps 16–18 (the Step-15 fence M1–M3 is *fully worked off*), the δ_1 chain is *decided* by Step 19 (kill under the derived measure), and the measure question is *decided at probe level* by Step 21 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), and the constructive BCOV derivation of the w_m normalisation itself is *executed* by Step 24 ($1/\det_j$ computed from three independent sources; nothing in w_m is declared any more at that level), so the named remaining target narrows to the *global BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor*; the cubic d-channel stays not needed and its physical-justification question ($c_d = 32 \times 60$; the $1920 = |W(D_5)|$ reading look-elsewhere-loaded and convention-contingent, v513) stays dissolved (Steps 19 and 21 do not revive it) — together with the continuum uplift of the Step-10 lattice witnesses (Xu's theorem for the abstract orbifold statement; the concrete seam quotient net and the interacting condensed $(E_8)_1$ net are Costello–Li territory). The route does *not* close `SEAM.EQUIV.01`; it *is* the keystone's second, quantitative route, named in its own right as `SEAM.EQUIV.TWISTOR.01` ([O]; the conditional MMST route is `SEAM.EQUIV.MMST.01`, closed modulo cited theorems; the parent closes if either route closes and stays [O] as an unconditional claim) — the same two-step shape as MMST (limit + maximal ideal) with the ideal quantitatively fixed, the deleting operator explicit, the limit state constructed, the index chain measured, the KLM triple witnessed, both faces of the inflow anchored, the collapse confirmed full-tensorially with one cubic door open, one level a theorem with its value pinned by the sector counter, the bulk-axion slot a construction with $\tilde{\lambda} = 6$ triply pinned, the back-reacted Ω_N closed-form with $(2\pi i)^2$-integral lockstep periods and the source charge $4 = |\mu_4|$ forced, the twisted KS measure (on the declared completion reading) landing on the unique quartic-free weighting, the a_0 uplift coupled to the centre count on four scales, the δ_1 chain decided — kill under the derived measure — the measure question decided at probe level for the declared reading (Step 21) and the w_m normalisation itself derived constructively (Step 24) — while the global quantisation (now narrowed to the global BCOV integral beyond the fibre zero-mode factor) stays open. `SEAM.EQUIV.01` stays [O] throughout; no marker moves.

The Woit bridge: chiral Wick rotation and the real structure [O]. One *external* programme is close enough in shape to deserve a named paragraph — named precisely so that proximity is not mistaken for confirmation. Woit's *Euclidean Twistor Unification* (`arXiv:2104.05099`) starts from the observation that the usual Wick rotation is problematic for *purely chiral* theories (the Euclidean continuation of a chiral fermion is not a chiral fermion; $SO(4) \cong SU(2) \times SU(2)$ splits what $SO^+(3, 1)$ holds together), and proposes the counter-move: formulate the theory *Euclidean-holomorphically on projective twistor space* $\mathbb{PT}$, and reconstruct the Lorentzian theory from a conjugation structure / boundary values — the real form is *recovered*, not assumed. TFPT's celestial/twistor route has, independently, assembled the very ingredients such a reconstruction needs: reflection positivity of the seam collar (v379; the Osterwalder–Schrader axiom, machine-witnessed), the OS reconstruction step itself (the cited AMT/OS selector in `FORM.SEAM.MMST.01`), the order-4 clock $\rho = \mathrm{diag}(i, 1)$ acting on the twistor sphere (v492), and the seam reflection (Step 20 makes the referent precise: the seam-circle *reflection*, not the deck/covering involution — the deck, whose freeness is topology v510, carries the $(-1)^F$ Kramers class instead). What is *missing* is stated exactly: a *global anti-linear map* $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \to \mathcal{A}(\mathbb{PT}/\Gamma)$ with $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$ on the *interacting* open+closed twistorial algebra, together with reflection positivity of the interacting BCOV+SDYM functional — the two inputs of the new central contract `WOIT.OS.TWISTOR.01` in `tfpt_research_contracts`. The α-stage status (Step 20, v519): *the missing global object now exists at the free level* — Θ with $\Theta^2 = +1$ and $\Theta\rho\Theta = \rho^{-1}$ on all four levels (sphere, $\mathbb{C}^4$, $\mathbb{Z}_8$ spin, Cl(16) Fock), with free RP on the seam system selecting the same family; Woit's two inequivalent real structures (ρ_{tw} and σ_{std}) are exactly the two families of that classification, in *different* contract slots. The β_1-stage status (Step 23, v522) sharpens the dictionary once more: the μ_4 clock is *Woit's euclidean rotation itself* (time-like: every seam mirror axis inverts it), the only gaugeable part of its tower is the GSO/fermion-parity $\mathbb{Z}_2$, and gauge-fixed RP holds under that typing — kill test (2)'s free shadow does not fire. The β_2-stage status (Step 25, v524) delivers the first *reconstructed* objects: the OS quotient of the free system is explicit — $(\mathcal{H}_{\mathrm{phys}}, \Omega)$ positive definite at both levels, the euclidean rotation a positive transfer step with spectral calculus and a reconstructed unitary rotation group (exact clock spectrum $\{1, \sqrt{2} - 1\}$ at $N = 8$), the compact-circle reconstruction thermal (KMS) with exactly the pre-declared silver witnesses. The β_3-stage status (Step 32, v565) closes the dictionary at the free/quotient level: Woit's Minkowski conjugation ($\mathbb{PT} \to \mathbb{PT}^*$) is constructed as the compatibility class of Hermitian dualities, its Minkowski branch carries the forced $(2, 2)$ signature and the forced OS cut on the seam circle, the kill branch (a clock-centralising family-A structure) is empty by algebra, and the unique induced seam member *is* the β_2 mirror $\theta_\perp$ — Θ_{phys} is the quotient shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality, with σ_{std} playing the role of the OS cut itself. *The interacting statement stays [O]* — kill tests 1 and 2 are discharged only at the free level, and until Θ + RP exist on $\mathcal{A}_{\mathrm{hol}}$, the Woit connection remains a strong analogy, not a mathematical integration; nothing in this paragraph is evidence for either programme, and no marker moves.

18 Honest status: bootstrap, not creation from nothing

- **What is genuinely shown [E]:** $g_{\mathrm{car}} = 5$ is an *overdetermined* fixed point of the E_8 closure (two independent Lie-theory conditions); the integer "8" in c_3 is the carrier Coxeter = E_8 rank; the discrete data closes a loop. The earlier landscape scan confirms $g_{\mathrm{car}} = 5$ is the *unique* value at which all hierarchies/family/gauge close — so the fixed point is unique, not one of many. **This uniqueness is now machine-formalised [E] in Lean 4** (Theorem A, `lean4-carrier-rigidity`): the division-free Pascal condition $2^g = g^2 + g + 2$ has the *unique* solution $g = 5$ (`carrier_rank_pascal_unique`; growth lemma for $g \geq 6$; $N_{\mathrm{fam}} = (2^4-1)/5 = 3$, $g+N_{\mathrm{fam}} = 8$), audited to use only the standard kernel axioms. The 2026-06-11 hardening round adds two formal modules: `AnchorLadder.lean` (the anchor power-sum law $p_n(a) = 2 + 2^n$ for *all* n, the ladder atoms $(3, 4, 6, 10)$, 240/8/248, and the binary ladder $p_{n+1} - p_n = 2^n$ as general theorems) and `PascalLadder.lean` (the odd-midpoint ladder identity $\sum_{k \leq K} \binom{2K+1}{k} = 2^{2K}$ for *all* K via Mathlib, plus the kernel-decidable bounded uniqueness of the K=2 carrier case) — the solves-side of the Pascal Ladder Theorem ([`verification/v108_pascal_ladder.py`]) is thereby [E].
- **What this is *not*:** a derivation from zero inputs. A self-consistency loop is consistent *at* a value; uniqueness is what makes it predictive (and that is supplied by the landscape scan). The continuous π is not produced by the loop — it is the Möbius/boundary geometry, the one irreducible primitive.
- **Conceptual gain:** the theory's foundation tightens from "two free axioms" to "*the* E_8-closure fixed point + the geometric π." That is the Möbius self-consistency you intuited: the seed is the

start of the cascade, and the cascade returns the seed's discrete content.

- **The current reduction state (2026-07-22; text consolidation, markers unchanged):** the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *one discrete symmetry-lift bit* (flag transitivity $\iff$ $\tau = i$; equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v499/v506/v507/v510/v512), and π — with the [C] identifications R1/R2 of v499 as the remaining conditional layer. `AX.P2.01` stays a declared axiom and `QGEO.REALIZE.01` stays [C]/[O]: this line describes the *reduction state*, not an abolition.

Part IV
Open items and the E_8-slice scan

19 The honest open-items list

This is the complete current ledger of what is *not* fully closed, graded by how hard the gap is.

#	Open item	Grade	Hardening path / what closes it
1. SM core — finite computations, no new physics			
1a	absolute quark c-scale (ratios closed)	[O]	quark *ratios* are integer Plücker readouts on the derived selector stratum (v49/v69/v71); only the *absolute* $U_f^\star$ normalisation remains — an anchor
1b	θ_{12} solar angle	[E]	**texture closed (below):** explicit $\mu\tau$-symmetric texture verified; only $\varepsilon{=}\frac{3}{4}\varphi_0^{\rm ret}{\approx}c_3$ (0.23%) stays conditional
2. Scheme layer — standard RG, no new input			
2a	$m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$	[C]	RG threshold matching $R_{\rm SM}$; m_H on the closed UV quartic $1/16\pi^2$
2b	hadron pole spectrum (singlet)	[C]	singlet algebra $\mathcal{A}_{\rm sing}$ explicit; dynamics already closed (gap $6\log\frac{3}{2}$)
3. Conditional — named analytic hypotheses			
3a	gravity: local eq. parameter-free; ambient redundancy	[E]/[C]	full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ (v358/v359); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure a [C] redundancy (v369)
3b	strong CP / QFT closure	[C]	reflection positivity (RP) is the one remaining input
3c	Λ branch label (rec vs car)	[C]	one discrete choice; typed via $(8\pi)^2$, value not fixed
4. Axioms — not eliminable, only hardenable			
4	P1 (c_3), P2 ($g_{\rm car}{=}5$)	[O]	more Lean; P2 algebra already machine-verified; remain inputs
5. Frontier — typed transfer targets, never compiler powers (see frontier doc)			
5	η_B, m_p/m_e, Koide, dark matter, ambient QG measure	[C]	transfers now runnable solvers with kill tests (v371–v375); ambient measure a [C] redundancy (v369); not forced (scan results below)
6. Cosmology late-time pipeline — not simplified by the compiler			
6	C_ℓ transfer, $a_{\ell m}$ seam, baryogenesis	[C]	frontier-doc machinery; $a_{\ell m}$ "good CMB world, not yet this CMB world" (SMICA test)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Net. Items 1 are finite arithmetic; 2 are standard RG; 3 hinge on named hypotheses (RP, the Λ branch, the UV completion); 4 are the two irreducible axioms; 5–6 are honestly open. The inflation sector (the only cosmology piece the compiler *does* sharpen) is now a parameter-free prediction ($M{=}c_3^{7/2}\bar{M}_{\rm Pl}$, document 2).

Item 1b closed: the explicit solar-angle Majorana texture [E]

The last open SM angle is now realised by an *explicit* texture, not a fitted number. Take the $\mu\tau$-symmetric Majorana matrix (tri-bimaximal at $\varepsilon{=}0$)

$$M_\nu(\varepsilon) = \begin{pmatrix} -\eta(\varepsilon) & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \eta(\varepsilon) = \frac{2\sqrt{2}}{\tan 2\theta_{12}} - 1, \quad \sin^2\theta_{12} = \tfrac{1}{3}(1-2\varepsilon). \tag{1}$$

$\mu\tau$-symmetry forces $\theta_{23} = 45°$ and $\theta_{13} = 0$ *exactly*; diagonalisation at the TFPT solar deviation $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$

gives

$$\boxed{\sin^2\theta_{12} = \tfrac{1}{3} - \tfrac{\varphi_0^{\rm ret}}{2} = 0.306747, \quad \theta_{23} = 45^\circ, \quad \theta_{13} = 0} \quad \textbf{[E]} \tag{2}$$

Honest residual ([verification/v9_neutrino_texture.py]). The texture and the angle are exact. The seam identification is $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret} = \frac{3}{4}\big(\frac{1}{6\pi} + \frac{3}{256\pi^4}\big) = c_3 + \frac{9}{1024\pi^4}$: the *leading geometric term is exactly* $c_3 = \frac{1}{8\pi}$, and the 0.23% residual is purely the $\varphi_0^{\rm ret}$ seed tail (document 2). So $\varepsilon = c_3$ holds at leading order; the full $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ is what enters the texture. The reactor angle θ_{13} ($\sin^2\theta_{13} = e^{-5/6}\varphi_0^{\rm ret}$, document 2) enters through a separate $\mu\tau$-breaking term and is not part of this minimal solar texture. **Scope:** this realises the TFPT solar shift in the $\mu\tau$-symmetric *limit*; it is *not* the full PMNS matrix. The θ_{23}=45° value is the symmetric-limit centre, *not* an octant prediction — current global fits (NuFIT 6.0) leave the octant unresolved — and the reactor angle plus the combined matrix require the separately stated $\mu\tau$-breaking channel.

20 The E_8-slice scan: results (honest hit / miss)

All seven maximal slices of 248 were scanned for the unassigned quantities. The rule was a meaningful (low-complexity) compiler/slice expression matching to $< 0.5\%$; matches that need a free fitted factor are reported as *misses*.

Slice / target	Result	Verdict
$A_4{\times}A_4 = SU(5)^2$	$\sin^2\theta_W\vert_{\rm GUT} = 3/8$ sits in the $SU(5)$ hypercharge embedding; the 12 coset states are the proton-decay X, Y bosons	[E] (known)
$F_4{\times}G_2$ / Koide	Koide $Q = 2/3 = \vert\mathbb{Z}_2\vert/N_{\rm fam}$ (sheet over families); $J_3(\mathbb{O})$ has dim 27=26+1, 3×3 octonionic "generations" on the diagonal	[C] structural
$D_8 = SO(16)$	$128 = (16,4) \oplus (\overline{16},\overline{4})$ — *same* spinor as $D_5{\times}A_3$; ν_R (seesaw) lives in the 16; **no new dark-matter state** (only the 4th μ_4 direction)	[E]
$A_8 = SU(9)$, 9=3×3	$\Lambda^3(3,3) = (10,1){\oplus}(8,8){\oplus}(1,10)$: the $(8,8)$=64 dominates — naive family×colour **fails**	miss
$A_8 \supset SU(5){\times}SU(4)$	$\Lambda^3(5{+}4)$ contains the SM **10**: a genuine chiral-family-unification *lead*, but not TFPT-forced	lead
$E_6{\times}A_2$ / Jarlskog	cross term $2(\mathbf{1}^\top Ra)N_{\rm fam} = 162 = 248{-}78{-}8$ exact; $J \sim \lambda_Y^6$ only to right *order*, no clean hit	weak
E_8 theta shells	root shells $240, 2160, 6720, \ldots$; $31{=}2^{g_{\rm car}}{-}1$ is *not* a clean shell ratio	inconcl.
$m_p/m_e = 1836.15$	no clean E_8/compiler expression (a QCD-confinement ratio); apparent hits need a free factor	**miss**
$\eta_B \approx 6.1{\times}10^{-10}$	sits *between* $c_3^6{=}4{\times}10^{-9}$ and $c_3^7{=}1.6{\times}10^{-10}$; right order, no integer power	order only

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Genuine new readings the scan produced

- **Koide** $= \vert\mathbb{Z}_2\vert/N_{\rm fam}$**.** The empirical Koide value 2/3 is exactly the sheet over the family count — a clean (if simple) compiler reading, promoting Koide from "near 2/3" to "democratic $\vert\mathbb{Z}_2\vert/N_{\rm fam}$."
- E_8 **as the optimal 8-bit code.** E_8 is the optimal 8D lattice (kissing number 240). In byte framing $256{-}248 = 8 = \operatorname{rank} E_8$ and $256{-}240 = 16 = \dim S^+$ — a clean information-theoretic reading of the two load-bearing E_8 numbers.
- **Dark matter = no new state.** The $SO(16)$ scan shows the 128 is the *same* matter spinor; there is no spare E_8 singlet to be a WIMP. Any DM candidate must be the determinant-line axion (frontier doc), not a new E_8 rep — a useful *negative*.
- $SU(9) \supset SU(5){\times}SU(4)$ **chiral lead.** Λ^3 puts the SM **10** in the antisymmetric; the cleanest unexploited route to a family-unification reading, flagged for future work.

Honest negatives

m_p/m_e has *no* clean E_8/compiler derivation (it is a confinement ratio — the apparent "hits" all need a free fitted factor); η_B matches only at the order-of-magnitude level ($\sim c_3^{6.x}$); the Λ rec/car branch is *not* selected by the theta series; the Jarlskog invariant is right only to order λ_Y^6. These are reported as open, not forced.

21 Following the $SU(9)$ lead: cross-slice consistency

The one scan lead worth following is $A_8 = SU(9) \supset SU(5) \times SU(4)$. Splitting $9 = (5,1) \oplus (1,4)$ the E_8 branching $248 = 80 \oplus 84 \oplus \overline{84}$ decomposes *exactly* as:

piece	$SU(5) \times SU(4)$ content	reading
80 (adj)	$(24,1)+(1,15)+(5,\bar{4})+(\bar{5},4)+(1,1)$	$SU(5)_{\text{GUT}} + SU(4)_{\text{fam}}$ gauge
$84 = \Lambda^3 9$	$(10,1)+(10,4)+(5,6)+(1,4)$	the SM **10** as family-singlet + family-**4**
$\overline{84}$	$(\overline{10},1)+(\overline{10},\bar{4})+(\bar{5},6)+(1,\bar{4})$	the conjugate matter

(Dimension checks: 24+15+20+20+1 = 80; 10+40+30+4 = 84; total 248.)

The genuine content: A_8 and $D_5 \times A_3$ agree on the family [E]

The $SU(4)$ here is *the same* A_3 family group as in the construction slice $D_5 \times A_3$ — the four punctures μ_4 of $\mathbb{P}^1 \setminus \mu_4$. The $SU(5)$ is the Georgi–Glashow GUT sitting inside the carrier $SO(10) = D_5$ ($SU(5) \subset SO(10)$). So the two maximal slices are *consistent*: both read the family as $SU(4)$=A_3, and A_8 additionally exposes the $SU(5)$ matter content — the SM **10** appears as $(10,1) \oplus (10,4)$, i.e. a family-singlet **10** plus a family-**4**, where the **4** of A_3 is exactly TFPT's "$4 = N_{\text{fam}} + 1$" (three families + the μ_4 direction).

Honest limit: chirality is still the index, not E_8

$84 \oplus \overline{84}$ is *vectorlike* ($10 \oplus \overline{10}$ etc.): E_8 being non-chiral, the slice cannot by itself produce three *chiral* generations. The chirality is the boundary index $\text{Ind}(D^+_{\text{fam}}) = 3$ — the *same* input TFPT already uses. So $SU(9)$ is a consistent re-reading (matter in $SU(5)$, family in A_3), not an independent derivation of $N_{\text{fam}} = 3$. Each $SU(5)$ pair ($\mathbf{10} + \bar{\mathbf{5}}$) is anomaly-free, so the embedding is clean. [C]

Appendix: computational verification

The audit/bridge identities of this document (marked [E], [E]) are re-derived from $\{c_3, g_{\text{car}}\}$ and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the μ_4 glue E_8=$(D_5 \oplus A_3)+\mu_4$ (load-bearing).
- `v4_flavor_matrix.py` — the spectral selector $\det R$=$h(D_5)$=8, minors $\{2,3,5\}$.
- `v5_e8_cascade.py` — the cascade D_n=60$-$2n: endpoints, exponent rungs, IR tail, variance.
- `v6_bootstrap.py` — the self-consistency loop: $\mu^2-5\mu+4=0$, $g_{\text{car}}=5$ three ways, $8=\text{rank}\, E_8$.

The E_8-slice/atlas readings are explicitly [O] (audit-level), not part of the pass/fail suite. Run `python verification/run_all.py`; full map in `verification/README.md`.